

Earthquakes and Communal Conflict: Village-Level Evidence from Indonesia

Billy Thandy Tulungen^{a,c,*} Muhammad Halley Yudhistira^{a,b}

Jahen Fachrul Rezki^{a,b}

^aDepartment of Economics, Faculty of Economics and Business, Universitas
Indonesia, Depok 16424, Indonesia

^bInstitute for Economic and Social Research (LPEM), Faculty of Economics
and Business, Universitas Indonesia, Jakarta 10430, Indonesia

^cGovernment of North Sulawesi Province, Manado, Indonesia

*Corresponding author: `billy.thandy21@ui.ac.id`. Other authors:
`m.halley@ui.ac.id` (Yudhistira), `jahen.fr@ui.ac.id` (Rezki).

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Abstract

Do earthquakes raise or suppress communal conflict? Combining seven waves of Indonesia's village census (PODES, 2005–2024) with USGS ShakeMap intensities in a staggered difference-in-differences design, we find the evidence is compositional. Reported communal conflict falls while theft rises, and the two differ significantly under both fixed-effect structures ($p = 0.035$ under the stricter). Estimated event by event over 19 earthquakes, theft's divergence from the definition-stable communal outcome is 0.095 (bootstrap standard error 0.034), or 0.061 counting each earthquake once. Among the nine in ten treated villages without prior conflict, exposure raises onset by 0.70 points against a 2.1 percent rate.

Keywords: earthquakes, communal conflict, property crime, collective action,
Indonesia, difference-in-differences

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1 Introduction

Strong earthquakes do not simply make Indonesian villages more or less orderly. They change the kind of disorder villages report. After a village’s first damage-level earthquake, reported communal conflict falls while theft rises sharply, on the same survey form, recorded by the same respondent, in the same year. Assault does not move and robbery falls. The central result is not that disasters restore order. It is that the form disorder takes changes: coordination-intensive violence recedes while stealth property crime climbs, a pattern most consistent with a shock that degrades the village’s capacity for organised collective action.

This paper asks a question whose sign economic theory does not settle: does a strong earthquake raise or suppress communal conflict in the villages it strikes, and through which channels does it act? Large physical shocks are a recurring feature of life in hazard-prone developing countries, and communal violence is a development outcome in its own right: it destroys assets, deters investment, and erodes the local cooperation on which public goods provision depends ([Barron, Kaiser, & Pradhan, 2009](#)). The post-disaster period is also precisely when governments, security forces, and aid agencies decide how to allocate scarce resources between reconstruction and conflict prevention.

The empirical setting is Indonesia, the world’s most seismically active large economy and a country with a long record of localized communal violence ([Barron et al., 2009](#)). We combine seven waves of the Village Potential Survey (PODES, 2005–2024), a census of Indonesian villages, with USGS ShakeMap measures of ground shaking. Strong exposure is defined as Modified Mercalli Intensity (MMI) VI or above, the level at which ordinary masonry buildings begin to crack. Treatment timing follows the survey calendar: a village is first treated at the first PODES wave enumerated after its first $\text{MMI} \geq \text{VI}$ event. The baseline design treats first exposure as an absorbing state because earthquake damage, reconstruction, and emergency-state presence outlast the shaking itself. Two features of the sample bound what the comparison villages can be. The sample excludes villages outside the official medium/high seismic-hazard zones, so comparison villages come from the same broad risk set rather than from places with no meaningful earthquake exposure. And we estimate a same-earthquake design that compares villages shaken

above the damage threshold ($\text{MMI} \geq \text{VI}$) with villages that felt the same shock at sub-damaging intensity (MMI in $[4, 6)$), which corroborates the sign of the main estimate.

Within these zones, first strong exposure lowers reported communal conflict by 1.5 percentage points (standard error 0.004) against the treated villages' 6.1 percent pre-exposure rate, a decline of about a quarter. An episodic design in which treatment can switch off (de Chaisemartin & D'Haultfoeulle, 2026) gives a similar fresh-exposure effect of -1.5 percentage points. Both designs show the same time profile: the effect is concentrated in the wave of exposure and the wave after, with later lags not precisely estimated.

That average conceals a split we take seriously rather than relegate. The decline loads on the minority of villages that had already reported a communal-conflict episode. Among the roughly nine in ten treated villages with no prior conflict, a matched within-earthquake comparison finds that exposure *raises* the onset of communal conflict by 0.70 percentage points (standard error 0.22) against a counterfactual onset rate of 2.1 percent, a relative increase of about a third. The two arms are close in relative magnitude and opposite in sign. For most treated villages the shock does not pacify; it makes first-time conflict more likely, from a low base.

The finding that carries the paper is therefore compositional, and it is also the better-identified one. The same PODES forms that record communal conflict also record theft, robbery, and assault. Estimated on a single common sample of 211,082 village-waves, with the outcomes stacked and outcome-interacted fixed effects so that the standard errors account for the correlation of a village's outcomes, theft rises by 5.8 percentage points (standard error 0.9) against a 41 percent base while communal conflict falls by 1.5 points (standard error 0.4), robbery falls by 1.4 points, and assault does not move. A test of the equality of the theft and communal effects is rejected ($F = 64.63, p < 0.001$), as is equality of theft and the definition-stable *warga* outcome ($F = 51.14, p < 0.001$). The contrast survives a change of scale: re-estimated as Poisson, so that each coefficient is a proportional rather than an absolute response, the signs and the ranking are preserved.

The divergence also survives inference that treats the earthquake, rather than

the village, as the unit of assignment. Estimating the theft-minus-*warga* gap event by event over 19 usable earthquakes and bootstrapping whole earthquakes, the treated-village-weighted divergence is 0.095 (bootstrap standard error 0.034; 95 percent confidence interval [0.020, 0.155]) and the equal-event mean is 0.061 ($p = 0.046$). The pooled level decline does not clear the same bar: it holds when earthquakes are weighted by the villages they treat (-0.024 , standard error 0.010) but not under an equal-event mean (-0.015 , standard error 0.015). This is why we rest the paper on the contrast rather than on the level.

That contrast is what lets us discriminate among explanations. It is hard to reconcile with a uniform reporting artifact, with a general fall in disorder, or with a simple opportunity-cost story in which income destruction raises all forms of violence. It is more consistent with a production function for collective violence: motive is necessary, but so are inter-group contact, mobilization, and the ability to evade monitoring. The asymmetry within property crime is the part a rising-motive account alone does not deliver, since with encounters and guardianship unchanged a stronger motive for predation would raise theft and robbery together, and robbery falls. Direct measurement points the same way: high-involvement public-purpose *gotong royong*, the mutual-help collective action the questionnaire illustrates with night-watch rotations, falls by 3.1 to 3.2 percentage points after exposure, while emergency mutual aid, which needs no standing coordination, does not.

This paper contributes to the economics of conflict in two senses. First and foremost, we provide quasi-experimental evidence on how a large physical shock affects communal conflict, and we discipline the competing explanations for that effect by observing not only *whether* the shock moves conflict but *which margin of disorder* it moves. The cross-national record disagrees even on the sign: [Brancati \(2007\)](#) and [Nel and Righarts \(2008\)](#) find that disasters raise conflict risk, while [Slettebak \(2012\)](#) finds, if anything, less, and newer localized evidence finds that disasters can lower conflict where they strike but raise it nearby through spatial displacement ([Amarasinghe, Raschky, Zenou, & Zhou, 2026](#)). The reviews do not settle it either. [Koubi \(2019\)](#) finds no robust and general effect of climatic shocks on conflict onset and attributes the disagreement to differences in mechanism and context, whereas the hierarchical meta-analysis of [Burke, Hsiang, and Miguel](#)

(2015), pooling 55 studies, finds that deviations from moderate temperature and precipitation systematically raise conflict risk. What both call for is the same thing, evidence on the mechanisms that link a physical shock to violence, and that is what we provide. What these studies cannot observe is *why* a shock moves conflict in one direction rather than another, or whether a local decline is genuine suppression or mere relocation. Evidence of this kind, on which channels are and are not consistent with the data, is rarely available in this literature, and it speaks to the debate between opportunity-cost (Becker 1968; Dube and Vargas 2013) and group-structure accounts in which conflict turns on how a population is divided rather than on how poor it is (Esteban, Mayoral, and Ray 2012; Ray and Esteban 2017), within the organising framework of Blattman and Miguel (2010). We also confront the displacement concern directly: exploiting the geography of each shock, we find no detectable rise in conflict among nearby never-treated villages, which bounds, without eliminating, the concern that the suppression we estimate is merely the spatial relocation that Amarasinghe et al. (2026) document (Section 5.5).

Second, we document a methodological pitfall that travels well beyond this setting. In survey panels with unequally spaced waves, calendar-year exposure lags misaligned with the survey enumeration month both conceal the true response and manufacture a spurious one. Estimated on an identical sample, dating exposure by calendar year attenuates the two-wave effect from -0.022 to -0.003 and renders it insignificant, while producing a significant coefficient one lag later that the aligned specification shows to be zero (Table 20). This is a general caution for the growing body of work that merges the timing of physical shocks onto periodic household or village surveys, not a concern specific to our data. In showing it, we also provide the first village-level quasi-experimental estimate of the earthquake–conflict relationship for Indonesia.

What this design can and cannot establish. We state the binding limits here rather than distribute them through the paper. The pre-treatment record is mixed and we do not read either pre-period as clean. The average pre-treatment coefficient is small and insignificant for both outcomes, but a joint Wald test over all pre-treatment group-time cells rejects under wave fixed effects for the

composite ($\chi^2(15) = 42.9$, $p < 0.001$) and, less strongly, for *warga* ($\chi^2(15) = 29.4$, $p = 0.014$); the same test does not reject once province-by-wave effects absorb the province-specific trajectories that Section 4.1 identifies as its economic source. The differenced outcome on which the composition result rests has a pre-trend of its own, dominated by theft's, and it rejects ($p < 0.001$); we report this in Section 5.7.1 and rest the compositional claim on the shock-level divergence, which is estimated event by event, rather than on the village-level dynamic specification. We do not read the theft magnitude as a standalone causal estimate: its own pre-trend does not pass (-0.043 , $p = 0.04$), and theft can rise from weaker guardianship, from aid inflows, or from changed reporting. On mechanisms, the direct proxies we can measure are themselves trending and none is clean, and the sharpest direct test of the guardianship version of the account fails: theft rises by more, not less, where no security post existed to lose (Section 5.7.3). We therefore hold the capacity account as the leading reading rather than an established one, and treat Section 5.7 as reduced-form discrimination among channels rather than a mediation decomposition. Whether earthquakes also raise the motive for predation is something our data cannot show, and the pattern we find does not require it. Finally, no independent violence series covers the cohorts that identify these estimates: ACLED's Indonesia coverage begins in 2015, and Appendix P reports why the one database whose window reaches them cannot serve as an external check.

The paper proceeds from design (Sections 2–4) to results and the limits of the evidence (Section 5); Section 6 concludes.

2 Conceptual Framework

Strong earthquakes can plausibly raise or lower conflict, depending on which mechanism dominates. The framework here centres on two competing channels.

The inputs of collective violence. Communal violence draws on four inputs: grievance or economic stress; the opportunity for inter-group contact; the capacity to mobilize and coordinate a group; and, working against them, the intensity of monitoring by the state and community, whose capacity shapes whether grievances

escalate into organised violence (Besley & Persson, 2011). These inputs are complementary: organised violence needs grievance, contact, and mobilization capacity present together and monitoring weak, so removing any one of the capacity inputs suppresses it regardless of how strong the grievance is (Blattman & Miguel, 2010; Esteban et al., 2012; Ray & Esteban, 2017). A strong earthquake damages the venues and roads on which inter-group contact depends and absorbs survivors' time in recovery, so the assembly and coordination that mobilization requires become harder to supply. It may also deepen grievance through asset destruction and income loss, though our data cannot show whether it does, and the argument below does not need it to.

The organisational capacity a shock may degrade is not dedicated to violence. The same standing arrangements that let a village assemble a crowd also let it staff a night watch, so degrading them withdraws deterrence from property crime at the moment they withdraw mobilization from collective violence. In Indonesia these are the same institutions: the village night watch, *siskamling* and its *poskamling* posts (Barker, 1998). That observable guardianship deters property crime, and does so locally, is established by Di Tella and Schargrodsky (2004), who find a large deterrent effect of police presence confined to the blocks where officers were posted. Tajima (2014) argues that communal violence in Indonesia erupted where the withdrawal of authoritarian security exposed communities that had come to depend on the state rather than build arrangements of their own; his shock, the collapse of authoritarian rule, withdrew external monitoring while leaving village mobilization intact, so disorder rose, whereas an earthquake withdraws the mobilization as well, so the same account predicts a decline. A strong earthquake also disperses households and damages dwellings, which lowers the chance that an offender encounters a resident. Both movements are the classical inputs of the routine activity approach (Cohen & Felson, 1979), in which the dispersion of activity away from households raises property crime with motive held fixed. Temporary external presence may rise as responders and security forces arrive, which the counterinsurgency evidence suggests can suppress organised violence (Berman, Shapiro, & Felter, 2011), but it is directed at visible disorder, at crowds and fighting, and not at stealth taking from emptied houses, so it reinforces the

first movement without offsetting the second.

These consequences sort acts by the inputs each requires, and the sorting is the identification idea behind the mechanism analysis in Section 5.7. Collective violence needs both coordination and confrontation, so it should fall. Theft needs neither: it is a stealth act, deterred by guardianship rather than prevented by any coordination constraint, and an emptied dwelling is a better target rather than a lost one, so both movements push it up. Lapsed guardianship pushes robbery up, while scarcer encounters push it down, so theory signs only the comparison, that robbery must rise by less than theft, and leaves its level to the data. Assault is unsigned for the same reason with an added one: its encounters are largely among co-resident acquaintances, which dispersal thins far less than it thins offender-stranger encounters, so we count a null there as uninformative rather than as a success. The four acts therefore separate without any assumption about whether the motive for predation has changed, which is what makes their composition a test of capacity and not of grievance.

Income-destruction channel. The canonical opportunity-cost model (Becker, 1968) predicts that any shock reducing expected income lowers the cost of violence. The canonical development evidence for that channel is Miguel, Satyanath, and Sergenti (2004), who instrument economic growth with rainfall across 41 African countries and find that a negative growth shock of five percentage points raises the likelihood of civil conflict by about one half in the following year. Our shock is physical rather than a growth shock, and on the collective margin we recover the opposite sign, which is what the composition evidence below is meant to explain. Applied to earthquakes, Dube and Vargas (2013) show for Colombia that income shocks shift the calculus of predation through two opposing forces: a fall in wages lowers the opportunity cost of appropriation, while a rise in contestable rents raises its return, and the net sign depends on which force dominates. The analogous earthquake prediction is that asset destruction and labor-market disruption raise conflict; the channel implies a *positive* reduced-form effect.

Contact disruption and emergency coordination. The competing channel runs the other way: contact disruption suppresses potential violence regardless of

grievance, and disaster response can bring a temporary surge of external monitoring that raises the cost of organising violence (Berman et al., 2011); whether this external presence, rather than durable village policing, is what changes is an empirical question we return to below. Prior work shows that disasters can redirect social participation and preferences toward recovery-oriented coordination: Bai and Li (2021) find, for the same 2004–2007 Indonesian earthquakes, that seismic exposure raised community social participation, and Cassar, Healy, and von Kessler (2017) find that the 2004 tsunami raised prosocial behaviour in rural Thailand, alongside greater risk aversion and greater impatience. In our setting, however, the observed PODES proxies do not support a durable generalized social-capital surge; as Section 5.7.3 shows, the stronger evidence points to disrupted communication, contact, and mobilization capacity rather than to rising solidarity. And households in recovery mode concentrate resources on reconstruction rather than conflict, which itself changes the opportunity-cost calculus in the opposite direction from the income-destruction prediction (Dube & Vargas, 2013).

Why a temporary shock can have persistent effects. The shaking lasts seconds, but a strong earthquake is a temporary shock to the ground and a persistent shock to the village’s physical and institutional state. Structural damage at MMI VI and above is not undone when the shaking stops: houses, markets, roads, and places of worship stay damaged until reconstruction, which in rural Indonesia takes years. Exposure is therefore a state change rather than a pulse. This motivates the treatment definition in Section 4, where we model first exposure as an absorbing state, estimate the episodic design alongside it, and test directly whether the effect decays.

Testable implications. The framework yields five predictions, each tested below. First, communal conflict should fall, because it is the only act in our data that requires both of the inputs the shock degrades (Section 5).

Second, and most sharply, the four acts should separate by input rather than by motive. We separate coordination from confrontation by the responses of four acts to the same shock: theft should rise as guardianship lapses, communal conflict should fall as mobilization becomes harder to supply, robbery should rise by less

than theft because the encounter it needs becomes scarcer while the guardianship it evades also lapses, and assault is unsigned (Section 5.7.1). A story in which the shock simply raises the motive for predation predicts theft and robbery moving together, so the robbery margin distinguishes that account from ours. It does not, however, separate ours from an account combining a raised motive with physical displacement, which delivers the same four signs with no role for the night watch. We take that account seriously and test against it below.

The framework’s comparative static has already been observed with the opposite sign on the input. [García-Hombrados \(2020\)](#) studies the 2010 Chilean earthquake and finds property crime *falls* durably in exposed municipalities, attributing it to a strengthening of community life and the adoption of community-based crime prevention. That is the same dual-use input moving the other way, and it is consistent with evidence that disasters can raise collective participation ([Bai & Li, 2021](#); [Cassar et al., 2017](#)). Our contribution is not that earthquakes raise theft, which is not general, but that the sign of the theft response tracks the sign of the community-capacity response, which PODES lets us observe. [Cromwell, Dunham, Akers, and Lanza-Kaduce \(1995\)](#) make the same point for Hurricane Andrew, where informal guardianship rose after the storm and crime fell.

Third, the capacity inputs should be observably degraded. This is the prediction our data test least well. Collective participation is measured directly only from 2018, formal policing does not move significantly, and the community security post trends throughout the panel, so each direct proxy is either short, flat, or confounded (Section 5.7.3). We therefore treat the composition of crime, not the input measures, as the evidence that carries this channel. Fourth, because communal violence is fought *between* groups, the contact and mobilization inputs are more available where the population is more polarized, so the conflict-reducing effect should be larger where pre-treatment group polarization is higher (Section 5.7.5). Fifth, the two channels differ in timing and intensity: income destruction should produce its largest effect at short lags, fading as incomes recover, whereas contact disruption and coordination effects should persist; and because structural damage rises sharply with shaking, the effect should be larger where shaking was more intense. We test this with the same-earthquake band comparison (Section 5.6); the threshold ladder

does not settle it, for reasons given in Section 5.3. The same logic applies to the shock rather than to the village, and with a sharper implication. The inputs a village draws on to organise are not confined to the village, so what should matter is how much inhabited land a given earthquake shook at damaging intensity, not the Richter magnitude of the event or the intensity any one village experienced. The prediction is therefore that the effect is carried by large-footprint earthquakes and is absent for small ones, which implies in turn that an unweighted average over events should be attenuated toward zero relative to an average weighted by exposed villages. We test this at the level of the earthquake in Section 5.6.

3 Data and Measurement

3.1 Conflict Outcome: PODES

The main outcome variable is a binary indicator of communal conflict from Indonesia’s Village Potential Survey (PODES, *Potensi Desa*), a census of all villages conducted by BPS (Statistics Indonesia) every three years. PODES asks the village head (*kepala desa*) whether any intergroup conflict or mass brawl occurred in the village in the past twelve months.

It is a census, not a sample: every village answers the same question in every wave, so coverage does not vary with media attention or accessibility, the central weakness of event-based conflict data in rural areas. What it misses is also clear: it is binary (no intensity margin), retrospective over twelve months (no exact dates), and self-reported (a village head may under-report events that reflect poorly on the village). Idiosyncratic under-reporting that is uncorrelated with treatment attenuates the estimate toward zero, making our estimates conservative; treatment-correlated reporting is the serious threat, and we test it directly with a four-part reporting-artifact analysis in Section 5.7 and an external event-data comparison in Appendix H.

The analysis uses seven PODES waves enumerated between 2005 and 2024, each fielded in May of its label year (2005, 2008, 2011, 2014, 2018, 2021, and 2024). We begin in 2005, the first wave in which the mass-brawl conflict outcome is defined consistently. The analysis window of 2005–2024 covers six major seismic

events of national significance: the 2004–05 Sumatra-Andaman sequence, the 2006 Yogyakarta earthquake, the 2009 Padang–Pariaman earthquake, and several subsequent events in Lombok (2018), Sulawesi (2018), and West Java (2022).

Because the conflict question’s wording changes across waves, we construct a systematic crosswalk that harmonizes the definitions; the exact wording and variable coding for each wave are documented in the replication package. Our primary outcome is the broadest available measure of communal conflict in each wave, an indicator equal to one if the village reports any intergroup conflict or mass brawl. Because the specific categories that make up this measure expand from 2008 onward, comparability across waves is a real concern, and we address it directly in two ways: with a narrower subtype whose wording is stable across all seven waves, namely a civilian-versus-civilian brawl, which we call the *warga* outcome throughout, and with a dedicated robustness check in Section 5.6.

Village boundary harmonization and analysis sample. Indonesia underwent extensive administrative subdivision (*pemekaran daerah*) in recent decades, the number of villages growing from roughly 68,000 to 84,000, which poses a fundamental panel-construction challenge. We address it with a chain-linkage crosswalk that maps every village in each wave back to the boundary it occupied in 2005, its “2005 parent” village. We call the resulting panel the *2005-boundary panel* throughout. It is built from BPS Master File Desa records and supplemented by GPS spatial joins and manual verification where name matching fails. Because villages created by later pemekaran are mapped back to their 2005 parent polygon, administrative splits cannot mechanically enter or exit the sample. Appendix G documents the procedure, match rates by wave, and two crosswalk-robustness re-estimations. Controls are baseline (PODES 2003, enumerated August 2002) village characteristics, entered in levels for the Callaway–Sant’Anna nuisance model and interacted with wave dummies in the two-way fixed-effects specifications; after dropping observations with missing controls or outcomes, the analysis sample is $N = 292,712$ village-waves for the absorbing two-way fixed-effects specification and $N = 227,045$ for our primary Callaway–Sant’Anna estimator, which identifies villages first shaken in waves 3–8 (2008–2024).

3.2 Seismic Exposure: USGS ShakeMap MMI

The treatment variable is ground-shaking intensity, measured on the Modified Mercalli Intensity (MMI) scale, a twelve-point scale (I to XII) of how strongly an earthquake is felt and how much damage it does at a given location. We take MMI from USGS ShakeMap, which interpolates instrument and felt-report data into a fine grid of shaking values for each earthquake (Worden et al., 2018). For each event, we compute the average MMI across the grid cells that fall inside a village’s boundary; across events within a calendar year, we keep the maximum, so the measure reflects the most severe shaking a village experienced that year.

The event catalogue is drawn from the USGS Comprehensive Catalog (ComCat) for Indonesia and covers **1990–2024**, comprising 72,745 events at $M \geq 4.0$. It deliberately extends fifteen years before the first analysis wave. The absorbing cohort assignment is defined by whether a village was *ever* exposed, including before the panel opens, so a village shaken in the 1990s that was invisible to a shorter catalogue would be assigned to a spuriously late cohort or, worse, to the never-treated comparison group while in fact carrying a decade-old treatment. Appendix J reports the sensitivity of the headline estimate to the catalogue start year.

Treatment is defined two ways: (1) **continuous**: the village-year MMI value; and (2) **binary**: $D_{mmi6} = \mathbf{1}\{\text{MMI} \geq \text{VI}\}$, corresponding to “Strong” shaking per the EMS-98 scale at which widespread damage to ordinary masonry structures begins (Wald, Quitoriano, Heaton, & Kanamori, 1999). Exposure is mapped to the first PODES enumeration following the earthquake, that is the first wave whose May enumeration date falls after the event. Because the PODES conflict item covers the preceding twelve months, this mapping can leave the first treated wave only partially exposed when the earthquake falls inside that recall window; Section 5.6 audits the event dates and reports a clean-post restriction that drops those cases. Robustness to the intensity cutoff is assessed with alternative MMI thresholds (V and VII) in Section 5.6; robustness to the intensity *measure*, using an MMI derived independently from ShakeMap Peak Ground Acceleration, is reported in Appendix O.

For the staggered difference-in-differences estimators, which compare villages

by *when* they are first treated, we identify for each village the first calendar year in which its shaking reaches MMI VI or above, and map that to the corresponding PODES survey wave. Treatment is therefore first *observed* strong-shaking exposure during the ShakeMap catalogue period, not necessarily a village’s first strong earthquake in history. Village fixed effects absorb fixed differences associated with historical seismicity, though not any part of a past shock’s effect that itself evolves over the panel. Villages that never experience such shaking over 2005–2024 form the never-treated comparison group.

3.3 Sample Restriction: Seismic Hazard Zones

The analysis is restricted to villages in Indonesia’s Kawasan Rawan Bencana (KRB) Medium or High seismic-hazard zones, mapped by the Center for Volcanology and Geological Hazard Mitigation (PVMBG/ESDM) and consistent with Indonesia’s official national seismic-hazard assessment ([Irsyam et al., 2020](#)). Limiting to KRB zones ensures that all included villages have non-trivial earthquake exposure, so the not-yet-treated comparison villages are drawn from the same seismic risk set rather than from areas that face no earthquake hazard at all. The restricted sample covers Sumatra, Java, Bali, Nusa Tenggara, Sulawesi, Maluku, and Papua.

A natural concern is whether the hazard classification is itself endogenous to the earthquakes we study. It is not: the KRB classification is a hazard assessment based on fault and subduction-zone geometry rather than the realized damage of any particular event ([Irsyam et al., 2020](#)); the restriction enters the design only through village fixed effects, so identification comes from the timing of exposure within a fixed set of villages, not from the map’s boundaries; and dropping the restriction entirely and re-estimating on the full national sample leaves the effect intact. Appendix J reports the map’s vintage, the cohorts it predates, and the full-sample estimates.

Appendix Figures 6 and 7 map the ever-treated villages by first-exposure cohort and place the cohorts on the survey calendar, each anchored by a well-known destructive event.

3.4 Alternative Conflict Measure: ACLED

For robustness, we use ACLED (Armed Conflict Location and Event Data Project) event data geo-matched to village boundaries, available for 2015–2024. ACLED covers riots, violence against civilians, and battles involving civilian groups. Because temporal overlap with PODES is limited to three waves (2018, 2021, 2024), ACLED is used only as an external comparison, not as a primary outcome or a validation of PODES.

3.5 Descriptive Statistics

Table 1 presents summary statistics for the primary analysis sample. The pooled conflict probability is 2.6 percent; roughly 1.2 percent of village-waves experience $\text{MMI} \geq \text{VI}$ in the prior wave. The low baseline matters for reading magnitudes later: communal conflict is a rare event, so a treatment effect of a few percentage points is large in relative terms. The pooled mean, however, understates the relevant benchmark: ever-treated villages reported conflict at 6.1 percent across their pre-treatment waves, roughly twice the pooled rate, and Section 5 interprets all estimates against this treated-cohort baseline rather than the pooled mean.

4 Empirical Framework

4.1 Choosing Among Specifications

The counterfactual this design needs is a village in the same province observed in the same survey wave, and that is where the economics points rather than a statistical convenience. Indonesia’s seismic zones lie disproportionately outside Java, in post-conflict provinces on their own declining conflict trajectories. Village fixed effects absorb the level of such a trajectory but not its slope, so a comparison that pools across provinces credits the earthquake with a decline the province was already experiencing. The preferred specification therefore absorbs village and province $\times$ wave fixed effects and excludes the always-treated cohort.

Raw cohort paths, plotted in Appendix Figure 4, are consistent with both features of that structure. Treatment cohorts differ substantially in the level of

Table 1: Descriptive Statistics: Medium and High Seismic-Hazard Zones

Variable	Mean	Std. Dev.	Min	Max	<i>N</i>
<i>Panel A: Outcomes</i>					
Any communal conflict (composite)	0.026	0.158	0	1.000	303,457
Communal conflict, definition-stable (<i>warga</i>)	0.016	0.126	0	1.000	303,457
<i>Panel B: Treatment</i>					
Ever exposed $\text{MMI} \geq \text{VI}$ (absorbing)	0.272	0.445	0	1.000	303,457
Strong shaking, $\text{MMI} \geq \text{VI}$ (window t-1)	0.012	0.109	0	1.000	303,457
MMI intensity (mean-then-max, window t-1)	2.298	1.903	0	8.330	303,457
<i>Panel C: Baseline Control Variables (PODES 2003, enumerated August 2002)</i>					
Any household electricity	0.945	0.225	0	1.000	294,357
Paved road access	0.562	0.494	0	1.000	294,357
Internet available	0.032	0.175	0	1.000	294,357
Police post present	0.102	0.301	0	1.000	294,357
Health centre present	0.414	0.490	0	1.000	294,357
Number of primary schools	2.480	2.365	0	32.0	294,357
Distance to district capital (km)	44.1	54.6	0	985.0	292,712

Sample: KRB Medium/High seismic-hazard zones, PODES waves 2005–2024, native 2005-boundary baseline (43,351 villages $\times$ 7 waves). Panel C reports the baseline controls used throughout: PODES 2003 village characteristics, entered in levels for the Callaway–Sant’Anna nuisance model and interacted with wave dummies in the two-way fixed-effects specifications. Control *N* falls below the outcome *N* because baseline characteristics are missing for a small number of villages.

reported conflict, which is what village fixed effects absorb, and the never-treated path is close to flat with a slight downward drift after 2018, which is the regional trend province-by-wave effects are meant to absorb. Those paths are unadjusted and the smaller cohorts are noisy, so we read them as motivation for the fixed-effect structure rather than as evidence on parallel trends, which the tests below address directly.

Staggered designs admit many defensible specifications, and choosing among them after seeing the estimates invites exactly the criticism such designs attract. Our preferred specification is therefore selected on the economically relevant counterfactual set out above, not on how the diagnostics come out. The pre-trend test then serves as validation rather than as the selection criterion, and it validates: the joint test of the pre-treatment leads does not reject for either outcome ($p = 0.64$ composite, $p = 0.70$ *warga*), whereas under wave fixed effects the Callaway–Sant’Anna joint Wald test over pre-treatment group-time cells rejects for both ($p < 0.001$ and $p = 0.014$). These are different statistics, for a reason given in Section 4. A reader who prefers a mechanical rule reaches the same design, since this is also the simplest specification that satisfies the test. The choice is

conservative on magnitude besides: of the two, it yields the *smaller* estimate.

We report the estimators below on the wave-fixed-effects panel as well, both because they are the conventional benchmarks and because the contrast between the two is itself informative.

4.2 Estimator: Callaway and Sant’Anna (2021)

A central causal estimator is the group-time average treatment effect from [Callaway and Sant’Anna \(2021\)](#), hereafter CS(2021):

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g], \quad (1)$$

where $G = g$ denotes the cohort of villages whose first treatment wave is g , and $Y_t(0)$ is the potential outcome the same villages would have had at wave t absent any earthquake. Because it never uses already-treated villages as a comparison group, it avoids the contamination that biases conventional two-way fixed effects when treatment timing is staggered ([Callaway & Sant’Anna, 2021](#); [Roth, Sant’Anna, Bilinski, & Poe, 2023](#)). We aggregate $ATT(g, t)$ to a single average effect across post-treatment periods using doubly-robust inverse-probability weighting ([Sant’Anna & Zhao, 2020](#)), taking villages not yet treated at each point in time as controls. Standard errors are analytical, based on the Callaway–Sant’Anna influence function; few-cluster checks use a wild cluster bootstrap (9,999 Webb-weight replications, seed 42).

Why absorbing treatment? An earthquake is an episodic physical event, so treating a village as exposed forever after its first strong shock requires justification: the framework of [Section 2](#) predicts hysteresis, the precisely estimated response is concentrated in the first two waves while later estimates are imprecise, and the episodic design of [Section 4.4](#) lets treatment switch off; [Appendix J](#) gives the full justification.

Uneven wave spacing. Because PODES waves are unevenly spaced (gaps of 3, 3, 3, 4, 3 and 3 years) while the estimator requires a regular time grid, we index

waves by their ordinal position rather than by calendar year. Indexing by calendar year would produce fractional time labels and drop most treatment cohorts.

Unbalanced panel and software. PODES does not observe every village in every wave, so the estimation panel is unbalanced. We use the default pairwise-balancing rule of the `csdid` implementation (Callaway & Sant’Anna, 2021), under which each $ATT(g, t)$ is computed on the villages observed in both of the two periods it compares, rather than restricting the whole analysis to villages observed in all seven waves. This retains cohorts that a fully balanced restriction would drop and is the panel analogue of the estimator in Callaway and Sant’Anna (2021). All CS(2021) estimates in the paper use `csdid` version 1.72; the replication package records the version of every user-written command.

Cohort composition. Villages first exposed before or during 2005 are always-treated in the 2005-boundary panel and are excluded from CS(2021) by the not-yet-treated control requirement. Cohorts G3–G8 (first treated 2008–2024) identify the estimator, with group-level ATTs for all six cohorts reported in Appendix Table 16. Because state capacity, reconstruction regimes, and conflict history differ across the period, treatment effects need not be homogeneous across cohorts; the estimand applies to cohorts entering treatment between 2008 and 2024, and we do not extrapolate it to the excluded early cohorts.

4.3 Benchmark Estimator: TWFE

As a benchmark, we estimate a two-way fixed-effects linear probability model (TWFE-LPM):

$$y_{vt} = \alpha D_{treat,vt} + \gamma' \mathbf{X}_{vt} + \mu_v + \lambda_t + \varepsilon_{vt}, \quad (2)$$

where $D_{treat,vt} = \mathbf{1}\{t \geq g_v\}$ is a binary absorbing treatment indicator equal to one from the first treatment wave onward, $\mathbf{X}_{vt}$ is a vector of seven village characteristics measured at the PODES 2003 baseline (household electrification, paved road access, internet availability, police post, health-centre presence, number of primary schools, and distance to the district capital), entered as levels in the doubly-robust estimators and interacted with wave dummies in the two-way specifications, μ_v are village

fixed effects, and λ_t are wave fixed effects. Standard errors are clustered at the village level following [Abadie, Athey, Imbens, and Wooldridge \(2023\)](#).

We also estimate a distributed-lag TWFE with exposure lags aligned to May-to-May windows, so each lag falls fully before the conflict reference window (full specification in [Appendix J](#)). Alongside CS(2021) we report the interaction-weighted event-study estimator of [Sun and Abraham \(2021\)](#).

The distributed-lag results, and the timing artifact that unequally spaced survey panels create when exposure windows are misaligned, are documented in [Section 5](#). CS(2021) remains the primary heterogeneity-robust staggered estimator for its explicit treatment of staggered timing. Two terms recur below and we fix them here. The preferred *counterfactual specification* is the province-by-wave two-way fixed-effects design of [Section 4.1](#); the primary *heterogeneity-robust staggered estimator* is CS(2021), which runs on the wave-effects panel because it cannot absorb interacted fixed effects. Every estimate below is labelled by the design it uses.

4.4 Complementary Estimand: Episodic Exposure (dCdH)

Because earthquakes are episodic physical events, we complement the absorbing design with an estimand in which treatment can switch off. We estimate the de Chaisemartin and D’Haultfoeuille ([2026](#)) estimator with a non-absorbing treatment indicator: $D_{interval,vt} = \mathbf{1}\{\exists \text{ month in interval}[t-1, t) : \text{MMI} \geq 6\}$, which switches between zero and one across waves. The dCdH Effect₁ captures the ATT one wave after switching to $D_{interval} = 1$, that is, the effect of fresh exposure in the current inter-wave window regardless of the village’s exposure history. Together, the two designs answer two distinct questions: what is the effect of having entered the post-earthquake state (absorbing), and what is the effect of being freshly shaken (episodic)?

4.5 Identification Assumptions

Physical exogeneity. The location and intensity of earthquakes are determined by fault geometry, seismic-wave propagation, and local geophysical conditions, which are plausibly orthogonal to short-run village-level social and political conditions, conditional on the hazard-zone restriction, village fixed effects, and pre-

treatment trajectories (Gignoux & Menendez, 2016; Kirchberger, 2017). The shaking a village realizes is nonetheless a nonrandom function of an exogenous shock, since villages closer to active faults face systematically higher expected exposure (Borusyak & Hull, 2023); the hazard-zone restriction addresses this directly by comparing villages that share a common expected exposure band, and village fixed effects absorb the remaining fixed differences in fault proximity.

Parallel trends. Under the CS(2021) estimator with covariates, the relevant assumption is *conditional* parallel trends: after accounting for observable pre-treatment differences in growth and infrastructure trajectories, the potential untreated outcomes of eventually-treated cohorts evolve in parallel with those of not-yet-treated villages (Sant’Anna & Zhao, 2020). We assess pre-trends in two ways, and they give a mixed verdict. Pre-treatment coefficients are cumulative deviations from the omitted wave immediately before first exposure, the standard event-study convention. The average pre-treatment coefficient is small and insignificant for either outcome (-0.0077 , $p = 0.33$ composite; -0.0038 , $p = 0.56$ *warga*). A joint Wald test of all pre-treatment group-time $ATT(g, t)$ coefficients, which does not net offsetting signs against each other, is less reassuring: it *rejects* for the composite outcome ($\chi^2(15) = 42.9$, $p < 0.001$) and, less strongly, for the definition-stable *warga* outcome as well ($\chi^2(15) = 29.4$, $p = 0.014$). Two things bound what that rejection can imply. It is specific to wave fixed effects, though the comparison cannot be made with the identical statistic: `csdid` cannot absorb interacted fixed effects, so the province-by-wave evidence comes from the joint test of the event-study leads rather than from the group-time Wald statistic. That test does not reject once province-by-wave effects absorb the province-specific trajectories that Section 4.1 identifies as its economic source ($p = 0.64$ composite, $p = 0.70$ *warga*), and the saturated Sun–Abraham event study, which can absorb them, agrees ($\chi^2(5) = 4.69$, $p = 0.46$). And the pre-treatment drift the data do exhibit runs *against* the finding rather than toward it: the unconditional group-time average pre-trend is positive ($+0.0107$, $p = 0.01$), as is that of the specification least exposed to bad-control concerns ($+0.0151$, $p < 0.01$), and a positive pre-existing drift biases a subsequent decline toward zero instead of manufacturing it. That

second argument is bounded, and we state its limit here rather than leave it to be found: it holds against a drift that *continues* into the post-treatment period, not against a transitory elevation that *reverts* by the time of exposure. What applies to the reverting case is a different residue, which Section 5.6 prices: conditioning on a passed pre-test is itself a selection step, and it displaces the immediate estimate by about 0.004, again toward zero rather than away from it. We nonetheless do not treat either pre-period as clean; we carry the *warga* outcome as the definition-stable series against which the composite is checked, and, for reasons of statistical power we take up below, rest the paper’s sharpest evidence on the cross-outcome composition contrast, whose identifying assumption we state explicitly in Section 5.7.1 rather than treat as absent. The always-treated cohort, first exposed in 2005, is excluded, so every identified cohort is first exposed between 2008 and 2024. The 2005 wave itself remains in the panel as a pre-treatment period for those cohorts, and it is the one wave in which the composite outcome uses the broader mass-brawl gate rather than the post-2008 communal subtypes. We therefore do not rest the result on the composite alone: the definition-stable *warga* series and the identical-wording mass-brawl gate move in the same direction (Table 18). We therefore do not rest identification on an unconditional parallel-trends claim. We treat conditional parallel trends as an assumption to probe rather than assert: Section 5.6.2 documents the covariate trajectories being conditioned on and shows that the direction of the residual growth imbalance is conservative for our finding; we report HonestDiD sensitivity bounds (Rambachan & Roth, 2023), under which the composite sign is robust on the Sun–Abraham event study but not on the primary estimator (Section 5.6.2), a gap that reflects a different target parameter and a different control group rather than a difference in evidence.

No anticipation. Seismic events at the frequency of PODES surveys (every 3 years) are unforecastable. Villages cannot systematically adjust conflict behaviour in anticipation of a future earthquake.

Measurement and reporting validity. The outcome is reported by the village head on the PODES form, and the same official leads the local disaster response, so one might worry that a decline in *reported* communal conflict reflects reduced

reporting capacity rather than reduced conflict. We treat this as a threat to identification, not merely to interpretation. The central defense is behavioural: theft is recorded by the same respondent on the same form, and it *rises* sharply while communal conflict falls, a composition no uniform reporting shock can produce. Section 5.7 develops this argument in full, adds three further observational checks, and reports placebo items from the same form (village-head age and sex); neither responds meaningfully, though the age placebo is marginal ($p = 0.09$).

Resolving a pre-period signal. An earlier calendar-year coding of first exposure produced a marginally significant one-wave pre-period coefficient. Diagnostic analysis traced this to wave-boundary timing: because PODES is enumerated in May, a village first shaken in, say, October 2010 was mechanically coded to treatment wave 2011 even though the event postdated the wave-2011 conflict reference window, misplacing onset by one wave for about 21 percent of ever-treated villages. Re-timing first exposure to the first survey wave enumerated *after* the initial $\text{MMI} \geq \text{VI}$ event removes that calendar artifact and returns the *average* pre-trend to insignificance (composite $p = 0.33$; *warga* $p = 0.56$). The averaged diagnostic is not the whole story, however: a joint Wald test over the pre-treatment group-time cells still rejects for the composite outcome ($\chi^2(15) = 42.9$, $p < 0.001$) and also, less strongly, for the definition-stable *warga* outcome ($\chi^2(15) = 29.4$, $p = 0.014$). We therefore carry the pre-trend as an open caveat rather than a settled diagnostic. Section 5.6 discloses the individual leads in full and reports the HonestDiD sensitivity bounds (Section 5.6.2), which are the binding constraint on what this design can claim, rather than the pre-trend tests.

Pre-treatment balance. Within the KRB Medium/High sample, eventually-treated and never-treated villages were closely comparable at the 2005 wave, before first exposure for every identified cohort: nine of ten normalized differences fall below the 0.25 threshold (Imbens & Wooldridge, 2009), and the one exception, the health-centre count at 0.378, is not load-bearing, since adding it interacted with wave moves every estimate by less than one percent (Appendix Table 21). The imbalance is visible but it is not doing the work. Remaining level differences are absorbed by village fixed effects, and the pre-trend tests in Section 5.6 address

differential trends.

5 Results and Discussion

5.1 The Preferred Specification

Table 2 reports the specification selected by the rule of Section 4.1: village and province $\times$ wave fixed effects, always-treated cohort excluded. First exposure to damage-level shaking lowers the probability that a village reports communal conflict by 1.5 percentage points against a pre-exposure mean of 6.1 percent, a reduction of roughly a quarter. The estimate is 1.4 points without covariates and 1.5 with them, so it does not rest on the control set.

The event-study rows show where the effect sits in time. It is present in the wave of exposure (-0.0169) and the wave after (-0.0208), and the three subsequent lags are jointly indistinguishable from zero ($p = 0.38$). Because the absorbing indicator averages over later waves in which villages have already recovered, it understates the effect at its peak; the pooled estimate is best read as smaller than the immediate response, because it averages across horizons.

Both outcomes pass the joint pre-trend test in this specification ($p = 0.64$ composite, $p = 0.70$ *warga*), as the economic reasoning of Section 4.1 anticipates. The definition-stable *warga* outcome moves in the same direction throughout but does not attain conventional significance here. Its pre-exposure mean is 3.6 percent against 6.1 for the composite, so it carries roughly half the power; we read it as corroborating in sign rather than as independent confirmation.

Because treated villages enter the window from elevated conflict levels, the obvious competing account is mean reversion, and it is worth stating the floor it leaves. Letting villages with different fixed 2005-baseline conflict follow their own wave trends, which is precisely the design that allows conflict-prone villages to revert along their own trajectory, still leaves the effect at -0.0104 (standard error 0.0026). Reversion can absorb part of the decline but not the whole of it. What it does bound is where the decline lives: the effect loads on the fewer than one in ten villages with a prior conflict episode, so the claim we defend is suppression where conflict already occurred, not prevention of onset.

Table 2: Preferred Specification: Province $\times$ Wave Fixed Effects, Always-Treated Cohort Excluded

	Any communal conflict	<i>Warga</i> (definition-stable)
Absorbing exposure, no covariates	−0.0137*** (0.0039)	−0.0024 (0.0031)
Absorbing exposure, baseline covariates	−0.0148*** (0.0040)	−0.0031 (0.0031)
Event study: wave of exposure	−0.0169*** (0.0060)	−0.0038 (0.0047)
Event study: one wave after	−0.0208*** (0.0065)	−0.0051 (0.0051)
Pre-trend joint p (leads $-4, -3, -2$)	0.638	0.700
Pre-exposure mean	0.0613	0.0365
Effect as % of pre-exposure mean	-24.2	-8.5
Observations, covariate rows	227,045	227,045
Observations, no-covariate row	234,465	234,465

All columns include village and province $\times$ wave fixed effects, with the always-treated cohort excluded, so no already-treated village serves as a control. Standard errors clustered at the village level. Covariates are PODES 2003 baseline characteristics interacted with wave dummies. The two event-study rows come from a single regression that also includes leads at $-4, -3$ and -2 and lags at $+2, +3$ and $+4$; the latter are jointly indistinguishable from zero (composite $p = 0.38$), so the effect is confined to the two waves following exposure. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

5.2 The Conventional Benchmark: CS(2021)

Table 3 reports the benchmark for two outcomes. Panel A gives the CS(2021) aggregate ATT: -0.0150^{***} for the composite “any communal conflict” indicator ($SE = 0.0043$, $N = 227,045$) and a smaller, marginally significant -0.0060^* ($SE = 0.0034$, $p = 0.075$) for the definition-consistent *warga* indicator. Panel B gives the Sun–Abraham (2021) interaction-weighted estimator, with significant immediate effects ($t_0 = -0.0199^{***}$ composite; -0.0075^* *warga*). Neither outcome has a clean

joint pre-trend under wave fixed effects, and Section 5.6 takes that up.

We report the two columns together rather than as a headline plus a robustness check, because each trades one property for another. The composite is the more powerful but its component categories expand after 2008. *Warga* is definition-stable across all seven waves but rarer, and on the level effect that imprecision binds: it agrees in sign with the composite and is not significant under the preferred specification (-0.0031 , standard error 0.0031). We therefore do not present it as a second level estimate. What it delivers is an outcome whose survey definition is fixed, which is what the composition contrast of Section 5.7.1 requires and where it is decisive.

Economic magnitude. First strong exposure lowers the communal-conflict probability by 1.5 percentage points, with a 95 percent interval of $[-2.34, -0.66]$ points. The benchmark is not the pooled sample mean of 2.6 percent but the treated villages' own counterfactual: ever-treated villages reported conflict at 6.1 percent across their pre-treatment waves, roughly twice the pooled rate, because earthquakes strike the more conflict-prone, faster-growing parts of the hazard zone. The estimand is accordingly local to hazard-zone villages that are actually struck. The identifying variation comes from cohorts G3–G8 (first treated 2008–2024), principally the Padang 2009 earthquake and later events. Post-treatment dynamic ATTs are negative at all six estimable horizons and individually significant at t_{+0} , t_{+1} and t_{+3} ; the rest are negative but imprecise, so the longer-run path shows no reversal without establishing persistence.

Group-level heterogeneity. No single cohort drives the effect: five of six estimable cohort ATTs are negative, and the one positive cohort, G6, is predominantly Aceh villages beginning from a near-zero floor less than a decade past the Helsinki peace agreement. Dropping G6, or Aceh altogether, leaves the estimate unchanged or larger (-0.0172^{***} ; Appendix Tables 16 and 35).

Table 3: Primary DiD Estimates: Effect of First MMI $\geq$ VI Exposure on Communal Conflict
2005-boundary panel, 2005–2024, KRB Medium/High

	Warga conflict (consistent def.)	Any communal conflict (composite)
<i>Panel A: Callaway–Sant’Anna (2021), not-yet-treated control</i>		
Overall ATT (simple)	−0.0060* (0.0034)	−0.0150*** (0.0043)
Pre-trend, average lead (p)	−0.0038 (0.56)	−0.0077 (0.33)
Pre-trend, joint Wald (p)	$\chi^2(15) = 29.4$ (0.014)	$\chi^2(15) = 42.9$ (0.000)
<i>Panel B: Sun–Abraham (2021) IW, never-treated control</i>		
Effect at t_0	−0.0075* (0.0039)	−0.0199*** (0.0049)
Effect at t_{+1}	−0.0057 (0.0041)	−0.0173*** (0.0052)
Pre-lead t_{-2} (placebo)	+0.0077	+0.0138**
<i>Panel C: Earthquake-level inference (stacked-event design, 19 events)</i>		
Treated-village-weighted ATT	−0.0101	−0.0242
event block-bootstrap SE [p]	(0.0062) [0.050]	(0.0102) [0.009]
Equal-event mean, secondary [p]	−0.0079 [0.50]	−0.0148 [0.34]
Earthquake events (clusters)	19	19
Baseline mean (pooled)	0.016	0.026
Pre-treatment mean, ever-treated	0.036	0.061
Observations	227,045	227,045

Notes: Panel A reports analytical influence-function SEs and Panel B village-clustered SEs in parentheses. The average pre-trend does not reject, but a joint Wald test of all pre-treatment group-time ATT(g, t) coefficients rejects for both outcomes under wave fixed effects (the aggregated event-study leads oscillate in sign rather than trending toward the post-treatment decline); the test is passed once province-by-wave effects are absorbed (Section 4.1). Panel C treats the earthquake, not the village, as the unit of assignment: the treated-village-weighted ATT is stacked across the 19 earthquakes and its SE comes from an event-level block bootstrap that resamples whole earthquakes (9,999 draws); the naive village-clustered SE for this estimand understates uncertainty because treatment is driven by 19 shocks. The equal-event mean weights each earthquake equally (12 of 19 negative). Conley spatial-HAC SEs at 50–150 km cutoffs and kabupaten-level clustering leave the composite reduced-form estimate significant at the 1 percent level (Section 5.6); under a deliberately conservative province-level wild-cluster bootstrap (few clusters), the pooled CS(2021) staggered ATT does not reject at conventional levels ($p = 0.16$), which we report rather than suppress; the corresponding two-way fixed-effects composite estimate does reject under the same bootstrap ($p = 0.040$, Section 5.6), so the two figures describe different estimators, not a changing p -value. Pre-treatment mean = mean outcome among ever-treated villages before first exposure. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Benchmark TWFE, the Episodic Estimand, and Estimator Comparison

The two-way fixed-effects benchmark (Appendix Table 33) tells the same story and is not a negative-weighting artifact: a Goodman–Bacon decomposition assigns 23.8 percent of the weight to forbidden comparisons, but they return -0.0205 against -0.0173 for the clean ones, pulling in the same direction. The episodic estimator of de Chaisemartin and D’Haultfœuille (2026), which identifies the effect only when a village switches into fresh exposure, gives -0.015^{***} ($SE = 0.004$) with dynamics decaying from -0.015 to -0.005 , matching the absorbing event study; its joint placebo does not reject ($p = 0.056$), though the nearest individual placebo, 0.010 , is significant with the opposite sign. Full decompositions, distributed lags, the timing artifact we document for unequally spaced panels, and the placebo panel are in Appendix J.

Table 4 collects every estimator (plotted in Appendix Figure 8). Across heterogeneity-robust and conventional estimators, and across absorbing and episodic treatment definitions, the estimate ranges from -0.013 to -0.020 and every confidence interval lies below zero; the imputation estimator of Borusyak, Jaravel, and Spiess (2024) gives -0.0132 (standard error 0.0035) under a stronger identifying assumption. The question across estimators is not which number to believe but whether any overturns the others, and none does. Our preferred *specification* remains the province-by-wave design of Section 4.1; CS(2021) is the estimator we carry through the robustness battery because it has the largest set of published diagnostics.

Table 5 makes the same point on a common sample and control set and adds the design that holds the earthquake itself fixed. Six estimates built on different identifying assumptions range from -0.0148 to -0.0208 , or between 24 and 34 percent of the 6.1 percent pre-exposure mean. The stacked row is the most demanding of the six, since treated and control villages experienced the same event in the same wave, so it cannot be explained by which earthquakes occurred, when, or by anything common to the wave of enumeration. It lands at -0.0153 . That position should not be read as agreement on magnitude: comparing villages struck by the same earthquake at damaging versus sub-damaging intensity targets

the *incremental* effect of crossing the threshold, smaller than the total effect by construction. We do not call it the strongest evidence, because its own pre-lead rejects at 5 percent ($+0.015$, $p = 0.03$) and its earthquake-clustered standard error is far larger.

Intensity gradient. The framework predicts a dose-response, and the effect does strengthen from $\text{MMI} \geq \text{V}$ to $\text{MMI} \geq \text{VI}$ before flattening. We do not claim to have identified one: changing the threshold also changes which cohorts and earthquakes enter the treated group, so the ladder is not a pure intensity contrast holding composition fixed, and no regression discontinuity is available at MMI VI because the cutoff is a classification convention rather than an assignment rule (Appendix D). We rest the intensity argument on the same-earthquake stacked design instead.

Table 4: Estimator Comparison: Effect of Earthquake Exposure on Communal Conflict (nb2005 Baseline)

	Estimate	SE	Controls	N
<i>Panel A: Absorbing first exposure ($D_{treat} = \mathbf{1}\{t \geq g\}$, window-aligned)</i>				
TWFE (D_{treat})	-0.0198***	(0.0033)	Yes	227,045
TWFE (D_{treat} , balanced)	-0.0181***	(0.0032)	No	234,465
CS(2021) overall ATT	-0.0150***	(0.0043)	Yes	227,045
BJS imputation (Borusyak et al.)	-0.0132***	(0.0035)	Yes	227,045
Sun–Abraham IW, effect at t_0	-0.0199***	(0.0049)	No	234,465
dCdH, Effect ₁ (absorbing)	-0.0199***	(0.0049)	No	190,752
<i>Panel B: Episodic exposure ($D_{interval}$, non-absorbing, can switch from 0 to 1 and back to 0)</i>				
TWFE ($D_{interval}$)	-0.0140***	(0.0026)	Yes	292,712
dCdH, Effect ₁ (switching)	-0.0150***	(0.0040)	No	249,237
Ever-treated villages (absorbing, estimable)			3,207	
Switcher villages (episodic, Effect ₁)			3,868	
Pre-trend checks			CS Pre_avg n.s.; SA t_{-2} lead +0.0138**; dCdH joint placebo n.s.	
<i>Notes:</i> Outcome: any communal conflict. All rows use the survey-window-aligned exposure timing (first PODES wave enumerated after the event). Point estimates differ across rows because of sample (balanced vs. unbalanced), controls (baseline-by-wave covariates where feasible; no-controls rows use specifications for which the estimator’s own first-step regression is infeasible with controls at the baseline wave), and estimand (overall ATT vs. effect at t_0 vs. switcher effect). The Sun–Abraham and absorbing dCdH rows round to the same estimate and standard error at four decimals by coincidence, not by duplication: the underlying values are -0.019853 (SE 0.004883) and -0.019858 (SE 0.004905) respectively, on different samples. CS(2021): doubly-robust, not-yet-treated control, analytical SEs. BJS: Borusyak, Jaravel & Spiess imputation estimator, efficient under homogeneity but requiring parallel trends in all periods (a stronger assumption than CS, which requires it only post-treatment). Sun–Abraham: interaction-weighted, never-treated control cohort. dCdH: de Chaisemartin & D’Haultfoeulle. SE clustered at village level. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$				

Table 5: One Effect, Six Estimates: Convergence Across Identification Strategies

Design	Estimate (SE)	% of pre- exposure mean	Pre-trend joint p	N
Absorbing TWFE, province $\times$ wave FE	-0.0148*** (0.0040)	-24.2	0.638	227,045
Finite duration (two waves)	-0.0163*** (0.0035)	-26.6	0.638	227,045
Event study: wave of exposure	-0.0169*** (0.0060)	-27.6	0.638	227,045
Event study: one wave after	-0.0208*** (0.0065)	-33.9	0.638	227,045
Callaway-Sant’Anna (2021)	-0.0150*** (0.0043)	-24.5	—	227,045
Stacked within-earthquake	-0.0153*** (0.0039)	-25.0	0.030	149,130

Rows 1–5 are estimated on the same sample (KRB Medium/High, always-treated cohort excluded) with baseline PODES 2003 controls; standard errors clustered at the village level. Pre-trend joint p tests leads at -4 , -3 and -2 against zero. The pre-exposure mean is 0.0613. Callaway-Sant’Anna uses wave fixed effects, since the estimator does not accommodate province $\times$ wave absorption; its pre-trend is reported separately via `estat pretrend`. The stacked design compares villages exposed at $\text{MMI} \geq \text{VI}$ to villages that felt the *same* earthquake at $\text{MMI} \in [4, 6)$, with event $\times$ village and event $\times$ wave fixed effects over 24 earthquakes; clustering by earthquake widens its standard error to 0.0093. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

5.4 Event Study

Figure 1 presents the event study on the preferred specification. The diagnostics discussed in the rest of this subsection are those of the CS(2021) event study on the conventional wave-fixed-effects panel, plotted in Appendix Figure 10, because that is the panel on which the pre-trend question binds. Pre-treatment coefficients there are cumulative deviations from the omitted wave immediately before first exposure. The average lead is small and insignificant for both outcomes ($p = 0.33$ composite, $p = 0.56$ *warga*), and the leads show no monotone trend building toward

the post-treatment decline. But the averaged diagnostic conceals a significant individual lead two waves before exposure that is marginally significant at the 10 percent level ($+0.013$, $p = 0.07$), and a joint Wald test over all pre-treatment group-time cells rejects for the composite outcome ($\chi^2(15) = 42.9$, $p < 0.001$) and for the definition-stable *warga* outcome as well ($\chi^2(15) = 29.4$, $p = 0.014$). We therefore do not read either pre-period as a clean pass. The pre-test has roughly even odds of missing a violation large enough to matter here (Section 4), and the HonestDiD bounds of Section 5.6.2 show that the aggregate post-treatment estimate carries almost no margin against residual violation. We therefore carry the definition-stable *warga* measure throughout, on two grounds: it is fully comparable across all seven waves, and its pre-trend is the milder of the two. We do not read it as a second estimate of the level effect, for which it is underpowered and, as Section 5.6.2 shows, far less robust than the composite to departures from parallel trends. Under wave fixed effects both outcomes reject the joint Wald test ($p = 0.014$ for *warga* against $p < 0.001$ for the composite), and under the preferred province-by-wave specification both pass ($p = 0.70$ and $p = 0.64$).

We also ask whether the test had the power to detect a violation in the first place (Roth, 2022), and two points follow. The calculation refers to the pre-test the **pretrends** package implements, which flags a violation when any individual pre-treatment coefficient is significant, and not to the joint Wald test reported above. First, a pre-trend large enough to survive the test would displace the estimate *toward* zero, not away from it, so it would understate rather than manufacture the negative composite effect, the same direction as the growth imbalance in Appendix N. Second, and less comfortably, the *warga* effect is small relative to the violation the test would miss half the time: a linear pre-trend of $+0.0038$ per wave on the Callaway–Sant’Anna estimates accumulates over two waves to more than the pooled *warga* ATT of -0.0060 . We therefore do not read “*warga* passes its pre-trend test” as evidence in its own right, and rest the definition-stability argument on the cross-outcome composition contrast of Section 5.7.1, which rests on a different assumption rather than on none, and which we state there.

That the longer-run estimates show no reversal is the direct check on the absorbing-treatment assumption of Section 4: the post-earthquake state, not the

shock window alone, carries the effect.

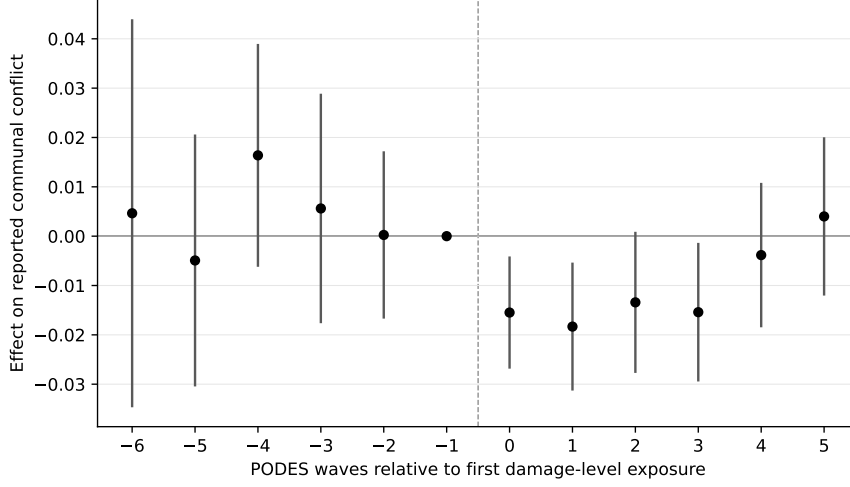


Figure 1: Event Study: Effect of First $\text{MMI} \geq \text{VI}$ Exposure

Note: Interaction-weighted event study of Sun and Abraham (2021), estimated with village and province $\times$ wave fixed effects and never-treated villages as the control cohort, with pointwise 95% confidence intervals. Outcome: any communal conflict (composite); sample: the 2005-boundary panel, always-treated cohort excluded. This is the only specification available to us that both absorbs the province-by-wave counterfactual selected on economic grounds in Section 4.1 and is robust to treatment-effect heterogeneity across cohorts and event time. The Callaway–Sant’Anna estimator cannot deliver the first, because `csdid` admits no interacted fixed effects; a conventional dynamic two-way specification delivers the first but not the second. The event-time indicators are fully saturated, so every relative period except the omitted wave before exposure enters the regression and no post-treatment cell is pooled into the reference category. A joint Wald test of the pre-exposure leads does not reject ($\chi^2(5) = 4.69$, $p = 0.46$). The imputation estimator of Borusyak et al. (2024) agrees on the pre-period under the same fixed effects (leads of +0.009, +0.015 and −0.002, none significant) but cannot estimate the effect there with useful precision (standard error 0.028 against 0.006 here), because village and province-by-wave effects together absorb nearly all of the variation its imputation step relies on. The Callaway–Sant’Anna event study on the wave-fixed-effects panel, whose joint pre-trend test rejects, is reported in Appendix Figure 10. Event time $t = 0$ is the first treatment wave; treatment timing is window-aligned first exposure.

The pre-trend verdict turns on the fixed-effect structure rather than on the estimator, and we state the pattern in one place because it is the evidence behind the specification choice of Section 4.1. Under wave fixed effects the joint pre-treatment test rejects in every estimator family we run: Callaway–Sant’Anna ($\chi^2(15) = 42.9$, $p < 0.001$ composite; $\chi^2(15) = 29.4$, $p = 0.014$ *warga*), the imputation estimator of Borusyak et al. (2024) (leads of +0.028, $p = 0.001$ and +0.042, $p < 0.001$), a Sun–Abraham lead two waves before exposure significant at 5 percent, and a nearest placebo of 0.010 under the episodic estimator of de Chaisemartin and D’Haultfoeulle (2026). Under province-by-wave effects it passes in every family that can absorb them: the dynamic specification ($p = 0.64$

composite, $p = 0.70$ *warga*), the saturated Sun–Abraham event study of Figure 1 ($\chi^2(5) = 4.69$, $p = 0.46$), and the imputation estimator, whose three leads are individually insignificant there. That the verdict flips with the fixed effects and not with the estimator is what the economic argument of Section 4.1 predicts, and it is a stronger statement than the diagnostic alone could support: the specification is not merely the one that passes, it is the one that passes across estimators that make different assumptions about treatment-effect heterogeneity. What this does not buy is precision. The imputation estimator that clears the pre-test under province-by-wave effects cannot estimate the effect there (+0.013, standard error 0.028), because village and province-by-wave effects together absorb nearly all the variation its imputation step uses. The point estimates converge and the diagnostics agree on the specification; the cost is that the design has less power the more of the regional counterfactual we absorb, and we report the bounds of Section 5.6.2 rather than read any single pre-test as settling the matter.

5.5 Does the Decline Displace? Conflict in Neighbouring Villages

A within-village fall in reported communal conflict need not mean less conflict overall: disasters can *lower* conflict where they strike while *raising* it nearby, so total conflict rises through displacement (Amarasinghe et al., 2026). We test this directly, assigning each never-treated village to a mutually exclusive proximity ring by distance to the nearest treated village, switching on only once a neighbour is treated. Displacement predicts *positive* ring coefficients; we find the opposite, with all three rings negative under village and wave fixed effects and the nearest at -0.0101 . Under province-by-wave effects the rings attenuate to insignificance and one turns slightly positive (0.0016, $p = 0.69$), so what we can say is that no ring is ever significantly positive, not that the rings themselves are robust (Table 6; Appendix J.5). The local decline therefore shows no detectable offset from displacement onto neighbours; if anything it extends outward. The contrast with Amarasinghe et al. (2026), whose mobile armed actors reallocate toward lootable rents when a shock lowers the local prize, lies in the conflict technology: communal conflict is fought between resident groups over local grievances, though

our data show only no detectable displacement, not the reason for it. We read this as bounding, not eliminating, the displacement concern, since PODES cannot track individuals who relocate across village boundaries.

5.6 Threats to Identification and Robustness

This section asks whether the headline result survives the main threats to the design, and is organised around those threats rather than around specifications. The estimate stays negative and close to the headline across most margins, but not under every weighting and unit of assignment, and we collect the exceptions in the taking-stock subsection below. Appendix J reports every check in full; here we state the question, the answer, and the bearing on the main claim.

5.6.1 How Many Shocks? Stacked Event Design and Event-Level Inference

The design has 3,207 ever-treated villages, but they are not 3,207 independent assignments: villages in the same footprint receive the same shock, so village-clustered standard errors may overstate precision. The staggered variation is not driven by a handful of events. Assigning each village to the earliest earthquake exposing it to $\text{MMI} \geq \text{VI}$, **19 distinct earthquakes** with at least twenty in-sample first-treated villages each, from Yogyakarta 2006 and Padang 2009 to Pidie Jaya 2016, Lombok and Palu 2018, and Cianjur 2022, account for 94 percent of the staggered-treated villages (3,009 of 3,207; 44 smaller events supply the remaining 198).

Treating the earthquake rather than the village as the unit of assignment separates two estimands, and the distinction is the substantive point of this subsection. Let τ_e be the effect of earthquake e on the villages it treated. The *treated-village-weighted* average weights each τ_e by the number of villages that earthquake first exposed and answers what happened to the average exposed village; it is the event-level counterpart of every village-level estimate in the paper, since any village panel weights earthquakes that way. It is -0.0242 (standard error 0.0102, earthquake bootstrap interval $[-0.047, -0.007]$). The *equal-event* mean counts each of the nineteen earthquakes once and answers what the average earthquake

Table 6: Spatial Displacement: Conflict in Never-Treated Villages by Proximity to Treated Villages
Takeaway: no statistically significant rise in conflict near treated villages, so the local decline is not simply relocated.

	(1) Village+wave FE (village cl.)	(2) Village+wave FE (kabupaten cl.)	(3) Village+prov. x wave FE (kabupaten cl.)
Within 0–25 km	-0.0101*** (0.0029)	-0.0101** (0.0042)	-0.0024 (0.0054)
25–50 km	-0.0048** (0.0023)	-0.0048 (0.0033)	0.0016 (0.0041)
50–100 km	-0.0099*** (0.0017)	-0.0099*** (0.0029)	-0.0051 (0.0033)
Constant	0.0314*** (0.0010)	0.0314*** (0.0014)	0.0279*** (0.0019)
Baseline conflict mean	0.026	0.026	0.026
Observations	212,016	212,016	212,016

Sample: never-treated villages in the KRB Medium/High zone (villages whose nearest already-treated village is used to build a time-varying proximity ring). Ring dummies indicate the distance band (0–25, 25–50, 50–100 km) to the nearest treated village (any window-aligned first exposure at or before the current wave, including the always-treated 2005 cohort); rings are mutually exclusive and time-varying. Villages more than 100 km from any treated village are the omitted pure-control group. Column (3) adds province x wave fixed effects, absorbing the regional earthquake-year shock that drives ring exposure. Standard errors clustered as indicated in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

does to its footprint. It is -0.0148 (standard error 0.0150), not distinguishable from zero. Its point estimate coincides numerically with the village-level ATT reported elsewhere; the two are distinct estimands that happen to agree to four decimals, and only the village-level estimate is precisely estimated.

We claim the first estimand, because the question posed in the introduction is what strong shaking does to the villages it strikes. On the second the paper establishes no decline, and a reader who asks what the average earthquake does should read it as not demonstrating an effect. Two things bound that null. The twenty-village cutoff is a choice: lowering it leaves the treated-weighted average between -0.0242 and -0.0254 and rejecting at 1 percent at every cutoff, while the equal-event mean swings from -0.031 ($p = 0.02$) at five villages through -0.025 ($p = 0.07$) at ten to -0.004 ($p = 0.78$) at two (Appendix Table 22), which reads as an estimand that is poorly determined rather than as an absent effect. And the equal weighting masks a split the framework predicts, though we note the reading is post hoc: among the six earthquakes with the largest on-land damaging footprints the mean effect is -0.038 , with five of six negative at 10 percent, while among the thirteen smaller-footprint events it is -0.004 . What makes an event large here is footprint, not magnitude; the two largest-magnitude events in the window are deep offshore ruptures with small on-land footprints and near-zero effects. The pooled evidence is carried by the large-footprint earthquakes, and we regard that as the honest ceiling on what the staggered result can claim.

The compositional result does not depend on the choice. The theft-versus-*warga* divergence is positive and significant under *both* estimands (Section 5.7), which is one reason we rest the paper on the composition rather than on the level. Table 7 collects every event-level estimand, weighting and inference procedure; Figure 2 shows all nineteen event-specific estimates. Appendix J.4 gives the stacked construction, the two-way clustering, disjoint-control and aftershock-independence checks, all of which leave the estimate essentially unchanged.

The same-earthquake local design, and where the decline lives. A third counterfactual holds the shock itself fixed: for each earthquake, treated villages (mean $\text{MMI} \geq \text{VI}$) are compared with villages in the same footprint that

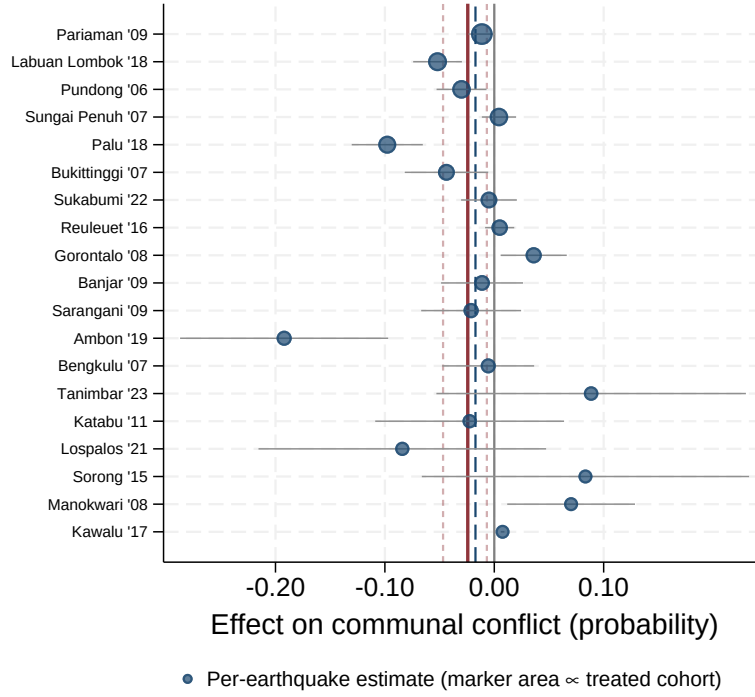


Figure 2: Event-Specific Difference-in-Differences Estimates

Note: Each row is one earthquake's own difference-in-differences estimate of the effect on communal conflict, with a 95% confidence interval. Rows are ordered by treated-cohort size and labelled by the earthquake's nearest place (USGS ShakeMap catalogue) and year; marker area is proportional to the treated cohort, counted as unique first-treated villages. The solid grey line is zero. The dashed navy line is the pooled stacked estimate (-0.0237), variance-weighted toward the large, sharply-identified shocks. The solid maroon line is the treated-village-weighted estimate (-0.0242), with its event-level bootstrap 95% confidence interval shown as dashed maroon lines. Twelve of the nineteen estimates are negative, and the effect is concentrated in, though not exclusive to, the major earthquakes.

Table 7: Event-Level Inference: All Estimands Side by Side

The effect is negative under every weighting, inference level, and control group, though its precision varies: strongest in the matched design and weakest under the equal-event mean.

	Estimate	Inference		p	Events
		SE	95% CI		
Panel A. Global staggered design (treated vs never/not-yet-treated controls)					
Pooled, variance-weighted (village clustered)	−0.0237	0.0038	—	<0.001	19
same, earthquake clustered	−0.0237	0.0100	—	0.03	19
Treated-village-weighted (event bootstrap)	−0.0242	0.0102	[−0.047, −0.007]	0.01	19
Equal-event mean	−0.0148	0.0150	—	0.34	19
Disjoint controls, treated-wtd (event boot.)	−0.0222	0.0111	[−0.047, −0.004]	0.02	19
Panel B. Same-earthquake local design (treated vs felt-controls, $MMI \in [4, 6]$)					
Unmatched pooled (village clustered)	−0.0153	0.0039	—	<0.001	24
same, earthquake clustered	−0.0153	0.0093	—	0.11	24
Unmatched equal-event mean	−0.0138	0.0088	—	0.13	24
Matched pooled (earthquake clustered)	−0.0175	0.0091	—	0.07	24
Matched treated-village-weighted (event boot.)	−0.0196	0.0105	[−0.043, −0.002]	0.02	23

Notes: Each row is a distinct way of aggregating and pricing uncertainty in the same effect. *Event bootstrap* resamples whole earthquakes with replacement (9,999 draws, seed 42) and recomputes the unique-village cohort weights each draw. The treated-village-weighted average weights each earthquake by its number of unique first-treated villages; the equal-event mean weights every earthquake alike; the pooled row up-weights the large, sharply-identified shocks. Panel A compares treated villages to never- and not-yet-treated controls (the total effect of exposure); Panel B compares them to villages that felt the *same* earthquake at sub-damaging intensity (the incremental effect of crossing the damage threshold), and its *Matched* rows additionally exact-match treated to felt-control villages on pre-treatment conflict and coarsened geography. Outcome is the composite communal-conflict indicator throughout. Village- and wave (or event-relative-wave) fixed effects and village-clustered standard errors unless noted; *same*, *earthquake clustered* rows use the identical point estimate with the standard error re-clustered at the earthquake-event level. Source do-files: 13_stacked_event, 20_event_bootstrap, 23_disjoint_controls, 16_local_stacked, 22_local_matched (all_nb2005.do).

felt sub-damaging shaking (mean MMI in $[4, 6)$). Because the controls also felt the earthquake, this identifies the *incremental* effect of crossing the damage threshold, smaller than the total effect by construction. The composite is -0.0153 , significant at 1 percent under village and kabupaten clustering but not under 24-cluster earthquake-level inference ($p = 0.11$); the equal-event mean over the same 24 events is -0.0138 ($p = 0.13$), with 14 of 24 event estimates negative. Its pre-period is not clean either: the lead two waves before exposure is $+0.0154$ ($p = 0.03$), so part of a same-window comparison can reflect mean reversion, and exact-matching on the full pre-period conflict path and coarsened geography sharpens rather than removes that caveat.

Splitting the matched sample by pre-period conflict is the informative step, and it cuts against the headline. Among the roughly nine in ten treated villages with *no* prior conflict, exposure *raises* the onset of communal conflict: the matched effect is $+0.0070$ (standard error 0.0022) against a counterfactual onset rate of 2.1 percent among matched controls, a relative increase of about a third. The aggregate local decline instead loads on the minority with a prior conflict history (-0.0265 , standard error 0.0166), exactly where reversion cannot be separated from a treatment effect. For the modal treated village, then, the shock does not pacify. We read the local design as corroborating the *sign* of the aggregate at most, not as independent identification, and we carry the onset result forward as a stated limit on what the level estimate describes. Table 8 reports the split, the pooled matched estimates under both clustering choices, and the raw onset rates behind them; Appendix J.6 reports the fixed-effect structure and the matching.

Table 8: Onset and Reversion in the Matched Within-Earthquake Design

	Estimate	(SE)	<i>p</i>	Observations
<i>Panel A. Matched within-earthquake difference-in-differences</i>				
No prior conflict episode	+0.0070***	(0.0022)	0.0014	137,453
Prior conflict episode	−0.0265	(0.0166)	0.1100	5,967
Pooled, village-clustered	−0.0175***	(0.0038)	0.0000	143,425
Pooled, earthquake-clustered	−0.0175*	(0.0091)	0.0672	24 events
Pooled, <i>warga</i>	−0.0061**	(0.0029)	0.0377	143,425
<i>Panel B. Onset rates, villages with no pre-exposure conflict</i>				
Comparison villages, post	0.0213			76,815
Treated villages, pre	0.0000			4,713
Treated villages, post	0.0280			7,391
Memo: villages with prior conflict, comparison post	0.0714			2,689

Panel A compares villages shaken at $\text{MMI} \geq \text{VI}$ with unshaken villages within the same earthquake, restricting to cells that contain at least one treated and one comparison village; 24 earthquakes have matched support. All specifications absorb stack-by-village and stack-by-wave fixed effects. Standard errors are clustered at the village level except in the earthquake-clustered row, which clusters on the earthquake and reports the number of clusters in place of observations. The split in the first two rows is by whether the village had ever reported a communal-conflict episode before exposure, so the first row is the onset margin and the second is the margin on which mean reversion can operate. Panel B reports raw rates on the same matched sample and is descriptive. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.6.2 Identification Threats

Treatment-outcome timing. PODES conflict at wave t covers the twelve months before May enumeration, so an earthquake inside that window gives a partial-exposure $t = 0$ outcome. An exact-date audit shows 74 percent of the nineteen stacked earthquakes are clean-post, and every major earthquake carrying the effect is clean-post. Restricting to clean-post cohorts leaves the no-controls full-panel estimate slightly larger (−0.0206 versus −0.0186) and leaves *warga* significant at 1 percent (−0.0072, standard error 0.0027), so the finding is not an artifact of outcomes measured across rather than after the shock (Appendix J.8).

Parallel trends. The pre-treatment record is mixed and we do not read either pre-period as clean. Section 4 states the assumption and the evidence in full; the summary is as follows. Two lead conventions give different readings and we report both rather than choose. Averaging the *cumulative event-study leads*, the pre-trend is small and insignificant under the headline control set (-0.0077 , $p = 0.33$; Table 3). Averaging the *group-time pre-treatment cells* instead, it is positive and rejects, both without covariates ($+0.0107$, $p = 0.01$) and under the headline baseline set ($+0.0159$, $p < 0.001$; Table 42, and the same convention underlies the $+0.016$ reported for the communal outcome in Table 12). We do not resolve this in the paper’s favour: the second convention is the less forgiving one and it rejects. A joint Wald test over all pre-treatment group-time cells likewise rejects under wave fixed effects ($\chi^2(15) = 42.9$, $p < 0.001$ composite; $\chi^2(15) = 29.4$, $p = 0.014$ *warga*) and cannot be computed under province-by-wave effects, because `csdid` cannot absorb interacted fixed effects; the joint test of the event-study leads, which can, does not reject there ($p = 0.64$ composite, $p = 0.70$ *warga*). The drift the data do exhibit is *positive*, which biases a subsequent decline toward zero rather than manufacturing it, and conditioning on a passed pre-test displaces the immediate estimate by about 0.004, again toward zero. Appendix J reports the individual leads, the Roth (2022) power calculation, and the covariate trajectories being conditioned on.

Is the effect a region-type artifact? Village fixed effects absorb region *levels*; the sharper test is region-specific *trends*. Adding province $\times$ wave fixed effects attenuates the no-controls full-panel absorbing effect by about 35 percent, from -0.0186 to -0.0120 , significant at 1 percent under village clustering but only marginal under kabupaten clustering, and the definition-stable *warga* margin does not survive (Appendix J.7).

Measured versus imputed exposure. Exposure is built by matching each village-year to a ShakeMap record, and a village-year with no matching record is assigned zero shaking. That is correct when no earthquake reached the village and incorrect when one did but the event has no ShakeMap footprint, in which case a treated village enters the comparison pool. Across the estimation sample 36.5

percent of village-waves carry an imputed rather than a measured first lag, and the measured share varies markedly across waves, from 43.5 percent in 2005 to 91.1 percent in 2018. Two checks bound what this can do. Restricting to village-waves whose exposure is measured leaves the composite at -0.0127 ($p = 0.02$), slightly smaller than the full-sample -0.0148 and of the same sign and order. Restricting instead to the waves from 2014 onward, where the measured share is highest, roughly doubles it to -0.0310 ($p < 0.001$), and the definition-consistent *warga* outcome is significant there as well (-0.0124 , $p = 0.04$), one of the few specifications in which it clears conventional levels. We do not read the second as a clean measurement experiment, because restricting to later waves also changes the cohorts, the calendar period and the post-exposure window. Both are consistent with attenuation from misclassification, but we stop short of asserting the sign of the bias, because the measured share varies systematically across waves and missingness correlated with era, geography or event size need not attenuate. What the checks establish is that the estimate survives when the imputed village-waves are removed.

Functional form. The estimate survives the measurement and functional-form choices. It is unchanged in sign and broad magnitude across four raster-to-village aggregation rules on all 1,423 event ShakeMaps, losing conventional significance only under the pixel-maximum rule; rebuilding exposure from an MMI derived independently from instrumental Peak Ground Acceleration reproduces it (CS -0.0129 , standard error 0.0039); and because communal conflict is rare, we verify that the linear probability estimates are not an artifact of the zero boundary, since parallel trends in levels and under a nonlinear transformation cannot in general both hold (Roth & Sant’Anna, 2023). A conditional logit gives the same sign and significance (Appendix J).

Cohort dependence, placebos and randomization inference. No single *cohort* drives the result: dropping each in turn leaves every estimate negative and significant, from -0.0103 to -0.0194 around the no-controls full-sample benchmark, and the equal-weight cohort average (-0.0182) confirms this is not a weighting artifact. Because two of the largest events fall in the same cohort, this is a leave-

one-cohort-out and *not* a leave-one-earthquake-out check, and we do not claim the stronger version (Appendix J). A forward-lag placebo is a clean null (-0.0045 , $p = 0.40$), and a spatial randomization that relocates each earthquake’s footprint to a random never-treated location produces no placebo draw at least as negative as the actual no-controls full-panel estimate (-0.0186) over 500 relocations, a one-sided $p = 0.002$.

HonestDiD sensitivity. This is the binding constraint, not the pre-trend tests. On the preferred specification the composite sign survives relative-magnitudes violations up to a quarter of the largest pre-period movement but not up to a half. The definition-stable *warga* target breaks down far earlier, at $\bar{M}$ between 0.05 and 0.10, but that figure is computed on the Sun–Abraham event study rather than on the preferred specification, and $\bar{M}$ is not a common yardstick across estimators, so it is not a bound on the headline (Appendix R).

5.6.3 Regional Trends and Forbidden Comparisons

Two objections apply to the absorbing design at once. Seismic Indonesia is not a random draw of the country: the hazard zones sit largely in the outer islands, including post-conflict provinces on their own downward path, and village fixed effects absorb the level of that path but not its slope. And the already-treated comparison problem of [de Chaisemartin and D’Haultfoeuille \(2020\)](#) carries 23.8 percent of the weight rather than a negligible share.

Table 9 addresses both. Province $\times$ wave fixed effects compare a village only against others in the same province and wave, absorbing any differential regional trajectory. Dropping the always-treated cohort, 9,856 of 43,351 villages, removes the units with no pre-period. It does not remove every forbidden comparison, and we say so: in a staggered design an earlier-treated cohort is already treated when a later one switches and can still serve as its control. Dropping the cohort barely moves that estimate, from -0.0186 to -0.0181 . What answers the residual concern is not the sample restriction but the estimators of Section 4, none of which uses an already-treated unit as a control. The joint pre-trend test clears in all four columns, which it does not under wave fixed effects, and the estimates do not

weaken. The result is concentrated rather than general: inter-village conflict falls by 1.7 percentage points, 45 percent of its pre-exposure mean, while inter-group conflict within the village is a precise null. Table 10 traces the estimate across fixed-effect structures. The kabupaten $\times$ wave column is reported for completeness but is not informative: it absorbs 62 percent of the remaining treatment variance, its standard error inflates by a factor of 1.42 against the baseline column, less than the 1.62 that loss alone predicts, and its interval still contains the province $\times$ wave estimate.

Table 9: Strong Shaking and Communal Conflict: Province $\times$ Wave Fixed Effects, Always-Treated Cohort Excluded

	(1) Any communal	(2) Mass-brawl gate	(3) Inter-village	(4) Inter-group
Strong shaking (MMI $\geq$ VI)	-0.0137*** (0.0039)	-0.0146*** (0.0041)	-0.0168*** (0.0037)	-0.0024 (0.0031)
Pre-exposure mean	0.0613	0.0657	0.0371	0.0365
Effect (% of pre-exposure mean)	-22.3	-22.2	-45.3	-6.5
Pre-trend joint p	0.607	0.540	0.197	0.299
Observations	234,465	234,465	200,970	234,465

Standard errors clustered at village level in parentheses. All columns include village and province $\times$ wave fixed effects. The always-treated cohort (villages already exposed when the panel opens, 9,856 of 43,351) is excluded, which removes the units that have no pre-period; earlier-treated staggered cohorts can still serve as controls for later ones, as Section 5.6 reports. Pre-trend joint p tests leads at -4 , -3 and -2 against zero. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Inter-Village Conflict Across Fixed-Effect Structures

	(1) Wave FE	(2) Prov $\times$ wave	(3) + drop always	(4) Kab $\times$ wave
Strong shaking (MMI $\geq$ VI)	-0.0165*** (0.0031)	-0.0165*** (0.0037)	-0.0168*** (0.0037)	-0.0095** (0.0044)
Observations	260,106	260,106	200,970	200,940

Outcome is inter-village communal conflict. Village fixed effects throughout. Column (3) is the primary specification. Column (4) absorbs 62 percent of the remaining treatment variance (6.40 to 2.43 percent); its standard-error inflation of 1.55 tracks the variance loss (predicted 1.62) and its interval contains column (3), so it is uninformative rather than contradictory. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Does the effect fade? An effect persisting undiminished for a decade is hard to reconcile with disruption of an inherently temporary capacity. Table 11 estimates the profile against months since the most recent damage-level shaking, with earthquake fixed effects so that each lag is identified within the same earthquake.

Inter-village conflict decays: strongest within the first year, still -0.0064 at 36 to 48 months, and indistinguishable from zero beyond roughly six years, with the decay test rejecting equality between recent and distant bins. Neither the composite nor the mass-brawl gate shows a comparable profile, so the decay is specific to the inter-village margin. Identification here is within-exposed and across elapsed time rather than exposed against unexposed, and the profile is not smoothly monotone.

Table 11: Decay of the Effect with Time Since Shaking, Earthquake Fixed Effects

	(1) Any communal	(2) Mass-brawl gate	(3) Inter-village
0–12 months	-0.0029 (0.0031)	-0.0009 (0.0033)	-0.0105** (0.0051)
12–24	0.0008 (0.0039)	0.0020 (0.0044)	-0.0018 (0.0026)
24–36	-0.0094 (0.0084)	0.0047 (0.0093)	-0.0096* (0.0057)
36–48	-0.0053 (0.0034)	-0.0038 (0.0036)	-0.0064*** (0.0024)
48–72	-0.0002 (0.0036)	0.0008 (0.0039)	-0.0008 (0.0029)
72–120	0.0032 (0.0023)	0.0041* (0.0025)	0.0015 (0.0016)
Decay test p (recent = distant)	0.2642	0.5557	0.0121
Observations	303,456	303,456	260,105

Months elapsed between the most recent $\text{MMI} \geq \text{VI}$ shaking and the survey. Village, province $\times$ wave, and EARTHQUAKE fixed effects throughout. Event fixed effects are essential rather than optional: without them each lag window is dominated by a different earthquake in a different wave, and the profile looks flat for reasons of composition rather than timing. Identification is therefore within-exposed and across elapsed time, not exposed versus unexposed. The 120+ month bin is collinear with the event effects and is absorbed. Decay test compares the two most recent bins against the two most distant. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5.6.4 Design, Clustering and Spatial Structure

The estimate survives the exposure, clustering and spatial choices, all detailed in Appendices J and B. It stays negative and significant at a broader $\text{MMI} \geq \text{V}$ (-0.0063^{***}) and a narrower $\text{MMI} \geq \text{VII}$ (-0.0174^{***}), though the ladder traces no dose-response (Section 5.3). Re-clustering at the kabupaten level keeps the composite significant at 1 percent, and the conservative 28-cluster province bootstrap leaves it at 5 percent while *warga* does not clear conventional levels. Repeated

exposure is rare (96.6 percent of treated villages are struck in only one window) and where it occurs we detect no additional effect (0.004, $p = 0.75$); statistical insignificance is not equivalence, so we read this as consistent with, not establishing, a first-shock state change. Spatial spillovers, [Oster \(2019\)](#) coefficient stability under which adding controls moves the estimate *away* from zero, and Conley spatial-HAC errors all leave the result intact.

5.6.5 Taking Stock

Several results are genuinely sensitive and we set them out together. HonestDiD is the binding constraint. The pooled CS(2021) ATT is -0.0150^{***} under the conventional wave-fixed-effects panel, and the two-way fixed-effects estimate under the preferred province-by-wave specification is -0.0148 , since `csdid` cannot absorb interacted fixed effects, so the choice the pre-trend diagnostic settles moves the magnitude by two ten-thousandths. On the definition-consistent *warga* outcome the pooled estimate is -0.0060 ($p = 0.075$); under the most conservative province-level wild-cluster bootstrap the composite clears 5 percent on the *two-way fixed-effects* benchmark ($p = 0.040$) while *warga* does not ($p = 0.255$). We state plainly what that test does to the primary estimator: the pooled CS(2021) ATT does *not* reject under the same bootstrap ($p = 0.16$). With 28 clusters this is the least powerful inference in the paper, but it is also the most conservative, and on it the level result is not established under the estimator our causal claims otherwise rely on. The stacked event design is carried by the larger earthquakes, and the equal-event mean is not distinguishable from zero. The parallel-trends evidence is mixed under the conventional panel and passes under the preferred one, with the residual composite signal a positive drift whose direction biases against the finding.

None of this overturns the negative sign or the compositional pattern, but together these mark the boundary of the evidence: a decline in reported communal conflict after strong shaking within hazard-zone villages, with no evidence of reversal but with only the first post-exposure horizon precisely estimated, and not a universal or mechanism-decomposed law.

5.6.6 Limitations

Population displacement. If shaking drives residents out, the pool for collective violence shrinks and the decline is mechanical rather than behavioural. We can test this only in part. PODES records population and household counts through 2011 and not from 2014 onward, so the 2008 and 2011 cohorts can be checked while the later cohorts, which supply most of the identifying variation, cannot. On the 130,052 village-waves with counts, exposure does reduce headcounts: log population falls by -0.0121 (standard error 0.0055) and log households by -0.0141 (standard error 0.0059), about 1.2 and 1.4 percent. That residential location is the margin a disaster moves durably, while income recovers faster than expected, is what [Deryugina, Kawano, and Levitt \(2018\)](#) document for Hurricane Katrina, and it is why we read headcounts rather than income as the first stage available to us.

Displacement is therefore real rather than absent, but far too small to carry the result. On the same restricted sample the communal-conflict effect is -0.0118 against a 5.1 percent pre-exposure rate, a fall of roughly 23 percent, so for a 1.2 percent loss of population to produce it mechanically the elasticity of reported conflict with respect to village population would have to be about nineteen. The PODES counts are a village-head report rather than an enumeration, but matching the 2010 Population Census to the 2011 wave for 35,628 in-sample villages gives a correlation of 0.99 in levels and logs and a log-log slope of 0.970 (standard error 0.001), so correcting for the implied compression would move the first stage from -1.2 to about -1.25 percent. The composition bounds the mechanical version directly: depopulation predicts that *every* category of reported disorder falls, yet theft rises sharply over the same villages and waves while communal conflict falls. That rules out depopulation as the sole account. It does not rule out selective displacement of particular groups, which neither PODES nor our design observes, and we flag that as an open threat rather than one we have closed ([Appendix J.5](#)).

Internal and external validity. Identification rests on parallel trends conditional on baseline covariates rather than on experimental assignment, and the sign of the primary staggered estimate does not survive relative-magnitudes violations larger than a quarter of the worst pre-period movement. Three weaknesses are

shared across the designs of Table 5: the conditional pre-trend test is passed only once province-by-wave effects are absorbed; treatment is assigned by nineteen to twenty-four earthquakes rather than by villages, so village clustering overstates precision; and the definition-consistent *warga* outcome moves in the same direction throughout but never attains conventional significance on the specification that passes the pre-trend test. We rest the sharper claim on the compositional evidence, the one result that survives earthquake-level inference under either weighting.

The estimand is local and intent-to-treat. The sample is restricted to villages inside Indonesia’s officially mapped medium and high seismic-hazard zones, and within it the treated villages are more conflict-prone and faster-growing, reporting communal conflict at 6.1 percent before exposure against a pooled 2.6 percent. The estimate describes what a first damaging earthquake does to a village actually struck in a seismically active setting, not the effect of a hypothetical shock on an arbitrary Indonesian village, and it should not be extrapolated to settings whose conflict institutions differ from Indonesia’s.

5.7 The Composition of Disorder: Evidence and Mechanisms

Why would earthquake exposure reduce communal fighting on average across affected villages? This section presents the evidence that carries the paper: the composition of disorder. The key observation is that PODES records not only communal violence but also village-level crime, and different theories of the earthquake–conflict link make sharply different predictions across these outcomes. We first present that evidence, then discuss the channels it selects. We are explicit throughout about what this evidence can and cannot do. It establishes which *margins* of disorder move together, which is enough to discriminate among competing theories of the sign, but it does not *identify* the underlying economic channel, because PODES does not measure household welfare, consumption, or labor-market outcomes. We therefore read the mechanism results as reduced-form patterns consistent with a capacity account rather than as proof of it, and we flag direct welfare and labor outcomes, for example from the National Socio-Economic Survey (Susenas), as the measurement that would convert this consistency into identification of the

economic channel.

5.7.1 The Central Result: Theft Rises Where Communal Conflict Does Not

Table 12 estimates the same staggered design on four outcomes that differ in two dimensions: whether the act is *collective* or *individual*, and whether it is *economic* or *violent*. Communal conflict (column 1) is collective violence; theft (column 2) is individual and economically motivated; robbery (column 3) is individual property crime with violence; assault (column 4) is individual interpersonal violence.

Table 12: Mechanism Discrimination: CS(2021) by Outcome
Window-aligned first exposure, 2005-boundary panel

	Communal conflict (collective, violent) (1)	Theft (individual, economic) (2)	Robbery (property, violent) (3)	Assault (individual, violent) (4)
CS(2021) overall ATT	−0.0150*** (0.0043)	0.0862*** (0.0107)	−0.0032 (0.0044)	−0.0058 (0.0061)
Pre-trend Pre_avg (<i>p</i>)	0.016 (0.00)	−0.043 (0.04)	−0.012 (0.27)	−0.030 (0.07)
Baseline mean	0.028	0.451	0.032	0.059
Waves	7	5	5	5
Observations	227,045	161,305	161,305	161,305
Controls	Yes	No	No	No

Notes: Callaway–Sant’Anna (doubly-robust, not-yet-treated control), window-aligned first exposure to $\text{MMI} \geq \text{VI}$, nb2005 KRB Medium/High sample. Column (1) is the main specification (baseline-by-wave controls; conditional parallel trends). Crime outcomes (PODES: any incident in the past year) are available in waves 2005, 2008, 2018, 2021, 2024 and are estimated without controls because lagged/baseline controls shorten the pre-period available for the pre-trend test in this narrower 5-wave subsample. SE clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To make the composition a formally tested result instead of a comparison of separate regressions, we estimate five outcomes on a single *common sample*: 42,840 villages observed over the five waves in which the crime module is fielded, the same absorbing treatment, and no controls, stacking the outcomes with outcome-interacted fixed effects so that the standard errors account for the correlation of a village’s outcomes (Table 14). Like the communal-conflict outcome, the crime outcomes are constructed from wave-specific PODES crime-module items, with the exact item code for each of the five waves listed in the replication package; a change in the module’s wording across waves therefore shifts theft and communal

conflict together on the common sample and differences out of the cross-outcome contrast we rely on.

The contrast carries its own identifying assumption, and we state it rather than present the design as assumption-free. Estimating the outcomes on a common sample with outcome-interacted fixed effects is equivalent to estimating the same difference-in-differences on a single differenced outcome, theft minus communal conflict. What it requires is parallel trends in that difference: absent the earthquake, the gap between reported theft and reported communal conflict in treated villages would have evolved as it did in villages not yet exposed, and exposure changes the reporting of the two items, if at all, only by a common amount. This is weaker than the level effect’s assumption in one specific respect. Any village-wave shock that moves both items equally differences out, which covers a uniform disruption of the village head’s reporting, post-disaster enumeration conditions, and mechanical depopulation. Three threats survive the differencing and we name them. A shock correlated with exposure that moves the two items differentially does not cancel: an incentive to document property losses for aid claims would inflate reported theft without inflating reported communal conflict, and the evidence against that account is that robbery, also a property loss on the same form, falls rather than rises. Because theft’s pre-exposure rate is roughly ten times the *warga* rate, a reporting shock proportional to each item’s level does not cancel in an absolute difference; the earthquake-level equivalence bound on *warga* excludes a proportional surge large enough to produce theft’s rise. And the differenced outcome has a pre-trend of its own. We test it directly, and it rejects: a joint test of the pre-exposure leads of the differenced outcome rejects under both wave and province-by-wave effects ($p < 0.001$). We therefore do not claim that the contrast dispenses with pre-trend concerns. It replaces the communal pre-trend with the difference’s, which is dominated by theft’s, and theft’s does not pass. Two things bound what that rejection can imply here. The crime module is fielded in five waves only, with a ten-year gap between 2008 and 2018, so event time on this sample is severely unevenly spaced and the leads being jointly tested are not comparable in calendar time. And if the pre-exposure deviation in theft is a trend that would have continued, the divergence we report is understated rather than manufactured; subtracting

the full deviation from the observed gap still leaves it positive. We report the rejection because it is the sharper point, and we rest the compositional claim on the shock-level divergence, which is estimated event by event and survives both weightings, rather than on this village-level dynamic specification.

On identical data, theft rises (+0.0576^{***}) while communal conflict falls (−0.0152^{***}): the two move in opposite directions (Figure 3, Panel A), and a test of their equality is decisively rejected ($p < 0.001$), with a gap of +0.073. What we lean on is this cross-outcome contrast, not the size of the communal decline on its own.

Because the composite communal measure changes definition in 2008, we anchor the contrast on the fifth outcome, *warga*, the civilian-versus-civilian communal subtype whose survey wording is stable across all waves. The *warga* coefficient on its own is −0.0054 (standard error 0.0027, $p = 0.050$), so on this sample it falls, marginally, where the pooled staggered estimator returned a null; its 95 percent confidence interval extends no higher than zero, against theft’s +5.8. A test of the equality of the two treatment effects is nonetheless rejected ($p < 0.001$). The composition result therefore does not depend on the composite outcome or on its 2008 definition change: the definition-stable communal outcome fails to rise with theft, and the equality of their effects is formally rejected. The *warga* decline itself is marginal and does not survive kabupaten clustering ($p = 0.14$), so we rest the argument on the contrast and not on that coefficient. Both contrasts survive spatial dependence: re-clustering at the kabupaten level, the equality of theft and each communal measure is still rejected (Table 14). The composite communal coefficient remains significant under kabupaten clustering (−0.0152, SE 0.0061) while the *warga* coefficient does not (SE 0.0037). Clustering is not shock-level inference, so we still do not read this table as fresh shock-robust evidence that communal conflict falls; that claim rests on the staggered estimators of Section 5. What the table does establish is the divergence: theft rises where communal violence, however measured, does not. Robbery, the property-*violent* act, does not rise (−0.0137^{***}), and assault is null. The composition is therefore not an artifact of comparing outcomes measured on different samples, waves, or control sets: individual economic predation rises exactly where collective and property-violent

acts do not. Nor is it an artifact of testing several outcomes at once. Correcting the five per-outcome p -values for multiple testing (Benjamini & Hochberg, 1995; Jones, Molitor, & Reif, 2019) leaves theft’s rise and the communal and robbery declines significant at the 5 percent level; only assault and *warga*, the two outcomes we do not read as moving, adjust to clear nulls (Table 14).

The contrast under the preferred specification. Section 4.1 prefers province-by-wave fixed effects for the level outcome, and consistency requires applying that standard to the contrast rather than only to the level. We do so, and report the result even though it narrows what we can claim. Re-estimating the stacked outcome-interacted model with outcome $\times$ province $\times$ wave effects in place of outcome $\times$ wave, theft falls from +0.0576 (standard error 0.0086) to +0.0147 (standard error 0.0101) and communal conflict from -0.0152 (standard error 0.0036) to -0.0079 (standard error 0.0043). The equality test between theft and communal conflict still rejects ($F = 4.43$, $p = 0.035$); the equality test between theft and the definition-stable *warga* outcome does not ($F = 2.20$, $p = 0.138$), and under the proportional Poisson version neither clears 5 percent ($p = 0.090$ and $p = 0.865$).

We do not attribute this to over-absorption, because the arithmetic does not support that reading. Province-by-wave effects absorb 33.5 percent of the treatment variance that survives village and wave effects, which predicts a standard-error inflation of 1.23; the observed inflation for theft is 1.17. The loss of precision is therefore fully accounted for, and the movement in the point estimate is not. Comparing each coefficient against its own wave-effects confidence interval, theft and communal conflict both move outside theirs, while robbery, assault and *warga* remain inside. A substantial part of both effects is accordingly between-province rather than within, which is what one would expect of a shock whose footprint is regional, but which we cannot separate here from province-specific trends in reporting or in disorder.

What survives the paper’s own specification rule is therefore the narrower claim: theft and communal conflict move in opposite directions and differ significantly under both fixed-effect structures, while the sharper contrast with the definition-stable *warga* series holds under wave effects and not under province-by-wave. We

state the claim at that width for the rest of the paper.

Because treatment is assigned by earthquakes, we also take the divergence to the shock level, the honest unit given the concentration of the communal effect in a minority of earthquakes documented in Section 5.6.1. For each of the 19 earthquakes we estimate the treated-cohort effect on theft and on *warga* and form the per-event divergence D_e , the difference between the two treatment effects for earthquake e (Table 13). The treated-village-weighted mean divergence is +0.095, and unlike the communal decline it is significant under earthquake-level inference (Figure 3, Panel B): an earthquake bootstrap that resamples whole shocks returns a 95 percent interval of [+0.020, +0.155] ($p = 0.015$), and no single earthquake drives it (leave-one-out range [+0.081, +0.114]). The divergence is positive in 10 of the 19 events, and the equal-event mean is positive and significant in its own right (0.061, $p = 0.046$), so the result does not depend on weighting toward the larger earthquakes. The compositional shift, in other words, is the part of the result that holds up at the level of the shock itself under either weighting, even where the communal decline alone is only marginal and, on the equal-event estimand, absent. The *warga* arm of this divergence is bounded from above, not merely statistically indistinguishable from zero: its treated-village-weighted effect is -0.0101 with an earthquake block-bootstrap 95 percent interval of $[-0.0242, +0.0001]$. The upper bound rules out, at conventional confidence, any *increase* in civilian communal violence larger than about 0.01 percentage points, well under one percent of the 3.6 percent pre-exposure *warga* rate. The finding that communal violence does not rise is therefore an informative equivalence bound, not a mere failure to reject.

Table 12 confirms the same pattern under the primary CS(2021) estimator, applied to each outcome over the waves in which it is observed.

The pattern is a gradient and is consistent across the two specifications. Theft *rises* sharply, by 8.6 percentage points under CS(2021) ($p < 0.001$, about 19 percent of its 45 percent base) and by 5.8 points on the common sample. Its own pre-trend does not pass, however (-0.043 , $p = 0.04$; Table 12). For that reason, and because theft can also rise from weaker guardianship, aid inflows, or changed reporting, we do not read the theft magnitude as a standalone causal estimate; what the comparison establishes is the *contrast* with communal conflict,

Table 13: Composition at the Shock Level: Theft versus *Warga* per Earthquake
The theft-versus-communal divergence, aggregated to the earthquake, clears earthquake-level inference under both weightings, where the communal decline alone does not.

	Theft	Warga	Divergence
Treated-village-weighted mean	0.090	-0.005	0.095
Equal-event mean	.	.	0.061
Equal-event two-sided p-value	.	.	0.046
Bootstrap standard error	.	.	0.034
95% CI lower bound	.	.	0.020
95% CI upper bound	.	.	0.155
Two-sided bootstrap p-value	.	.	0.015
Leave-one-out min	.	.	0.081
Leave-one-out max	.	.	0.114
Events with D__e greater than 0	.	.	10.000
Usable earthquakes (total)	.	.	19.000

For each of the 19 usable earthquakes (onset PODES wave 2008-2024, at least 20 in-sample first-treated villages) we estimate the absorbing treated-cohort effect on theft and on the definition-stable warga outcome against never-treated controls, with village and wave fixed effects, and form the per-event divergence: the theft effect for that earthquake minus its warga effect. The treated-village-weighted mean weights each earthquake by its number of unique first-treated villages; the equal-event mean weights every earthquake alike (blank cells are not defined for that row). The earthquake bootstrap resamples whole earthquakes with replacement (9,999 draws, seed 42) and recomputes cohort weights each draw; the leave-one-out rows give the range of the treated-weighted divergence when each earthquake in turn is dropped. Divergence is positive (theft rises faster than warga) in 10 of 19 events. Missing cells (Theft/Warga columns below row 1) are statistics defined only for the divergence.

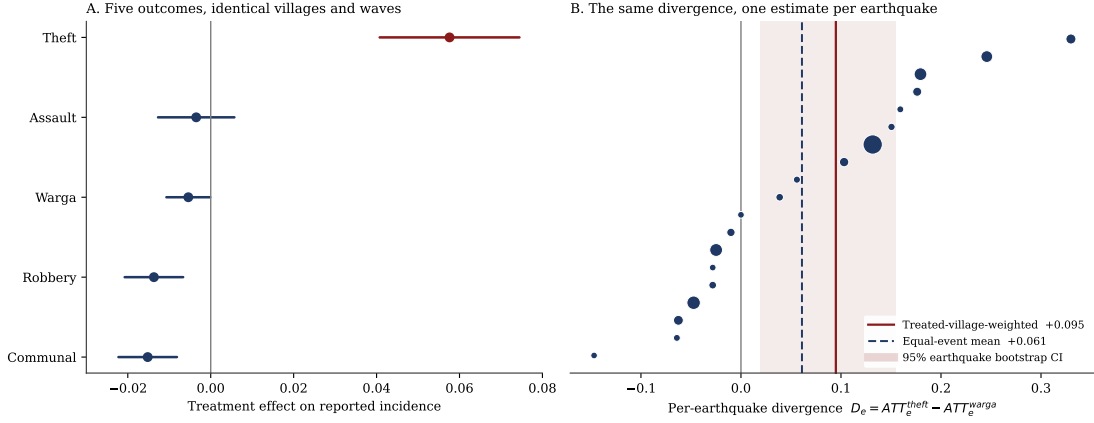


Figure 3: The Composition of Disorder

Theft rises where collective and property-violent acts do not, and the divergence holds when the earthquake, rather than the village, is the unit of inference.

Note: Panel A plots the absorbing treatment effect on each of five outcomes estimated on the single common sample described in this section, with village and wave fixed effects, no covariates, and 95 percent village-clustered confidence intervals; a coefficient of 0.058 is a rise of 5.8 percentage points. Theft is drawn in maroon. Panel B plots the per-earthquake divergence $D_e = ATT_e^{theft} - ATT_e^{warga}$ for each of the 19 usable earthquakes, sorted by magnitude, with marker area proportional to the earthquake’s number of unique first-treated villages. The solid maroon line is the treated-village-weighted mean, the dashed navy line the equal-event mean, and the shaded band the 95 percent earthquake block-bootstrap interval for the treated-village-weighted estimate (9,999 draws, seed 42). Individual earthquakes are not labelled here because the common-sample restriction changes the treated counts; the named per-event estimates for communal conflict appear in Figure 2 and the full event list in Appendix Table 37.

Table 14: Composition of Disorder on a Common Sample

Same villages, waves, and specification for all five outcomes; a formal test rejects equal treatment effects across outcomes, including against the definition-stable warga outcome.

	(1) Communal	(2) Theft	(3) Robbery	(4) Assault	(5) Warga
Absorbing D_{treat}	-0.0152*** (0.0036)	0.0576*** (0.0086)	-0.0137*** (0.0036)	-0.0035 (0.0047)	-0.0054** (0.0027)
FWER p (Westfall-Young)	0.00	0.00	0.00	0.44	0.10
FDR q (Benjamini-Hochberg)	0.000	0.000	0.000	0.450	0.062
Baseline mean	0.024	0.412	0.029	0.054	0.016
Observations	211,082	211,082	211,082	211,082	211,082

Each column is a separate absorbing regression (village and wave fixed effects, no controls) of the indicated outcome on window-aligned exposure D_{treat} , on the common sample of five crime-module waves (2005, 2008, 2018, 2021, 2024) restricted to village-waves where all five outcomes are non-missing. Communal is the collective violent composite conflict indicator; Theft is individual economic crime; Robbery is property crime with violence; Assault is individual violent crime; Warga is the civilian-versus-civilian communal subtype whose survey definition is stable across all waves. FWER p is the Westfall–Young step-down family-wise error-rate correction (500 village-block bootstrap resamples, seed 42); FDR q is the Benjamini–Hochberg false-discovery-rate value computed on the five per-outcome p-values. Formal cross-outcome composition tests (village-clustered, from the outcome-interacted stacked model on this same sample): theft vs. communal $F = 64.63$ $p = 0.000$; theft vs. warga $F = 51.14$ $p = 0.000$; theft vs. assault $F = 44.62$ $p = 0.000$. Kabupaten-clustered robustness: theft vs. communal $F = 10.05$ $p = 0.002$; theft vs. warga $F = 7.79$ $p = 0.006$. Standard errors clustered at the village level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

which does not require it, and we return to this below. Assault is null in both. Robbery does not rise: it is an insignificant -0.0032 under CS ($p = 0.46$) and negative on the common sample (-0.0137^{***}); either way it does not move with theft, so what climbs is specifically the non-violent, economic act. Communal conflict *falls*. Three inferences follow. First, the decline in communal conflict is not part of a general improvement in local order: property crime moves sharply in the opposite direction. Second, the theft rise is consistent with heightened post-disaster economic stress, though on its own it does not identify the underlying economic channel, since theft can also rise from weaker guardianship, abandoned dwellings, aid inflows, or changed reporting; we return to this limitation below. Studies of ordinary income shocks find the same asymmetry, with property crime responding more strongly than violent crime (Blakeslee & Fishman, 2018; Mehlum, Miguel, & Torvik, 2006), and Bignon, Caroli, and Galbiati (2017) recover the sign pattern we do, property crime rising while violent crime falls, after the phylloxera shock in nineteenth-century France. For Indonesia itself, Hardiman (2025) finds that rainfall-induced income shocks move crime victimisation entirely through property crime, with no detectable effect on other crime types. The asymmetry is also what a single degraded input predicts. Theft is a stealth act that a lapsed night watch stops deterring; mass fighting is a coordinated act that the same lapsed collective capacity stops enabling. Formal policing moves the same way, since police-post presence does not rise, so the two margins part company without any change in the motive for predation. We cannot separate that reading from a strengthening economic motive, because household welfare is unobserved here, and we return to the test that would separate them in Section 6. Whatever its source, this margin does *not* convert into interpersonal violence: assault, the closest individual analogue of fighting, does not move. The opportunity-cost-of-violence prediction (Becker, 1968; Dube & Vargas, 2013) fails to materialize even at the individual level. Third, given that individual violence is flat while theft rises, the decline must be specific to *organised*, *collective* violence. Mass communal fighting, unlike theft, is a coordination-intensive activity: it requires assembly, mobilization along group lines, and arenas of inter-group contact (DiPasquale & Glaeser, 1998; Esteban et al., 2012; Horowitz, 2001; Ray & Esteban, 2017; Yanagizawa-Drott, 2014). On

this evidence the pattern does not require a decline in the motive for conflict; the reading it selects is disruption of the social technology of mass fighting: the inputs that collective violence requires and individual theft does not.

Crime outcomes are available in five PODES waves and are estimated without covariates, so their pre-trend diagnostics are unconditional. Adding the standard baseline covariates leaves the theft point estimate similar ($+0.059$, $p < 0.001$) but, by shortening the pre-period, causes it to fail the pre-trend test; we therefore prefer the no-controls specification, whose pre-trend also fails (-0.043 , $p = 0.04$). Either way we treat theft as a discriminating comparison, not as an independent causal estimate. The role it plays does not depend on that test: rejecting a uniform collapse in reporting capacity requires only that the same respondent, in the same villages and the same recall window, recorded more theft while not recording more communal conflict. Because the crime module was dropped in 2011 and 2014, the immediate (t_0) theft effect is identified by the more recent cohorts (70 percent of ever-treated villages), while the Padang 2009 cohort enters only at longer horizons. The theft ATT is therefore weighted toward recent earthquakes and the conflict ATT toward all cohorts, yet the two move in opposite directions within every cohort configuration we can inspect, so the gradient is hard to attribute to cohort composition.

This is outcome discrimination, not mediation: the crime outcomes are alternative responses to the same shock, and their signs select among theories instead of decomposing the treatment effect. We read it as a shift in the *composition* of village-level social disorder, not as individual-level substitution, since PODES records incidence at the village level and cannot track perpetrators. Accordingly we estimate no channel-specific mediation shares, and we do not condition on theft in the conflict regression, since theft is itself a post-treatment outcome that would act as a bad control.

Could this be a reporting artifact? All four outcomes are reported by the same village head on the same PODES form, so one must ask whether post-earthquake administrative disruption could suppress *reporting* of communal conflict instead of conflict itself. Four features of the data argue against this. First,

the four items sit in the same security module of the questionnaire, with the same respondent and the same yes/no format, yet move in opposite directions; a disruption to reporting capacity would depress them together. Second, selective under-reporting of socially costly categories would depress the sensitive items together, and robbery does fall on the common sample (-0.0137^{***}), so that pattern alone cannot separate reporting from behaviour. What it cannot produce is theft moving the other way: under-reporting depresses items, it does not inflate them, and theft rises on the same form, from the same respondent, in the same wave. Third, salience runs against the artifact: a mass brawl is among the most memorable events in a village’s recent history, while diffuse petty theft is the harder item to recall; degraded recall would understate theft, biasing against the increase we find. Fourth, the conflict decline is negative at all six post-treatment horizons with no sign of reversal, far beyond any plausible horizon of post-disaster administrative disruption, though the longer-run estimates are imprecise. The ACLED cross-check (Appendix H) adds an independent, media-based lens with a nuance we report in full: ACLED mob-violence events tick up slightly in treated villages, but reading the 24 underlying event descriptions shows roughly two-thirds are vigilante or election-related actions against individual targets (mob justice against accused thieves, sorcery-accusation attacks), which are the documented sequelae of the scarcity margin our theft result is consistent with (Miguel, 2005), not inter-group mass fighting; the genuine inter-group episodes are a localized handful concentrated in Lombok. The externally-reported record, read at the event level, does not show the broad masked rise in communal fighting that the reporting-artifact story requires. A reporting story consistent with all four facts would have to be as specific as the behavioural one it replaces.

An empirical falsification test. The four arguments above are observational; we also test the artifact story directly. If post-earthquake conditions distorted what village heads report, the distortion should surface in other items filed by the same respondent on the same form. We construct two placebo outcomes with no plausible earthquake channel, the village head’s age and an indicator for a female head. The two specifications disagree, and the disagreement is itself informative.

Under wave fixed effects both placebos move and both reject their joint pre-trend test: the female-head indicator is -0.0099^{**} with pre-trend $p < 0.001$, and head age is -0.35^* with pre-trend $p = 0.004$. Under the preferred province-by-wave specification both pass their pre-trend test and the female-head placebo becomes a precise null (-0.0016 , pre-trend $p = 0.617$), while head age reverses sign to $+0.42$ years. Communal conflict on the identical subsample falls by 2.5 percentage points under wave effects and 1.5 under province-by-wave, significant at 1 percent in both. The naive TWFE benchmark does move on these placebos, but only because both trend strongly over the panel, which is exactly where staggered-timing weighting bias appears and the heterogeneity-robust estimator is required (Appendix I tabulates the full results). Under the estimator all causal claims here rely on, the same reporter, on the same form, shows no treatment-correlated movement in items an earthquake has no reason to move.

5.7.2 Channels Consistent with the Evidence

What follows is mechanism-consistent evidence, not a mediation decomposition: we do not claim to identify any channel's share. The organising logic is that a strong earthquake *may* raise the economic *motive* for disorder, through asset and income losses PODES cannot observe, while degrading the *capacity* to organise it collectively. The two outcomes resolve that tension differently. Individual theft, which needs only motive, rises. Communal violence, which also needs contact, coordination and the ability to evade monitoring, falls, because on this account the shock erodes those inputs faster than it may raise the motive.

The capacity-degrading channels. Three forces plausibly degrade the capacity for organised violence. *Contact disruption*: communal violence emerges from contested spaces and the inter-group contact central to relations in Indonesia (Bazzi, Gaduh, Rothenberg, & Wong, 2019; Lowe, 2021; Rao, 2019), which strong shaking can render unusable. We flag in advance that the one venue type we can count, places of worship, does not decline, so within this channel the evidence rests on communication rather than venue destruction. *Post-disaster monitoring*: a temporary surge of external actors raises the cost of organising violence, which the

counterinsurgency evidence suggests can suppress it (Berman et al., 2011), though the net effect of aid and presence is context-dependent and can be conflict-increasing where aid becomes a contestable prize (Crost, Felter, & Johnston, 2014; Nunn & Qian, 2014). This is the most tentative channel: PODES cannot measure the surge, and the one durable state-presence proxy it records, police-post presence, does not rise after exposure. *Recovery-period opportunity costs*: reconstruction can reallocate labour into construction and raise wages rather than collapse them (Kirchberger, 2017), lowering the return from conflict, opposite to the income-destruction channel (Dube & Vargas, 2013).

The rising theft signals active individual-predation incentives. Whether that reflects a lower return to honest work or a higher return to appropriation, the two income-side forces of Dube and Vargas (2013), neither requires inter-group coordination and both push individual predation upward. What the composition establishes is that these incentives, however generated, do *not* convert into collective violence. The -1.5 point communal estimate is therefore best read as the effect of degraded collective capacity: if the shock also raises the motive it is a net effect in which capacity dominates, and if it does not it is the capacity effect alone. Our data do not separate these, and the composition is the same under both. Susenas microdata would turn the motive from consistency evidence into a measured channel; because the opportunity-cost force predicts *more* conflict, a finding that welfare fell would sharpen this reading rather than weaken it.

5.7.3 Which Capacity Inputs Move: Communication, Access, and Social Capital

This subsection asks which of the framework’s inputs the shock moves, and we state the evidentiary standard once rather than at each item. Where an input can be used as a *pre-determined moderator*, the treatment is interacted with its pre-exposure value and the estimate carries the same identifying assumption as the main effect. Where an input is instead treated as an outcome in its own right, that is a second causal claim requiring its own parallel trends, and none of the input series PODES offers passes that test. We therefore lead with the moderator evidence and report the outcome regressions as descriptive.

Mobilisation capacity: the moderator test. Strong mobile-phone signal proxies the contact and mobilisation capacity on which collective violence depends (Yudhistira, Khairina, Dartanto, & Nuryakin, 2021); its role in collective action is established more broadly by Manacorda and Tesei (2020) for mobile coverage and protest in Africa and by Enikolopov, Makarin, and Petrova (2020) for social media and protest in Russia, where it works by lowering the cost of coordination rather than by spreading information. If that is the operative channel, the decline should be concentrated in villages that had signal to lose. It is. Interacting first exposure with signal presence at the 2005 baseline, a value fixed before any identified cohort is treated and held by 76 percent of villages, the interaction is -0.0201 (standard error 0.0071) on the composite outcome and -0.0129 (standard error 0.0056) on the definition-stable *warga* outcome, both under the preferred province-by-wave specification. The implied split is stark: among villages with no baseline signal the effect is $+0.0002$ on the composite, an unambiguous null, against -0.0199 among villages that had it.

Two features make this the strongest mechanism evidence in the paper. The moderator is pre-determined, so the test does not require parallel trends in the input itself, which is the standard route in this literature (Bai 2023; Amarasinghe et al. 2026). And it survives the province-by-wave specification, which most of the paper’s other margins do not.

It is not a decomposition, and one competing reading has to be priced. Baseline signal correlates with remoteness and development, so the interaction could be capturing either. Adding pre-exposure paved-road access, police-post presence and distance to the district capital as further interactions leaves the signal interaction at -0.0181 (standard error 0.0092) on the composite and -0.0129 (standard error 0.0075) on *warga*, the latter unchanged to four decimals; the distance interaction is null in both ($+0.0013$ and -0.0027). The road interaction is itself significant (-0.0111 and -0.0153), so more than one infrastructure input is doing work and we do not claim signal is the only one.

The same inputs as outcomes. Estimated as outcomes rather than moderators, the same series point the same way but none supports a causal reading. Mobile

signal falls by 3.1 percentage points after first exposure (CS -0.0312 , $p < 0.001$, against a 71 percent pre-exposure mean and despite secular coverage expansion), the count of village security personnel (*linmas*) rises (0.19 in asinh terms), and road *passability* does not move (four-wheel -0.5 , 95% CI $[-1.2, 0.3]$; year-round 0.6 , $[-1.1, 2.2]$) even as the paved-road *share* falls by 5.9 points. Every one of these series fails its own pre-trend test, the *linmas* series decisively (-0.12 , $p < 0.001$), so we report them as descriptive corroboration of direction and nothing more. Two further communication outcomes sit in Appendix K on the same footing: street lighting *rises* after exposure, which reconstruction explains more naturally than any capacity channel, and the second fails its pre-test.

The night watch is the guardianship input our reading leans on, and PODES records it: whether the village maintains a community security post (*poskamling*), asked in six of seven waves. Presence falls by 3.6 percentage points after exposure (CS -0.0355 , $p = 0.001$) against a 68 percent pre-exposure mean, the sign the capacity reading predicts. We do not read it causally: its pre-trend fails decisively (-0.143 , $p < 0.001$) and the pre-treatment movement is four times the estimated effect. Security infrastructure in these villages is trending for reasons the design cannot separate from exposure. We therefore report guardianship as the reading most consistent with the composition of crime, not as a channel we have identified, and note that every direct proxy PODES offers is either trending or measured too coarsely to carry the claim.

One further test discriminates between our reading and its closest rival, and it does not go our way. If the theft rise works through a lapsed night watch it should concentrate in villages that had one to lose; if it works through raised motive and emptied dwellings it should be roughly uniform in prior guardianship. Interacting the absorbing treatment with an indicator for a security post in the village’s last pre-exposure wave, a predetermined moderator, the interaction is negative and significant: theft rises by 7.2 percentage points where no post was present and by 2.5 where one was (-0.046 on the interaction, $p = 0.008$; $N = 14,380$ across 2,907 exposed villages), while the communal decline does not vary with prior guardianship at all ($p = 0.22$). The theft response is therefore *larger* where there was no watch to lose, the opposite of what the dual-use reading predicts. Either

villages with a standing post are simply more resilient on every margin, or the theft rise is driven by displacement and motive rather than withdrawn guardianship, and we cannot separate them. Two caveats attach: the test is identified off exposed villages only, since the moderator is undefined otherwise, and *poskamling* was a state-coordinated programme as much as a village institution (Barker, 1998), so its presence proxies state penetration as well as collective capacity. We report it because it is the sharpest test our data permit of the mechanism we favour, and it fails.

A rise in social cohesion (Aldrich, 2012) does not explain the decline either. Reading confidence intervals rather than significance (Abadie, 2020; Imbens, 2021; Romer, 2020), the PODES social-capital proxies show no solidarity surge: mutual-help participation, sports groups, Karang Taruna presence and the count of village institutions all have intervals that exclude movement of the magnitude a cohesion mechanism would require, and worship-venue counts are if anything marginally higher (0.008 in logs, $p = 0.10$), which counts against venue destruction and leaves the contact channel resting on mobile signal. Conceptually, denser associational life would if anything *raise* the capacity to organise violence (Satyanath, Voigtländer, & Voth, 2017), so evidence that these same earthquakes raised participation elsewhere (Bai and Li 2021; Cassar et al. 2017) is not in tension with a communal decline: it measures recovery-oriented participation, not the contact and communication capacity inter-group violence requires (Appendix L).

5.7.4 A Direct Test of the Capacity Channel

The channels above are inferred from what the shock does to inputs. PODES also measures collective capacity directly. From 2018 onward the survey asks whether villagers customarily engage in *gotong royong*, mutual-help collective action, and how widely they participate, on an identical three-point scale in 2018, 2021 and 2024. Two items are asked separately: collective action for public purposes, which the questionnaire illustrates with communal labour, night-watch rotations and village festivals, and collective action to help residents struck by misfortune such as death, illness or accident. The first is a direct measure of the capacity to organise the village; the second is a measure of disaster response.

Table 15 estimates the effect of first strong exposure on both. High-involvement public-purpose *gotong royong* falls by 3.1 to 3.2 percentage points against a mean of 86 percent, significant under both sets of fixed effects. The misfortune-response item does not fall; its extensive margin rises slightly. The two move in opposite directions, which is what the capacity reading predicts and what a general collapse in village functioning would not: routine collective organisation weakens while emergency mutual aid, which needs no standing coordination, does not.

We put weight on this cautiously. The item exists in only three waves, so only the 2021 and 2024 cohorts are identified within the window, 942 villages in all. A Callaway–Sant’Anna estimate on the same item corroborates the two-way specifications, at -0.0330 ($p = 0.008$), and the single pre-treatment lead that three waves permit does not reject (0.030 , $p = 0.38$), although a test built on one lead has little power. The calibration row of the table makes the power problem explicit: on this restricted sample the conflict effect itself is no longer significant under the preferred fixed effects. Two further limits bound the reading. The estimand is not the headline estimand: it averages the 2021 and 2024 cohorts only, so it covers different earthquakes over a different period and corroborates the direction of the effect without being a component of its magnitude. And the 2021 wave was enumerated in May 2021, during the pandemic, which depresses collective activity in level and falls in the first post-exposure wave for the 2021 cohort; wave fixed effects absorb that shock where it is common across villages, and it would bias the estimate only if restriction intensity covaried with earthquake exposure. We therefore read the result as corroborating the capacity interpretation, not as establishing it.

Table 15: Collective Capacity After Strong Shaking

Outcome	Wave FE	Province $\times$ wave FE	Pre-exposure mean	N
Public-purpose <i>gotong royong</i> , high involvement	−0.0309** (0.0138)	−0.0324** (0.0164)	0.863	78,312
any involvement	−0.0027 (0.0018)	0.0030 (0.0026)	0.994	78,312
Misfortune-response <i>gotong royong</i> , high involvement	−0.0112 (0.0099)	−0.0033 (0.0127)	0.911	78,312
any involvement	0.0023 (0.0020)	0.0059** (0.0025)	0.996	78,312
<i>Power calibration on the same restricted sample</i>				
Any communal conflict	−0.0394*** (0.0097)	−0.0148 (0.0112)	0.026	93,690

Gotong royong is measured on an identical three-point scale in PODES 2018, 2021 and 2024 (1 = present, most residents involved; 2 = present, few involved; 3 = no such custom); the item is absent or differently coded in earlier waves, so the window is three waves. Only cohorts first exposed in 2021 and 2024 are identified within it (942 villages, 1.8 percent of village-waves treated); villages exposed at or before 2018 are always-treated here and dropped. The calibration row shows that the conflict effect itself loses significance on this restricted sample under the preferred fixed effects, so these estimates are low-powered by construction. A Callaway–Sant’Anna estimate on the same item gives −0.0330 ($p = 0.008$), and the single pre-treatment lead the three waves permit does not reject (+0.030, $p = 0.38$), though a test built on one lead has little power (09_mech_direct_nb2005, master step B5.10). The 2021 wave was enumerated during the COVID-19 period, which depresses collective activity in level; wave fixed effects absorb that shock where it is common across villages. Standard errors are clustered at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.7.5 Suggestive Heterogeneity

Having examined the village-level inputs directly, we ask whether the effect is larger where those channels have more room to operate. Interacting first exposure with pre-treatment religious polarization deepens the effect by 1.8 percentage points per

standard deviation (-0.0175 , standard error 0.0065 , kabupaten clustering), and a gradient of the same sign appears on the definition-consistent *warga* outcome (-0.0076 , standard error 0.0037), while the monitoring interaction carries over to it only weakly (-0.0034 , standard error 0.0020 , against -0.0075 , standard error 0.0025 on the composite). That the inter-group measure tracks the inter-group moderator more closely than a generic monitoring channel is what a contact-and-mobilization account predicts.

We then apply to this gradient the same scale test that governs the rest of the paper, and it does not pass. Religious polarization is not orthogonal to how much conflict a place has to begin with. Baseline conflict rises from 2.1 percent in the least polarized tercile of kabupaten to 4.8 percent in the most polarized, a ratio of 2.3, and the kabupaten-level correlation between baseline conflict and religious polarization is 0.37 (0.39 within the estimation sample). A response that is common in proportional terms therefore delivers a larger *absolute* decline where polarization is high, with no separate role for polarization at all. Three tests bound how much of the gradient this accounts for. Adding an interaction of exposure with the leave-one-out kabupaten baseline conflict rate retains 40 percent of the polarization gradient and removes its significance (-0.0070 , standard error 0.0048), with the baseline-conflict interaction itself entering at -0.0141 (standard error 0.0053). Replacing that linear control with a flexible set of baseline-conflict tercile interactions, which imposes no functional form on the scale channel, leaves the polarization gradient at -0.0141 (standard error 0.0058). And estimating the interaction proportionally, as Poisson, gives -0.0675 with a 95 percent confidence interval of $[-0.269, 0.134]$, which does not reject a common proportional response.

The gradient is therefore not separately identified from the amount of conflict a place starts with: whether it survives depends on how baseline conflict is controlled for, and the proportional specification cannot distinguish it from nothing. We report it because it is the natural test of the mechanism, and we report its failure because the same standard governs the external-validation exercise we discard in Appendix P. We do not read it as evidence for the contact-and-mobilization account. That account rests on the composition contrast, which does survive the proportional test, and on the direct *gotong royong* measurement. Appendix C

gives the interaction tables, the polarization-versus-fractionalization collinearity, and the weaker ethnic counterparts in full.

6 Conclusion

A strong earthquake changes the composition of local disorder more than it changes its level, and the compositional result is the one this paper rests on. In medium and high seismic-hazard zones, a village’s first damage-level earthquake ($\text{MMI} \geq \text{VI}$) lowers reported communal conflict by 1.5 percentage points, about 25 percent of the treated villages’ 6.1 percent pre-exposure rate. Point estimates are more negative among villages with a prior history of conflict, but this heterogeneity is imprecisely estimated and particularly vulnerable to mean reversion; among villages that had never reported communal conflict, the shock does not suppress conflict onset and instead raises it by 0.70 percentage points (standard error 0.22) against a counterfactual onset rate of 2.1 percent, a relative increase of about a third. What rises instead is theft, which indicates that the decline in reported conflict does not represent a general improvement in local security.

The broad communal-conflict measure falls across the estimators, shaking thresholds, and exposure measures we use. Its robustness to departures from parallel trends is nonetheless limited: the HonestDiD identified sets exclude zero through $\bar{M} = 0.25$ and admit zero at $\bar{M} = 0.50$. The estimate is also concentrated in Indonesia’s larger earthquakes, and under shock-level inference it holds when earthquakes are weighted by the villages they treat ($p = 0.01$) but not under an equal-event mean ($p = 0.34$). The definition-stable *warga* series passes the joint pre-trend test under the preferred specification and agrees in sign with the composite, but it does not deliver a level estimate of its own: its coefficient is not significant under the preferred specification, and its HonestDiD breakdown value is an order of magnitude below the composite’s. Its role is to hold the survey definition fixed, and it is in that role, anchoring the composition contrast, that it carries weight. The theft-versus-*warga* divergence is what survives earthquake-level inference under both treated-village weighting ($p = 0.015$) and an equal-event mean ($p = 0.046$), the bar the pooled level estimate does not clear, and it is what lets us discriminate

among explanations for the conflict effect. It is not robust on every margin, and we say where it is not: under province-by-wave effects the theft-versus-communal contrast still rejects ($p = 0.035$) while the theft-versus-*warga* contrast does not (Section 5.7.1). What the estimates rule out is a general improvement in local security.

Every sign we estimate is consistent with two consequences of one shock. On the reading we favour, a strong earthquake degrades the village's capacity for organised collective action, and that capacity was doing two jobs at once. It supplied the mobilisation that communal violence requires, and it supplied routine guardianship, the night-watch rotations and communal patrols that deter property crime. Remove it and coordination-intensive violence falls while stealth theft rises, while acts that require an encounter do not: robbery falls by 1.4 percentage points on the common sample and assault does not move. That asymmetry is the part a rising-motive story alone does not deliver, since with encounters and guardianship unchanged a stronger motive for predation would raise theft and robbery together. Our most direct mechanism evidence points the same way. High-involvement public-purpose *gotong royong*, which the questionnaire illustrates with night-watch rotations, falls by 3.1 to 3.2 percentage points while emergency mutual aid does not, and police-post presence does not rise. We do not observe household income, so we cannot say whether the shock also raises the motive for predation; the theft rise is consistent with lapsed guardianship, with economic stress, or with both; Susenas microdata would separate them, and the sharpest test our data permit of the guardianship reading fails (Section 5.7.3). What we can say is that no change in motive is needed to generate the pattern we find. For development economics, the post-disaster period is a moment when the local technology of collective action is reshaped, and its inputs, the night watch among them, are things disaster governance can observe and restore.

The identified cohorts enter treatment between 2008 and 2024; villages already exposed before the panel begins are not directly estimated. Second, the crime-comparison evidence discriminates among mechanisms but does not decompose the treatment effect into income, guardianship, aid, and mobilization shares. Direct household evidence on welfare and labor-market margins, for example from

Susenas, would turn the motive side of the argument from consistency evidence into a measured channel. Spatial spillovers are tested directly in Section 5.6.4 and do not overturn the result.

The estimates are identified from rural and peri-urban villages in Indonesia’s medium/high seismic-hazard zones, a setting with dense village-level social organisation, a history of localized communal violence, and a state that mobilizes visibly after disasters. The production-function logic itself is portable: any large physical shock that destroys venues, absorbs residents’ time, and draws in external monitors should suppress coordination-intensive violence while leaving individual predation to respond to scarcity. What should not be extrapolated is the net sign. In settings where disasters displace populations into camps that concentrate rival groups, or where the state’s post-disaster presence is weak or itself predatory, the contact and monitoring inputs can move the other way, and the same framework predicts more violence rather than less, as in the legitimacy-and-coordination channel through which earthquakes *raised* conflict in imperial China (Bai, 2023). The Indonesian estimate is best read as one point in that mapping from shock characteristics to the inputs of collective violence, not as a general claim that disasters pacify.

One policy implication follows, stated modestly given the design’s limits. A quiet communal-conflict count after a disaster should not be read as evidence that no economic harm occurred, and disaster response should address livelihood and property security directly. We resist a second, tempting reading, that the decline signals eased inter-group tensions: if what falls is the *capacity* to organise violence rather than the grievances themselves, latent tensions could resurface as external monitoring and reconstruction wind down. Our design does not test that dynamic, and we flag it as a hypothesis for future work rather than a conclusion in its own right.

Data Availability

This is a restricted-access reproduction package, and we state precisely what can and cannot be shared.

Confidential and not redistributable. The village-level PODES microdata and

the BPS village-boundary product (*Peta Desa*) are confidential products of BPS-Statistics Indonesia and may not be redistributed by us. Researchers obtain them directly from BPS under its data-access terms. Because the derived analysis panels carry BPS village identifiers, they are withheld for the same reason and must be rebuilt from source.

Public or licensed, obtained from the provider. USGS ShakeMap MMI and PGA rasters are public domain; the PVMBG/ESDM seismic-hazard zone maps and the SNPK/NVMS incident database are public; ACLED and IPUMS International require each user to register and build their own extract under the providers' terms. Exact sources and download instructions for each are given in the deposit.

Provided in full. All analysis code is included: the programs that clean the raw sources, build the village panel, and produce every table and figure in the paper and appendix. A researcher with BPS access can rebuild the panel from the code and reproduce every reported number, and each number is traceable through the deposit's verification file to the program and log that produced it.

The replication package contains a complete data-availability and provenance statement, following the AEA Data Editor template, that classifies all thirteen sources by provider, licence, access route, and the code that consumes it.

CRediT Author Statement

Billy Thandy Tulungen: Writing – original draft, Writing – review & editing, Software, Formal analysis, Conceptualization, Validation, Project administration, Methodology, Data curation, Funding acquisition.

Muhammad Halley Yudhistira: Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization.

Jahen Fachrul Rezki: Writing – review & editing, Writing – original draft, Supervision, Investigation, Conceptualization.

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Declarations

The authors declare no competing interests. This study uses secondary and publicly available data and de-identified restricted survey aggregates; it did not require ethics-board approval.

Online Appendix

Earthquakes and Communal Conflict: Village-Level Evidence from Indonesia

Not for print; available online and in the replication package.

A Staggered DiD: Group-Level ATTs

Table 16: Callaway–Sant’Anna Group-Level ATT(g)

Takeaway: five of the six cohorts have negative ATTs; the only positive cohort is G6, the Aceh Pidie Jaya cohort, whose pre-treatment conflict rate sits at a post-conflict floor and can only revert upward.

Cohort	First treated wave	ATT	SE	z	p
<i>Panel A: Composite (any communal conflict)</i>					
G3	Wave 3 (2008)	−0.0086	0.0064	−1.33	0.182
G4	Wave 4 (2011)	−0.0121	0.0070	−1.73	0.084
G5	Wave 5 (2014)	−0.1281	0.0704	−1.82	0.069
G6	Wave 6 (2018)	+0.0322	0.0121	+2.66	0.008
G7	Wave 7 (2021)	−0.0526	0.0118	−4.47	0.000
G8	Wave 8 (2024)	−0.0244	0.0174	−1.40	0.160
Simple ATT	(all cohorts)	−0.0150	0.0043	−3.50	0.000
<i>Panel B: Warga (definition-consistent)</i>					
G3	Wave 3 (2008)	−0.0029	0.0053	−0.55	0.585
G4	Wave 4 (2011)	−0.0056	0.0054	−1.05	0.293
G5	Wave 5 (2014)	−0.0846	0.0569	−1.49	0.137
G6	Wave 6 (2018)	+0.0252	0.0097	+2.60	0.009
G7	Wave 7 (2021)	−0.0202	0.0088	−2.30	0.022
G8	Wave 8 (2024)	−0.0182	0.0166	−1.10	0.271
Simple ATT	(all cohorts)	−0.0060	0.0034	−1.78	0.075

Group-time ATTs from Callaway–Sant’Anna with baseline PODES 2003 covariates, doubly-robust weighting and not-yet-treated controls. G6, the Aceh Pidie Jaya cohort, is the only positive-ATT group; Appendix Table 35 shows that dropping it leaves the pooled estimate essentially unchanged. G5 contains 37 treated villages and its estimate is correspondingly imprecise.

B Alternative Clustering and Outcome Comparability

Table 17: Robustness: Kabupaten-Level Clustering

Takeaway: both outcomes remain significant at the 5 percent level under kabupaten clustering, though not under the more conservative province-level inference.

	Village cluster (32,435 clusters)	Kabupaten cluster (320 clusters)	Province cluster (28 clusters)
<i>Outcome: composite</i>			
D_{treat}	−0.0198*** (0.0033)	−0.0198*** (0.0065)	−0.0198** (0.0075)
t -statistic	−5.99	−3.06	−2.64
<i>Outcome: warga</i>			
D_{treat}	−0.0066** (0.0026)	−0.0066* (0.0036)	−0.0066 (0.0049)
t -statistic	−2.55	−1.82	−1.34
Observations	227,045		
Village + wave FE	Yes		

Point estimate is identical across columns by construction (same coefficient, only the clustering level of inference changes). All specifications include village and wave fixed effects and the baseline covariates interacted with wave dummies (see Table 33). Because the province level has few clusters, we also report wild cluster bootstrap p -values (Webb weights, 9,999 replications): $p = 0.040$ (conflict_any) and $p = 0.255$ (conflict_warga), consistent with the analytic inference. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 17 re-clusters standard errors at the kabupaten (district, 350 clusters) and province (28 clusters) level, for both conflict_any and conflict_warga. The point estimate is identical across columns by construction; only the level of inference changes. Village- and kabupaten-level clustering leave the composite estimate significant. Province-level clustering is a deliberately conservative check with few clusters and a correspondingly large loss of precision: the two-way fixed-effects composite still clears 5 percent analytically ($t = -2.64$) and under a wild-cluster bootstrap ($p = 0.040$), while warga does not ($p = 0.255$) and the pooled CS(2021)

ATT does not reject either ($p = 0.16$). This pattern, in which standard errors rise monotonically from village to kabupaten to province, is exactly what within-region spatial correlation in earthquake exposure and conflict predicts, and it does not overturn the headline result at any clustering level.

A subtler comparability concern is that the composite *conflict_any* outcome combines subtypes (warga, desa, suku) that become separately available only from 2008 onward, while the 2005 wave uses the mass-brawl (*perkelahian massal*) item directly. Table 18 addresses this on the absorbing windowed-first-exposure design. Column (1) reproduces the headline *conflict_any* result (-0.0198^{***}). Column (2) replaces the composite outcome with *conflict_warga*, the civilian-vs-civilian conflict indicator, which is constructed from a single consistent questionnaire item in every wave; the coefficient is smaller in magnitude but remains negative and significant at the 5 percent level (-0.0066^{**}), indicating the effect is not an artifact of the composite definition and is not driven solely by the inter-village or inter-ethnic subtypes that become available only from 2008 onward. Within the analysis window (2005–2024) the outcome definition is therefore consistent, and this measurement difference is not a threat to the reported estimates.

Table 18: Robustness: Outcome Variable Comparability

Takeaway: all three outcomes decline, including the enumerator’s gate question in column 3, whose wording is identical in every wave from 2005 to 2024. The sign of the result is therefore not an artifact of how conflict is defined. Part of its magnitude is: measured against its own pre-exposure mean, the gate outcome falls by about two thirds as much as the composite, because the composite narrows from the bare gate in 2005 to communal subtypes only from 2008.

	(1)	(2)	(3)
	Composite	Warga	Any mass brawl
D_{treat}	-0.0198*** (0.0033)	-0.0066** (0.0026)	-0.0161*** (0.0035)
t -statistic	-5.99	-2.55	-4.57
Observations	227,045	227,045	227,045
Village + wave FE	Yes	Yes	Yes

Notes: Standard errors clustered at village level in parentheses. All specifications use the absorbing D_{treat} (windowed first exposure) and include village and wave fixed effects plus the baseline-covariate-by-wave control set. Column 1, `conflict_any`, is the composite outcome (`warga + antar-desa + suku`) used in the headline result. Column 2 restricts to civilian-vs-civilian conflict, the subtype defined consistently across all PODES waves (see Section 3). Column 3 is the enumerator’s gate question itself, recording whether any mass brawl occurred regardless of type, and so also counts brawls against security or government personnel, student brawls, and other types that columns 1 and 2 exclude by construction. It is the only one of the three whose definition is identical in every wave from 2005 to 2024: in 2005 the composite is the bare gate, whereas from 2008 it is restricted to communal subtypes, so the share of gate-positive villages that the composite records as zero rises from 11.7 percent in 2008 to 33.7 percent in 2024. On the same estimation sample its mean is 0.0333 against 0.0273 for the composite, so the estimate in column 3 is 48.2 percent of its own base while column 1 is 72.6 percent of its own. We report it because a decline measured on a narrowing definition would overstate the effect, not because it resolves the pre-trend evidence discussed in Section 4, which it does not. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

C Heterogeneity in the Conflict Response

Having examined the village-level inputs of the contact and coordination channels directly, we ask a complementary question: is the conflict-reducing effect larger where those channels have more room to operate? Table 19 reports interaction

regressions that test whether the treatment effect varies with village proxies for these channels. Each column interacts the absorbing treatment with a standardized pre-treatment village characteristic, using the headline design (village and wave fixed effects, the baseline covariate set, and standard errors clustered at the village level).

Table 19: Mechanism Heterogeneity: Interaction Regressions

	(1) Monitoring	(2) Contact disruption	(3) State capacity
D_{treat}	-0.0195*** (0.0033)	-0.0194*** (0.0033)	-0.0199*** (0.0033)
$D_{\text{treat}} \times \text{Police (std.)}$	-0.0009 (0.0029)		
$D_{\text{treat}} \times \text{Road access (std.)}$		-0.0029 (0.0033)	
$D_{\text{treat}} \times \text{Dist. to capital (std.)}$			0.0026 (0.0023)
Observations	227,045	227,045	227,045
Adj. R ²	0.073	0.073	0.073

Standard errors clustered at village level in parentheses.

Absorbing windowed first-exposure treatment; village and wave fixed effects and the baseline covariate set in all specifications. Moderators standardized (mean zero, unit standard deviation). Baseline conflict incidence: 2.6%.

* p<0.1, ** p<0.05, *** p<0.01

The interactions in Table 19 do not sharpen the channels. On the composite outcome the conflict-reducing effect does not vary significantly with pre-existing police presence (interaction -0.0009 , $\text{SE} = 0.0029$), with paved-road access (-0.0029 , $\text{SE} = 0.0033$), or with distance to the district capital ($+0.0026$, $\text{SE} = 0.0023$). A police interaction does appear on the definition-stable *warga* outcome (-0.0075^{***} , Appendix Table 45), but it is specific to that outcome and to village-level clustering, so we do not build the monitoring channel on it. We read the road null as a null on this moderator, not as evidence against the contact channel. The relevant margin for that channel is the communication infrastructure whose proxies decline after exposure, not road access, which typically stays passable after moderate shaking.

The fourth prediction of Section 2 concerns group structure. Because communal violence is fought *between* groups, the contact and mobilization inputs it requires are more available where the local population is more polarized, so the shock's disruption of those inputs should reduce communal conflict more where polarization

is higher. Interacting the absorbing treatment with pre-treatment kabupaten *religious* polarization (Esteban-Ray, from 2000 census religious shares; [Montalvo and Reynal-Querol 2005](#), [Esteban et al. 2012](#)) bears this out: a one-standard-deviation increase deepens the effect by 1.8 percentage points (-0.0175). Because the moderator varies at the kabupaten level, we cluster the standard errors there; the interaction remains significant at the 1 percent level ($t = -2.69$, $p = 0.008$, 253 kabupaten clusters). This is the dimension along which Indonesia’s major communal episodes have historically run (Maluku, Poso), and it is where the polarization theory of conflict predicts the moderator should bind: polarization, rather than fractionalization, drives conflict intensity when the contested prize is *public*, such as religious or cultural dominance and local political control, rather than private and divisible, and when group cohesion is high ([Esteban et al., 2012](#); [Ray & Esteban, 2017](#)). Ethnic polarization ([Bazzi & Gudgeon, 2021](#)) points the same way in the continuous interaction but is not robust and we do not lean on it, consistent with religion proxying this public-prize axis of Indonesian communal contestation more cleanly than ethnicity. We do not, however, read this as isolating polarization from fractionalization. For religious divisions in Indonesia, where a large majority faces smaller minorities, the two indices are nearly collinear and a joint horse-race cannot separate them, with both moderating the effect at almost identical magnitude (interaction -0.0175 and -0.0165 , both $p < 0.01$). The ethnic counterparts are weaker but not uniformly absent: the ethnic polarization interaction is insignificant (-0.0092) while ethnic fractionalization is -0.0150 , significant at 5 percent (Appendix Table 46). We therefore read the gradient as evidence of an inter-group group-structure channel in the sense of [Esteban et al. \(2012\)](#), without attributing it to polarization rather than fractionalization specifically. To guard against selective emphasis, we estimate all five moderator interactions (police, road access, distance to the district capital, ethnic and religious polarization) on a common kabupaten-clustered specification (samples differ slightly by moderator coverage) and correct their p -values for multiple testing: only the police-monitoring and religious-polarization interactions survive, and these are the two the discussion emphasizes; the ethnic-polarization, road, and distance interactions do not. These patterns are suggestive at best. The moderators are

measured at the kabupaten level and moderate a village-level outcome, and, as Section 5.7.5 sets out, the religious gradient is not separately identified from the amount of conflict a place starts with: high-polarization kabupaten have 2.3 times the baseline conflict rate, and once that scale is controlled the gradient either loses significance or does not survive a proportional specification. We therefore do not read these interactions as evidence for the contact-and-mobilization account, and for the same reason we do not lean on the religious gradient against the reporting concern: a reporting shock proportional to each village’s underlying rate would itself vary with polarization through that correlation.

One further exercise speaks to the definition-stable *warga* outcome, whose pooled staggered estimate is negative but imprecise and is therefore weak evidence on its own. Because an average effect can mask heterogeneity, we ask whether the *warga* response is concentrated where the inter-group mechanism should bite. The pattern is there, subject to the identification caveat just stated. Interacting first exposure with pre-treatment religious polarization, the effect on *warga* deepens with polarization (interaction -0.0076 , standard error 0.0037), the same gradient we estimate for the composite outcome, whereas the monitoring interaction weakens markedly on *warga*: -0.0034 (standard error 0.0020) against -0.0075 (standard error 0.0025) on the composite (Appendix Table 45). The cleanest, definition-consistent measure of inter-group conflict thus responds to the inter-group moderator and not to the general monitoring channel, which is what a contact-and-mobilization account, rather than a generic restoration of order, would predict.

D Supporting Detail on Design and Robustness

This appendix holds detail supporting the design and robustness discussion of the main text: raw cohort paths, the case for the absorbing treatment definition, the hazard-zone restriction, the intensity ladder, and the full robustness battery.

D.1 Raw Outcome Paths by Cohort

Before any specification is imposed, Figure 4 plots the raw share of villages reporting communal conflict, wave by wave, separately for each treatment cohort and for the never- and always-treated groups. Cohorts differ in level: the 2021 and 2024 cohorts sit well above the never-treated line throughout, which is the composition fact that motivates village fixed effects. The never-treated path is close to flat and drifts down slightly after 2018, which is the trend that province-by-wave effects are meant to absorb. And the cohorts are of very unequal size, from 880 villages first exposed in 2008 to 37 in 2014, so the raw paths for the smaller cohorts are correspondingly noisy; the 2014 cohort is omitted from the figure for that reason.

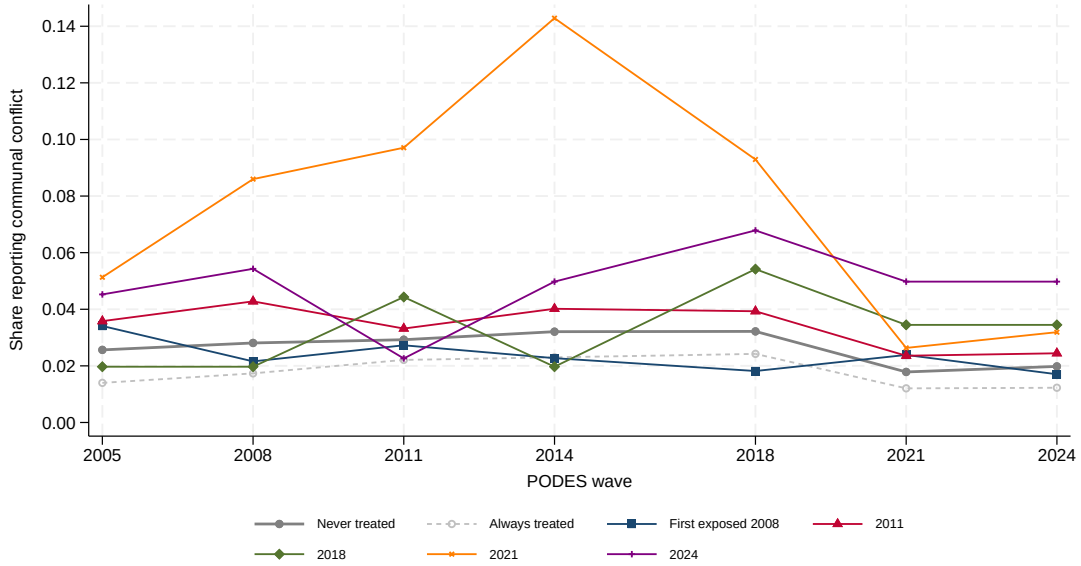


Figure 4: Raw Conflict Rates by Treatment Cohort

Note: Unadjusted means, no covariates and no fixed effects. Cohort sizes: never treated 30,288 villages; always treated 9,856; first exposed 2008, 880; 2011, 1,145; 2014, 37 (omitted); 2018, 1,145; 2021, 721; 2024, 221. The always-treated cohort is excluded from the preferred specification.

D.2 The Intensity Ladder and the Regression-Discontinuity Alternative

The conceptual framework predicts a dose-response relationship. We report the threshold estimates but do not claim to have identified one. Under the absorbing first-exposure design the effect strengthens from -0.0063^{***} ($SE = 0.0021$) at $MMI \geq V$ to -0.0198^{***} ($SE = 0.0033$) at $MMI \geq VI$, and then stops strengthening:

at $\text{MMI} \geq \text{VII}$ (“Very Strong”) it is -0.0174^{***} ($\text{SE} = 0.0035$), while the CS(2021) counterpart in the same table falls from -0.0150^{***} to -0.0054 and loses significance. The pattern is therefore not monotone across the full range.¹ We withdraw the dose-response reading for two reasons. First, the primary CS(2021) estimator does *not* separate the two higher bins: at $\text{MMI} \geq \text{VII}$ its estimate is -0.0054 and insignificant, against -0.0150^{***} at $\text{MMI} \geq \text{VI}$, reflecting the smaller and less precisely estimated cohort at that threshold (1,436 ever-treated villages against 3,207). Second, changing the threshold also changes which cohorts and which earthquakes enter the treated group, so the comparison is not a pure intensity contrast holding composition fixed. A cleaner test would compare villages struck by the *same* earthquake at different intensities; the stacked design of Section 5.6 does exactly that, and we rest the intensity argument on it rather than on the threshold ladder. We report the threshold estimates for completeness and draw no dose-response conclusion from them.

D.3 Event Study

D.4 Estimator Comparison

Table 4 collects every estimator for the primary baseline in one place (Figure 8 in the appendix plots the same estimates). Across heterogeneity-robust and conventional estimators, and across both the absorbing and episodic treatment definitions, the estimate stays negative and significant, ranging from -0.013 to -0.020 (the pooled CS ATT significant at 5 percent, the remaining estimators at 1 percent); the modest variation across rows reflects sample, controls, and estimand, not the sign or direction of the result, and every confidence interval lies below zero. A

¹A natural question is why we do not exploit the MMI VI threshold directly in a regression-discontinuity design. Two obstacles rule it out. First, MMI VI is a classification boundary chosen by the analyst, not an institutional trigger, so nothing in treatment receipt changes discontinuously there. Second, the running variable is not smooth in the neighbourhood of the cutoff. ShakeMap intensities reach villages through contoured raster bands, and the resulting village-event distribution is visibly heaped: 5.4 percent of the 1,961,850 in-sample village-event observations fall within 0.01 of an integer against roughly 2 percent under continuity, and MMI 6.0 is itself a mass point. The density is not so degenerate as to make local-polynomial estimation mechanically impossible, but combined with the first objection, that the cutoff is a classification convention rather than a treatment assignment rule, there is no discontinuity for such an estimator to recover. The band comparison reported in Section 5.6, which contrasts villages at MMI $[6, 7)$ with those at $[5, 6)$ struck by the same earthquake, is the defensible version of the same intuition and is unaffected by heaping within bands.

separate imputation-based estimator (Borusyak et al., 2024) gives a similar, more precisely estimated effect (-0.0132 , standard error 0.0035), at the cost of a stronger identifying assumption. Across estimators the question is not which number to believe but whether any of them overturns the others, and none does.

D.5 dCdH episodic estimator detail

The de Chaisemartin and D’Haultfoeuille (2026) episodic estimator identifies the effect only when a village switches into fresh exposure in the current interval, not the cumulative ever-treatment effect. It gives -0.015^{***} ($SE = 0.004$); the full estimates and placebo panel are in Appendix Table 34. Its joint placebo test does not reject at conventional levels ($p = 0.056$), but we report the individual placebos rather than the joint test alone: the nearest one is $+0.010$ ($SE = 0.005$), which is significant at 5 percent and carries the opposite sign to the effect. The episodic effect is comparable in magnitude to the absorbing estimates (TWFE -0.019 , pooled CS ATT -0.0150), as expected since it captures only fresh switching, not cumulative exposure. The two are equal at -0.015 , so this comparison does not discriminate between a persistent effect and one concentrated early: an absorbing average over horizons that are strongly negative at first and near zero later can match a fresh-exposure estimate without either implying persistence. The direct evidence on duration is the event study, which places the effect in the wave of exposure and the wave after and finds the later horizons individually imprecise (Section 5). The benchmark TWFE is larger in magnitude (-0.019), but it weights cohorts differently and is the estimator the heterogeneity-robust methods are designed to correct, so we do not read that contrast as evidence on duration. The episodic dynamics speak to duration directly: the effect falls from -0.015 at the first horizon to -0.008 and then -0.005 , matching the decay in the absorbing event study.

D.6 TWFE benchmark, distributed lag, timing artifact, Goodman-Bacon

The two-way fixed-effects benchmark, reported in full in Appendix Table 33, tells the same story as CS(2021) and is included only for comparability with the older literature; no causal claim rests on it alone. Its distributed lag traces the CS(2021) dynamic path: the effect is already significant one window after exposure (-0.0056^{**} , $SE = 0.0026$), deepens to -0.0229^{***} ($SE = 0.0035$) two windows later, and returns to -0.0010 ($SE = 0.0018$) by the third, so mass fighting responds quickly, builds, and fades. This distributed lag captures fresh exposure in specific May-to-May windows, whereas the absorbing CS(2021) design estimates the effect of entering the post-earthquake state; the fading of the episodic lag should therefore not be read as contradicting the absence of reversal in the absorbing post-exposure effect. The response is nonlinear in shaking intensity, not proportional to it, and it is not a knife-edge at MMI VI: weaker shaking produces smaller but nonzero reductions. The threshold ladder does not establish a monotone dose-response, for reasons we set out in Section 5.3. Dating exposure by calendar year rather than by the May-to-May survey window displaces the distributed-lag response by one wave. The two-wave effect of -0.0218 falls to -0.0028 and loses significance, while a coefficient of -0.0063 appears at three waves where the aligned specification gives zero. Both timings use the identical sample, so the difference is entirely one of dating (Table 20). We document this as a cautionary point for unequally spaced panels, not a feature of the data.

Table 20: Exposure Timing and the Distributed Lag: Calendar Year versus Survey Window

Exposure lag	Composite		<i>Warga</i>	
	Calendar-year	Survey-aligned	Calendar-year	Survey-aligned
Exposure, one wave before	−0.0017 (0.0018)	−0.0032 (0.0027)	−0.0019 (0.0015)	−0.0023 (0.0022)
Exposure, two waves before	−0.0028 (0.0025)	−0.0218*** (0.0034)	−0.0019 (0.0021)	−0.0074*** (0.0028)
Exposure, three waves before	−0.0063*** (0.0020)	−0.0000 (0.0017)	−0.0016 (0.0015)	+0.0011 (0.0013)
Observations	285,263	285,263	285,263	285,263

Each column is a distributed-lag two-way fixed-effects regression with village and wave fixed effects and standard errors clustered at the village level. The calendar-year columns date exposure by the calendar year of the earthquake; the survey-aligned columns date it by the May-to-May window that matches the PODES enumeration month, so that each lag falls entirely before the twelve-month conflict reference window. Both timings are estimated on the identical sample, so the columns differ only in how exposure is dated. Misalignment does not reverse any sign; it attenuates the response and displaces it by one lag. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Finally, the absorbing benchmark is not a negative-weighting artifact, although the reason is not that the forbidden comparisons are rare. A Goodman–Bacon decomposition on the current panel assigns 73.1 percent of the weight to treated-versus-never-treated comparisons and 3.1 percent to timing comparisons, leaving 23.8 percent to the forbidden comparisons that use already-treated villages as controls. That weight is not negligible, and we report it rather than the smaller figure an earlier version of this paper carried. It is, however, uninformative about the sign: the forbidden comparisons return -0.0205 against -0.0173 for the clean ones, so they pull the estimate in the same direction, not against it. We therefore address the concern by construction instead of by weight, and report a specification that drops the always-treated cohort entirely (Table 9). That restriction removes the units without a pre-period rather than every forbidden comparison, which

is why we rest the argument on the heterogeneity-robust estimators instead; the estimate strengthens instead of weakening (Appendix J, Figure 9).

D.7 Why the Absorbing Treatment Definition

forever after its first strong shock requires justification. Three arguments support it. First, the conceptual framework (Section 2) predicts hysteresis: the shock disrupts physical and institutional inputs to collective violence that remain disrupted until reconstruction, so the treated state may outlast the shaking. Second, the definition does not require the effect to persist, and the estimates indicate that it does not. The dynamic ATTs are concentrated in the wave of exposure and the wave after, with the three subsequent lags jointly indistinguishable from zero (Section 5), and the episodic estimator decays along the same path (-0.015 , then -0.008 , then -0.005 ; Appendix Table 34). That profile governs how the pooled estimate should be read, not whether the absorbing definition is admissible. Because the absorbing indicator averages over later waves in which villages have largely recovered, it attenuates the effect toward zero, so the pooled estimate is smaller than the response at its peak, because it averages across horizons. Third, the absorbing and episodic definitions answer different questions and we estimate both: the absorbing design identifies the average post-entry effect of being in the post-earthquake state, the episodic design (Section 4.4) the effect of fresh exposure regardless of history, and the two agree in sign and are close in magnitude (-0.019 versus -0.015).

D.8 Is the Hazard-Zone Restriction Endogenous?

earthquakes we study. The KRB zones we use are dated to 2010 in the source files; on that vintage they postdate the 2004–2009 events (Aceh, Yogyakarta, Padang) and predate the later ones (Aceh Singkil 2011, Pidie Jaya 2016, Lombok and Palu 2018, Cianjur 2022). The KRB classification is a seismic-hazard assessment based on tectonic setting, namely fault and subduction-zone geometry, in the same spirit as Indonesia’s national probabilistic seismic-hazard analysis (Irsyam et al., 2020), rather than the realized damage of any particular event, so a village’s hazard band reflects its tectonic setting rather than whether a specific earthquake happened to strike it. Second, the sample restriction defines a fixed set of villages and enters the

design only through village fixed effects, which absorb any time-invariant hazard level; identification comes from the timing of exposure within this fixed set, not from the map’s boundaries. We nonetheless treat the earliest cohorts, whose events predate the map, with the same caution we apply throughout, and the result is robust to dropping them one cohort at a time (Section 5.6). Two further facts show the map is not selecting the result. First, the restriction barely touches the treated group: virtually all strongly shaken villages already fall inside the hazard zones (the KRB and national samples contain essentially the same treated village-waves), so the map changes the comparison pool rather than which villages are treated. Second, dropping the hazard-zone restriction entirely and re-estimating on the full national sample leaves the effect intact. On all 441,805 village-waves, of which 138,348 lie outside the hazard zone, the absorbing TWFE estimate with baseline-by-wave controls is -0.0187 for the composite ($p < 0.001$) and -0.0063 for *warga* ($p = 0.014$), against -0.0202 and -0.0071 in-sample. Dropping the restriction attenuates the estimate mildly as the control pool becomes more heterogeneous, and does not remove it.

D.9 The Robustness Battery

The estimate survives the exposure, measurement, clustering, and spatial choices, all detailed in the appendix. The MMI VI cutoff is not special: the effect stays negative and significant at broader $\text{MMI} \geq \text{V}$ (-0.0063^{***}) and at narrower $\text{MMI} \geq \text{VII}$ (-0.0174^{***}), though neither exceeds the $\text{MMI} \geq \text{VI}$ estimate and the ladder traces no dose-response (Section 5.3). Measurement error in the raster-to-village mapping is not driving it: four aggregation rules across all 1,423 event ShakeMaps, and dropping the largest 5 percent of villages by area, leave the estimate stable in sign and broad magnitude (TWFE -0.0073 to -0.0209 ; CS -0.006 to -0.011 , losing conventional significance only under the pixel-maximum rule). Nor is it an artifact of ShakeMap’s blended intensity construction: rebuilding exposure from an MMI derived independently from instrumental Peak Ground Acceleration, holding the window-aligned timing fixed, reproduces the effect on every specification (CS -0.0129 , standard error 0.0039, against the ShakeMap- MMI -0.0150 ; Appendix O). Re-clustering at the kabupaten level (320 clusters) keeps the composite significant

at 1 percent and *warga* at 10 percent, and the conservative 28-cluster province inference still leaves the composite significant at 5 percent ($t = -2.64$), consistent with the wild bootstrap above, while *warga* does not clear conventional levels. The single-item *warga* outcome gives -0.0066^{**} (Appendices J and B).

The absorbing treatment definition also appears not to be costly on the dimension it most plausibly ignores, namely *repeated* exposure. Because it treats the first strong shock as a permanent state change, it could in principle miss the effect of a second one. Two facts bear on this. First, repeated exposure is rare: among the identified treated villages, 96.6 percent are struck above the damage threshold in only one survey window and just 3.4 percent in two or more (and a single window can itself contain more than one event). Second, where it does occur we detect no additional effect. Augmenting the absorbing indicator with a separate re-exposure indicator that switches on at a village’s second strong shock leaves the re-exposure coefficient small and insignificant (composite $+0.004$, $p = 0.75$; *warga* $+0.001$, $p = 0.93$), and a cumulative shock-count specification is negative but statistically indistinguishable from the single-shock effect (composite -0.004 per shock, $p = 0.21$). Statistical insignificance is not equivalence, and with so few re-exposed villages the re-exposure effect is imprecisely estimated; we therefore read these estimates as consistent with, but not establishing, a first-shock state-change interpretation, under which the absorbing design loses little.

On spatial spillovers, which could bias the never-treated counterfactual that carries most of the identification weight: adding neighbours’ treated share within 10–50 km as a spatial lag leaves the direct effect stable with small, insignificant spillover coefficients; a 25 km “donut” dropping never-treated villages with treated neighbours keeps the sign negative but loses precision (-0.0057 , $p = 0.236$), with an average pre-trend that does not reject ($+0.0034$, $p = 0.559$); province $\times$ wave fixed effects, the preferred specification, move the coefficient from -0.0198^{***} to -0.0148^{***} and leave it significant; and a forward-lag placebo is a clean null (-0.0045 , $t = -0.84$, $p = 0.40$), so the design shows no sign of anticipatory adjustment (Appendix J).

Selection on unobservables is an unlikely explanation: under the Oster (2019) coefficient-stability test, adding the controls moves the estimate *away* from zero,

so unobserved selection would have to run opposite to the observed selection on covariates to overturn the result. Nor is the effect an artifact of spatial correlation in the errors: Conley spatial-HAC standard errors (Conley, 1999) keep it significant at 1 percent (Appendix J).

E Pre-Treatment Balance

How comparable were eventually-treated and never-treated villages before any treatment occurred? Table 21 compares the two groups on ten village characteristics measured in levels at the 2005 wave, which pre-dates first exposure for every identified cohort. The purpose is not to demonstrate perfect random assignment; it is to show that, within the KRB Medium/High hazard-zone sample, treated and control villages were closely comparable before the shocks arrived.

Table 21: Pre-Treatment Balance at the 2005 Wave: Eventually-Treated vs. Never-Treated Villages, KRB Medium/High

	Eventually treated	Never treated	Norm. diff.
Communal conflict (binary)	0.038	0.026	0.072
Population	4099.779	3653.100	0.102
Number of households	1022.966	920.252	0.095
Paved road access	0.677	0.604	0.153
Household electricity	0.948	0.960	-0.054
Distance to district capital (km)	25.347	29.339	-0.169
Number of primary schools	3.225	2.670	0.197
Police post present	0.168	0.114	0.157
Number of health centers	0.722	0.455	0.378
Internet available	0.038	0.035	0.013

Village characteristics measured in levels at the 2005 PODES wave, which pre-dates first exposure for every identified treatment cohort. Eventually treated: villages with window-aligned first $\text{MMI} \geq \text{VI}$ exposure in 2008–2024 ($n = 3207$); never treated: villages in the same KRB Medium/High hazard zones with no $\text{MMI} \geq \text{VI}$ exposure through 2024 ($n = 30288$). Norm. diff. is the normalized difference of Imbens and Wooldridge (2009), $(m_T - m_C) / \sqrt{(s_T^2 + s_C^2)/2}$, a scale-free measure; the conventional threshold for concern is 0.25. This is a descriptive balance check, not a hypothesis test, so no standard errors or significance stars apply. Numeric formatting here (3 decimal places, no thousands separator) differs cosmetically from the hand-typeset version of this table.

Nine of the ten normalized differences (Imbens & Wooldridge, 2009) fall below the conventional 0.25 threshold. The exception is the number of health centres, at 0.378, which we flag rather than average away; the next largest are the primary-school count (0.197) and distance to the district capital (−0.169). Eventually

treated villages are larger than never-treated ones (4,100 against 3,653 residents, normalized difference 0.102) and better provisioned on several margins, which is what one would expect if seismically active districts differ systematically from the rest of the country. Baseline conflict differs by 1.2 percentage points (0.038 against 0.026, normalized difference 0.072).

Because the health-centre difference sits above the conventional threshold, we ask directly whether it is load-bearing. Adding the baseline health-centre count interacted with wave moves the estimate by less than one percent on every outcome, and trimming villages in the top decile of baseline health-centre provision leaves every estimate significant, with a largest movement of 10 percent on inter-village conflict. The imbalance is visible but it is not doing

F Additional Data Figures

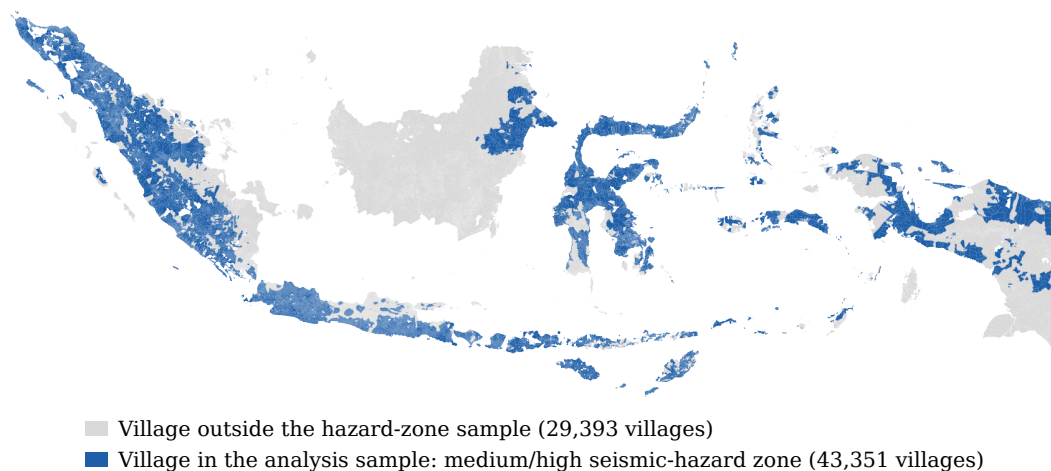


Figure 5: Analysis Sample: Medium/High Seismic-Hazard Zones

Note: Each village is drawn as its constant 2005 boundary polygon. Blue: the analysis sample, villages in the medium/high seismic-hazard zone (43,351 villages); light grey: the rest of Indonesia, outside the sample (30,001 villages). Hazard zones from the official seismic-hazard map of the Center for Volcanology and Geological Hazard Mitigation (PVMBG); village polygons from Statistics Indonesia (BPS).

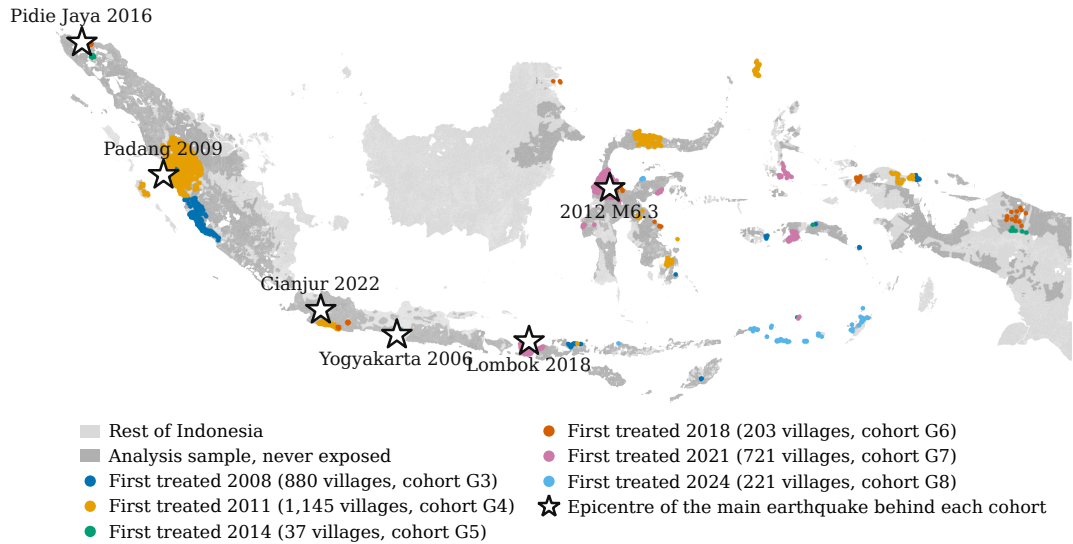


Figure 6: Ever-Treated Villages by First-Exposure Cohort, with the Main Earthquake Behind Each Cohort

Note: Colored dots are villages ever exposed to strong shaking (Modified Mercalli Intensity VI or above), colored by the survey wave at which they are first treated. Stars mark the epicentre of the earthquake that first exposed the largest share of each cohort's villages. The grey background shows the analysis sample (darker grey) within the rest of Indonesia (lighter grey). The 2021 cohort combines the Lombok (July 2018) and Palu (September 2018) earthquakes, both of which struck after the May 2018 enumeration. Colorblind-safe palette (Okabe-Ito). Shaking data from USGS ShakeMap.

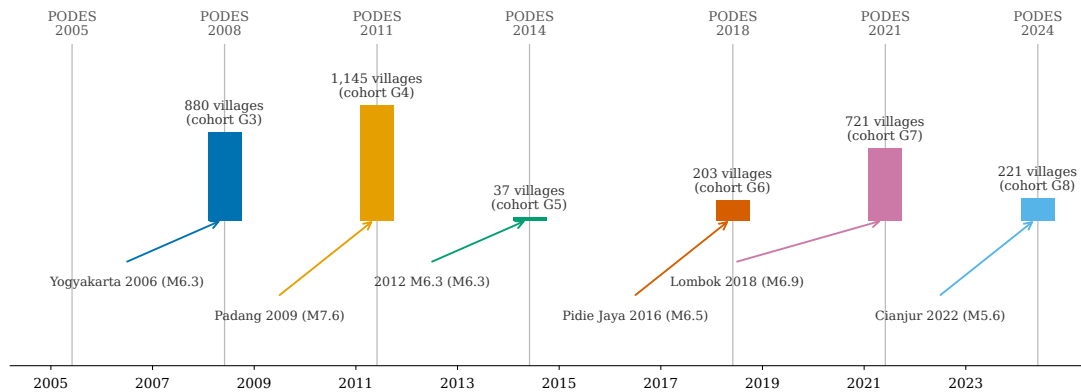


Figure 7: Treatment Cohorts on the Survey Calendar

Note: Vertical lines mark the May enumeration of each Village Potential Survey (PODES) wave. Bars are the identified treatment cohorts with their village counts (bar heights proportional). Arrows point from the main earthquake generating each cohort's first exposure (magnitudes from USGS) to the survey wave at which those villages enter treatment. A village joins a cohort at the first enumeration after its first strong-shaking event, so the 2018 Lombok and Palu earthquakes, which struck after the May 2018 enumeration, form the 2021 cohort.

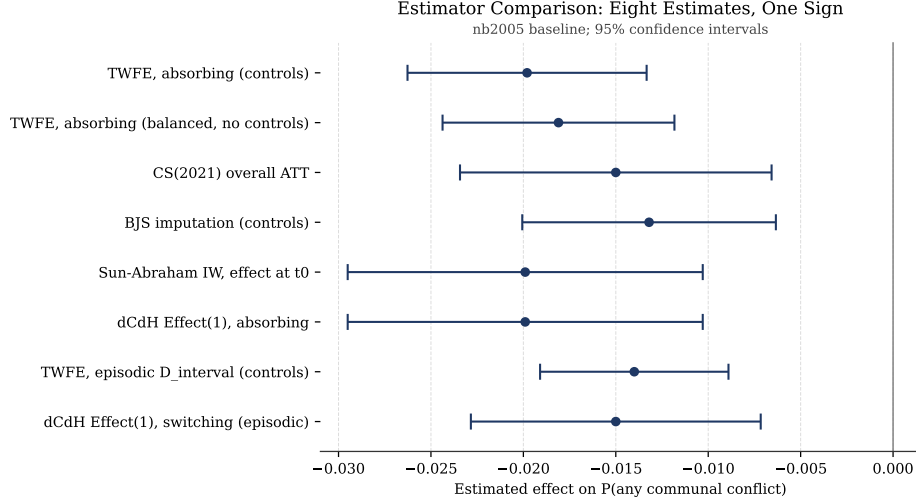


Figure 8: Estimator Comparison: Eight Estimates, One Sign

Note: Point estimates and 95% confidence intervals for the eight specifications in Table 4. Outcome: any communal conflict. Sample: the 2005-boundary panel, window-aligned exposure timing. The two TWFE rows are estimated with baseline covariates ($N = 227,045$) and on the balanced no-controls sample ($N = 234,465$); neither is the no-controls full-panel estimate of -0.0186 used as the reference point in the robustness discussion, which retains the always-treated cohort ($N = 303,457$). The takeaway is not the exact equality of coefficients across estimators but the stability of the sign and magnitude: all eight estimates are negative and significant (the pooled CS ATT at 5 percent, the rest at 1 percent) and lie in a narrow band of roughly -1.3 to -2.0 percentage points, so the result is not an artifact of a single estimator or treatment definition. These are not eight independent tests, since they draw on the same data and identifying variation; the point is robustness to estimator choice, not independent replication.

F.1 Event-Size Threshold and the Two Event-Level Estimators

The stacked event design of Section 5.6.1 keeps earthquakes with at least twenty in-sample first-treated villages, which yields nineteen events. The catalogue contains forty-seven earthquakes with at least two such villages. Because each stack is built on its own controls, an earthquake’s own difference-in-differences estimate does not depend on which other earthquakes are included, so the table below is computed from a single set of per-event estimates re-aggregated at each cutoff.

G Village Boundary Harmonization

The problem. Indonesia’s village count grew from roughly 68,000 to 84,000 in recent decades through administrative subdivision (*pemekaran*). A panel that ignores this either loses villages that change codes or treats offspring villages as

Table 22: Event-Level Estimands by Minimum Event Size

The treated-village-weighted average is unmoved by the cutoff; the equal-event mean is not.

Note: Each row re-aggregates the same per-earthquake difference-in-differences estimates over the earthquakes that meet the stated minimum number of in-sample first-treated villages. Outcome is the composite communal-conflict indicator. Each per-event estimate uses village and wave fixed effects on an event-relative window of $[-2, +2]$ against never-treated and not-yet-treated controls. The p -value for the treated-village-weighted average comes from an earthquake block bootstrap that resamples whole events (9,999 draws, seed 42); the p -value for the equal-event mean is a two-sided t -test on the event-specific coefficients. The twenty-village row reproduces the estimates reported in Table 7.

Minimum treated villages	Sample		Treated-village-weighted		Equal-event mean	
	Events	Villages	Estimate	p	Estimate	p
≥ 20 (paper)	19	3009	-0.0242	0.008	-0.0148	0.338
≥ 10	28	3130	-0.0251	0.003	-0.0253	0.070
≥ 5	33	3159	-0.0254	0.004	-0.0307	0.023
≥ 2	47	3198	-0.0243	0.006	-0.0045	0.777

new units, both of which would correlate mechanically with population growth and state expansion.

Table 23: From the 2005 Village Universe to the Analysis Sample

	Villages	%
Villages in the 2005 Master File Desa (universe)	69,957	
enumerated in PODES 2005	68,637	98.1
in KRB Medium/High hazard zones (risk set)	45,450	65.0
observed in all seven waves	42,333	93.1
less those with an incomplete conflict series	-231	
with a complete conflict series	42,102	92.6
Plus post-2005 units not linked to a 2005 parent	1,249	
Analysis sample	43,351	

The universe is the 2005 Master File Desa, not the set of parent identifiers in the constructed panel: the latter contains 1,249 units in the analysis sample (and 13,138 overall) that correspond to villages created after 2005 whose chain linkage did not resolve to a 2005 parent; these enter the panel under their own identifier and are retained, since the design needs only a consistent unit across waves, not a 2005 ancestor. The 231 villages observed in all seven waves but excluded have at least one wave with a missing conflict outcome, and the estimation sample requires a complete series. Percentages are shares of the universe except the last two, which are shares of the risk set.

Procedure. We harmonize all waves to constant 2005 parent boundaries with a chain-linkage crosswalk built from BPS Master File Desa (MFD) records: each wave’s village code is linked to its immediate predecessor code, and the chain is followed back to the 2005 parent. The chain is code-based throughout for the core waves, with a name bridge used only for the 2024 wave. The dominant method is a direct MFD chain (single- or multi-step, resolving to the 2005 code, 82 percent of wave-codes), complemented by the 2005 baseline self-match, an inferred unique-child map for residual parent codes, and a documented fallback for codes absent from the MFD vintage. Outcomes are aggregated to the parent by the maximum (binary outcomes) or sum (counts), so a parent reports conflict if any constituent village does; placebo and demographic variables use means (Appendix I).

Match rates. Wave-level match rates into the 2005 parent frame, verified from the raw wave files, are 100 percent (2005, the base year), 99.9 percent (2008), 98.4 percent (2011), 98.7 percent (2014), 95.9 percent (2018), 98.7 percent (2021), and 98.7 percent (2024); the balanced panel retains 93.2 percent of 2005-base villages. Unmatched codes are concentrated in newly created administrative areas whose MFD chains are broken and in territory reorganisations (notably new provinces in Papua).

Unresolved parents and the coverage ceiling. Broken chains have a second consequence worth stating explicitly. When the chain cannot resolve a post-2005 village to a 2005 parent, that village enters the constructed panel as its own parent identifier. The panel therefore contains 13,138 parent units that do not correspond to any village in the 2005 Master File Desa, and these units necessarily lack a 2005 observation and are dropped by the balanced-panel requirement. This is why Table 23 takes the 2005 Master File Desa as its denominator rather than the set of parent identifiers in the panel: the latter exceeds the 2005 village universe by roughly ten thousand units and would understate coverage. Of the villages missing only the 2005 wave, 637 are on Nias, which PODES 2005 did not enumerate after the 2004 tsunami and which Cassidy (2025) likewise excludes; a further 1,180 appear in the 2005 Master File Desa but were not enumerated, so their outcomes are genuinely absent.

We also asked whether the remaining unresolved parents can be folded back by name. A layered procedure matching on exact village name within subdistrict and then within district, followed by name-root matching, folds 2,220 of the 13,138 units and would raise coverage of the risk set from 93.1 to 94.7 percent. We do not adopt it: 6.6 percent of the links it produces are ambiguous, because several 2005 villages share a name within the same district, and a mislinked village injects measurement error into both treatment and outcome, which is worse than dropping it. The reported coverage is therefore a deliberate floor rather than a ceiling imposed by the data.

Boundary geometry vintage. The match rates above concern village identifiers, which are distinct from the underlying polygon geometry used for the spatial join to ShakeMap MMI and the KRB hazard zones. No BPS shapefile digitized contemporaneously with 2005 exists, so we build the primary boundary layer by dissolving the BPS 2014 administrative shapefile onto 2005 parent identifiers: a direct dissolve covers 90.5 percent of villages, and a layered fallback (borrowing, in order, from alternate-vintage Master File Desa codes, then from an earlier 2011-shapefile-based boundary layer, then from the 2003-parent boundary layer, each fallback validated against the province recorded in PODES GPS coordinates) raises cumulative coverage to 95.9 percent (65,794 of 68,637 villages). The alternate-vintage layer recovers 3,123 villages and the 2003-parent layer 528; the 2011-shapefile layer recovers none on this vintage, leaving 2,843 villages, 4.1 percent, without a polygon. The 2003-parent crosswalk check reported below ([Cassidy \(2025\)](#)-style robustness) uses an analogous boundary layer dissolved from the BPS 2011 shapefile. Because the treatment variable is a spatial join of raster MMI values to these polygons, using the most accurate digitized shapefile available, rather than a shapefile contemporaneous with the label year, is the more accurate choice for measuring exposure.

Why the analysis begins in 2005. The analysis window is 2005–2024, and all village units are harmonized to constant 2005-parent boundaries so that the panel tracks fixed geographic units through administrative subdivision. The analysis begins in 2005, the first wave in which the questionnaire asks specifically about

perkelahian massal (mass communal brawl), the outcome we study, so that this outcome is defined consistently throughout the analysis window.

Crosswalk robustness. Two re-estimations verify that the harmonization rules do not drive the result. First, a never-split sample drops every parent that ever maps from more than one raw village, eliminating any influence of the collapse rule: the TWFE estimate is -0.0160 ($t = -4.61$, $N = 208,306$) against -0.0181 ($t = -5.67$, $N = 234,465$) on the corresponding full sample, and the CS(2021) ATT is -0.0138 ($p = 0.002$). The subsample estimate is marginally smaller in magnitude and less precisely estimated, as one would expect after discarding an eighth of the observations, but it is not attenuated toward zero in any way that would indicate the collapse rule is generating the result. Second, dropping the parents whose chain ever used the fallback method leaves the estimate essentially unchanged (-0.0182 , $t = -5.69$). The harmonization machinery is not what produces the finding.

H ACLED Mob-Violence Events: Full Coding

What the ACLED Comparison Can and Cannot Show

ACLED offers a useful but imperfect external check on a village-head-reported outcome. It covers Indonesia only from 2015, so it overlaps PODES in the 2018–2024 waves and cannot validate the main 2009 Padang cohort. In the overlap, PODES communal conflict falls sharply (-3.97^{***} percentage points) while ACLED events rise slightly (Appendix Table 24); but reading the 24 treated-village mob-violence events individually shows that two-thirds (16 of 24) are not inter-group communal fighting but vigilante, moral-policing, or crime-accusation actions against individual targets, several of them mob responses to alleged theft, the scarcity-channel sequela of the rising-theft result (Miguel, 2005). The divergence is what the two measurement technologies predict and weakens the reporting-artifact reading: ACLED’s media-driven counts rise where post-disaster coverage concentrates, whereas PODES is a census independent of media attention. ACLED therefore neither replicates nor overturns the village-level result; it captures a different, media-visible tail of disorder. The full event coding appears in Appendix H.

Appendix P reports why the SNPK/NVMS incident database (2005–2014, [Barron, Jaffrey, and Varshney 2016](#)), the one violence series whose window reaches the 2009 Padang cohort, cannot serve as an external check here: its reporting expands sharply over the sample period, and the estimates it yields reverse sign when the same specification is estimated in proportional rather than level form.

Table 25 lists all 24 ACLED mob-violence events (precision-1 geolocation) occurring in treated village-waves, extracted and coded into four categories from the ACLED event descriptions: **V** = vigilante, moral-policing, or crime-accusation action against individual targets; **E** = election-related crowd violence; **IG** = genuine inter-group clash; **O** = other (neither). The full ACLED notes for every event ship with the replication package.

Table 24: External Comparison: ACLED vs. PODES Conflict Outcome, PODES Waves 2018–2024

	(1) PODES communal	(2) ACLED any	(3) ACLED mob violence	(4) ACLED violent demonstration	(5) ACLED VAC	(6) ACLED battles
D_{treat}	-0.0397*** (0.0097)	0.0047** (0.0022)	0.0027* (0.0016)	0.0024 (0.0015)	0.0017 (0.0013)	-0.0002*** (0.0001)
Baseline mean	0.024	0.002	0.001	—	—	—
Observations	100,485 (identical subsample)					
Village + wave FE	Yes					

Notes: TWFE on the three PODES waves overlapping ACLED coverage (2018, 2021, 2024); within this subsample identification comes from cohorts first exposed in 2021 and 2024. ACLED outcomes are built by a scripted event-level pipeline: events filtered to Riots, Violence against civilians, and Battles with exact geolocation (highest geo-precision only), point-in-polygon matched to village-parent boundaries, and assigned to the exact 12-month May-to-May PODES conflict reference window. *Mob violence* is ACLED's closest analogue to communal violence but also includes vigilante mob justice, moral-policing raids, and looting mobs. SE clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 25: ACLED Mob-Violence Events in Treated Village-Waves, Coded

#	Date	Province (district)	Cat.	Event summary (from ACLED notes)
1	2018-01	Yogyakarta (Bantul)	V	Crowd forces cancellation of church charity event alleged to proselytize
2	2020-05	SW Papua (Sorong)	IG	Two groups of drivers clash at terminal after a mugging
3	2020-05	West Java (Cianjur)	IG	Inter-village clash with machetes; cause unclear
4	2020-05	C. Sulawesi (Buol)	V	Rioters punch village head enforcing COVID prayer restrictions
5	2020-09	NTB (Bima)	V	Crowd burns house of man accused of sorcery
6	2020-10	Yogyakarta (Bantul)	O	Intra-organisational brawl among youth-organisation members
7	2020-12	Maluku (Ambon)	IG	Residents stone Papuan student dormitory before planned rally
8	2020-12	Papua (Mamberamo)	E	Intoxicated crowd seizes ballot box, attacks police
9	2021-04	NTB (C. Lombok)	IG	Two groups clash over family land dispute
10	2023-05	NTB (W. Lombok)	V	Crowd damages house of convicted rapist
11	2023-06	W. Papua (Manokwari)	V	Crowd beats journalist reporting market fire
12	2023-06	Bengkulu (Mukomuko)	V	Palm-oil company workers and farmers scuffle over theft accusations
13	2023-07	W. Papua (Manokwari)	IG	Two sub-district groups clash with weapons after assault
14	2023-07	W. Papua (Manokwari)	V	Intoxicated youths assault civilians (trigger of #13)
15	2023-07	Riau (Rokan Hulu)	V	Women torch cafe alleged to host immoral activities
16	2023-10	NTB (Mataram)	IG	Monjok-Karang Taliwang neighbourhood clash (long-standing feud)
17	2023-10	NTB (Mataram)	IG	Same feud, following day
18	2023-12	NTB (C. Lombok)	V	Villagers beat man accused of stealing billboard lights
19	2023-12	NTB (C. Lombok)	IG	Inter-village clash escalating from #18 (theft accusation)
20	2023-12	Yogyakarta (Sleman)	E	Clash with party volunteers; one death
21	2024-01	SW Papua (Sorong)	O	Prison break; guards attacked
22	2024-01	Maluku (Ambon)	V	State-company staff beat journalist reporting accident
23	2024-02	NTB (C. Lombok)	E	Candidate supporters damage rival supporter's house
24	2024-03	Bengkulu (Mukomuko)	V	Residents raid and close inn suspected of prostitution
Totals: V = 11, E = 3, IG = 8 (4 in Lombok-Mataram), O = 2.				

Source: ACLED yearly files 2016–2024, precision-1 Riots/Mob-violence events point-in-polygon matched to native 2005 village polygons and assigned to exact May-to-May PODES windows; treated village-waves under the windowed absorbing first-exposure design. Category codes: V vigilante/moral-policing/crime-accusation (individual target); E election-related; IG inter-group clash; O other.

I Reporting-Artifact Placebo Outcomes

Table 26: Placebo Outcomes from the Same PODES Form: Village-Head Age and Sex 2005-boundary panel, window-aligned first exposure; waves 2005–2024 excluding 2011

	Placebo outcomes		Same-sample
	Head age (years)	Female head	Communal conflict
<i>Panel A: wave fixed effects</i>			
CS(2021) ATT	−0.3545*	−0.0099**	
	(0.1948)	(0.0049)	
pre-trend χ^2 p	0.004	< 0.001	
TWFE	−0.6197***	−0.0110***	−0.0247***
	(0.1670)	(0.0040)	(0.0038)
<i>Panel B: province $\times$ wave fixed effects</i>			
TWFE	+0.4247**	−0.0016	−0.0148***
	(0.2092)	(0.0047)	(0.0040)
pre-trend joint p	0.479	0.617	0.462

Placebos are village-head age and a female-head indicator from the raw PODES apparatus module (six waves; 2011 excluded, coding unverifiable), aggregated to constant 2005-parent boundaries. Under wave fixed effects both placebos move and both reject their joint pre-trend test. Under province-by-wave effects both pass the pre-trend test, the female-head placebo becomes a precise null, and the head-age coefficient reverses sign to +0.42 years, about five months, which is not the direction a reporting-artifact story predicts. The panel-B column for communal conflict is the preferred specification of Section 5.1. Standard errors are clustered at the village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The placebo behaves differently under the two fixed-effect schemes, and the difference is itself informative. Under wave fixed effects it fails: both village-head age and the female-head indicator move, the latter significantly (-0.0099 , $p = 0.044$), and both reject their joint pre-trend test. Under the province-by-wave effects that Section 4.1 selects, both pass their pre-trend test ($p = 0.48$ and $p = 0.62$), the female-head placebo becomes a precise null (-0.0016 , $p = 0.73$), and the head-age coefficient reverses to +0.42 years, about five months, which is not the direction a reporting-artifact story would predict.

This is the same pathology the conflict outcome shows, and it has the same

source. Both apparatus variables trend strongly across the panel, head age from 45.0 to 48.3 years and the female share from 0.027 to 0.072, and those trends differ by region. A staggered design that absorbs only wave effects attributes part of such a trend to treatment whenever cohorts enter at different points on it; absorbing province-by-wave effects removes it. That the placebos fail exactly where the conflict pre-trend fails, and pass exactly where it passes, is independent support for the specification rule rather than a separate problem.

J Additional Robustness Detail

This appendix reports the full results for the robustness exercises summarized in Section [5.6](#).

J.1 Catalogue Start Year

The absorbing cohort is defined by ever-exposure, so it depends on how far back the earthquake catalogue reaches. Table [27](#) rebuilds the first-exposure cohort from three start years and re-estimates both headline specifications. The result is robust in sign and significance throughout, but the magnitude attenuates monotonically as the catalogue shortens: shortening it to 2000 moves 1,373 villages out of the always-treated cohort and into the comparison group, and shrinks the estimate by 15 percent under TWFE and 18 percent under Callaway–Sant’Anna. We retain the 1990 start, which is the only one under which cohort assignment is not contaminated by unobserved pre-panel exposure, and report the shorter windows rather than resolve the question by assertion.

Table 27: Sensitivity to the Earthquake Catalogue Start Year

Catalogue start year	Absorbing TWFE (SE)	Callaway–Sant’Anna (SE)	N (CS)	Always-treated villages dropped
1990 (baseline)	−0.0202*** (0.0033)	−0.0150*** (0.0043)	227,045	9,856
1995	−0.0193*** (0.0033)	−0.0143*** (0.0042)	231,049	9,263
2000	−0.0172*** (0.0032)	−0.0122*** (0.0041)	236,005	8,483

Each row rebuilds the first-exposure cohort using only earthquakes from the stated year onward, then re-estimates the absorbing TWFE (baseline controls interacted with wave, village and wave fixed effects) and the primary Callaway–Sant’Anna specification. Shortening the catalogue moves villages shaken in the 1990s out of the always-treated cohort and into the comparison group, which raises the estimation sample and attenuates the estimate by 15 percent (TWFE) and 18 percent (CS). All six estimates are significant at the 1 percent level. Standard errors are clustered at the village level. The 1990 start is retained because it is the only one under which cohort assignment is not contaminated by unobserved pre-panel exposure; the attenuation under shorter catalogues is reported rather than resolved.

J.2 Sensitivity Under the Preferred Specification

The sensitivity analysis reported above is anchored to the wave-fixed-effects panel, which is the specification the parallel-trends rule of Section 4.1 does not select. That matters here more than it usually would, because the relative-magnitudes yardstick is computed *from* the pre-treatment leads: a specification whose pre-period is noisier mechanically permits larger hypothetical violations and so breaks down sooner. Re-running the analysis on the preferred specification is therefore not a robustness check but a correction of the anchor.

Under province-by-wave effects the largest deviation between consecutive pre-treatment leads falls to 0.0117, which is 0.69 times the effect being defended, against 0.037 and 2.2 times under wave effects. The identified set accordingly excludes zero through $\bar{M} = 0.25$ and admits zero at $\bar{M} = 0.50$ (Table 28), where the earlier analysis broke down between $\bar{M} = 0$ and $\bar{M} = 0.25$. The result is still not robust to large violations, and we do not claim otherwise; what changes is that

the estimate now survives a violation a quarter as large as the worst movement observed in the pre-period, rather than failing at any positive violation at all.

Table 28: HonestDiD Under the Preferred Specification

$\bar{M}$	Lower	Upper	Excludes zero
Original	-0.024	-0.004	Yes
0.00	-0.023	-0.004	Yes
0.25	-0.026	-0.001	Yes
0.50	-0.031	0.003	No
0.75	-0.037	0.009	No
1.00	-0.043	0.016	No

Relative-magnitudes identified sets (Rambachan & Roth, 2023) for the preferred specification of Section 5.1. $\bar{M}$ is the permitted post-treatment violation as a multiple of the largest deviation between consecutive pre-treatment leads. In this specification that yardstick is 0.0117, which is 0.69 times the effect being defended; under the wave-fixed-effects specification it was 0.037, or 2.2 times the effect, which is why the breakdown point moves. The identified set excludes zero through $\bar{M} = 0.25$ and admits zero at $\bar{M} = 0.50$.

J.3 Covariate Overlap

The doubly-robust estimator reweights control villages by their estimated propensity to be treated, so it requires that the treated propensity distribution be contained in the control one. Table 29 reports the standard diagnostics cohort by cohort. Support is not a binding problem: in no cohort, and on none of the four boundary anchors examined, does more than half a percent of controls sit above the largest treated propensity, against the five percent threshold at which support violations are normally flagged.

Weight concentration is the milder concern. In three of the twenty-two cohort-anchor cells the most heavily weighted one percent of controls carries more than a fifth of the total control weight, the 2014 cohort most visibly, where only 37 villages are treated. Those are the cells in which the estimator leans hardest on a small number of comparison villages. We do not drop them, since the leave-one-cohort-out check of Section 5.6 shows that no single cohort carries the result, but we report the concentration rather than let the doubly-robust machinery obscure

it.

Table 29: Covariate Overlap by Cohort

Cohort wave	Treated villages	Controls	Max treated propensity	Controls above treated support	Max weight	Weight in top 1%
2008	880	32,615	0.084	0.02%	0.14	2.6%
2011	1,145	31,470	0.077	0.03%	0.09	2.2%
2014	37	31,433	0.175	0.02%	0.70	24.9%
2018	203	31,230	0.028	0.12%	0.04	4.3%
2021	721	30,509	0.156	0.25%	1.77	7.9%
2024	221	30,288	0.030	0.32%	0.18	5.5%

Overlap diagnostics for the doubly-robust Callaway–Sant’Anna estimator, cohort by cohort. The propensity score is fitted on the baseline covariates within each comparison. “Controls above treated support” is the share of control villages whose propensity exceeds the largest value observed among the treated, the standard support violation; it never exceeds 5 percent in any cohort, on any of the four boundary anchors examined (22 cohort-anchor cells in total). “Weight in top 1%” is the share of total control weight carried by the most heavily weighted one percent of controls; it exceeds 20 percent in three of those cells, which is where the doubly-robust estimator leans hardest on a small number of comparison villages. No cohort is dropped on overlap grounds.

J.4 Event-Level Inference: Robustness Detail

This appendix expands the event-level inference summarized in Section 5.6.1. Because control villages are reused across stacks, we cluster two-way on earthquake event and original village; this leaves the standard error essentially unchanged ($SE = 0.0101$, $p = 0.030$), which tells us that the binding dimension for inference is the number of distinct earthquakes rather than the reuse of controls.² A more direct check settles the reuse question: rebuilding the design so that each control village belongs to exactly one earthquake, by drawing controls only from never-treated villages and randomly assigning each to a single shock, removes the reuse entirely and leaves the estimates essentially unchanged (treated-village-weighted -0.0222 ,

²With only 19 clusters on the event dimension the two-way variance matrix is itself imprecisely estimated and required the [Cameron, Gelbach, and Miller \(2011\)](#) non-positive-definite adjustment, so we read it only as corroborating the event-clustered inference rather than as a sharper test.

earthquake-level bootstrap $p = 0.02$). Because the stacked design already draws controls from the whole hazard zone rather than from the immediate neighbourhood of each shock, this random partition splits the same national pool rather than substituting distant controls for local ones. Nor does the result depend on a single lucky allocation: repeating the random partition 300 times, the pooled estimate is tightly distributed around -0.017 (standard deviation 0.001, 90 percent of draws between -0.019 and -0.015 , every draw negative). Reuse is therefore not what makes the estimate marginal; the small number of distinct earthquakes is. Those 19 are genuinely independent shocks rather than one rupture counted many times: under a standard aftershock window (epicentres within 100 km and origin times within 180 days) they fall in separate seismic episodes, so treating them as independent units for inference is appropriate.

On the large-versus-small split of the 19 event-specific estimates: among the six events with the largest first-treated cohorts (at least 150 villages each, and up to 824) the mean effect is -0.038 , five of six are negative, four of them at the 5 percent level and the fifth at 10 percent ($t = -1.83$), with the strongest suppression reaching -0.098 . We are equally candid that the sixth is positive, though far from significant ($+0.004$), and that the concentration in large-footprint events means six earthquakes carry most of what the pooled estimate reports. Among the thirteen smaller events the mean is close to zero (-0.004), seven of thirteen negative, indistinguishable from noise.

J.5 Spatial Displacement: Ring-Design Detail

This appendix expands the displacement test of Section 5.5. Two non-displacement readings fit the negative ring coefficients. These ring villages typically felt the same earthquakes at sub-VI intensities, and we cannot rule out that felt shaking itself moves conflict, since the direct low-intensity contrast could not be constructed on this catalogue (Section 5.6.2); but the ring effect is not merely each village's own sub-VI dose, since conditioning on own continuous shaking leaves it essentially unchanged (0–25 km -0.0076 ; 50–100 km -0.0082 , both $p < 0.01$). And part reflects regional co-movement: with province $\times$ wave fixed effects the ring coefficients attenuate, the nearest staying negative and the middle becoming a precise zero

(+0.002, $p = 0.63$), none significantly positive. The ring coefficients could in principle be aggregated into a village-weighted total spatial effect, but we do not report or lean on such a magnitude: the ring contributions use village and wave, not province $\times$ wave, fixed effects, and partly reflect the felt-shaking channel, so the total is best read as the spatial extent of suppression, not a precise displacement estimate. This complements the spatial-lag check of Section 5.6.4: neither produces the positive sign displacement would require. Amarasinghe et al.’s own estimates by disaster type cut both ways and we report both directions. Earthquakes, the hazard closest to ours, show the largest immediate local decline among the six hazards they separate, which is consistent with the place-boundedness reading, though that coefficient is imprecise (-0.023 , standard error 0.021). The outward spillover in the same column is positive and significant (0.029, standard error 0.014), and it is the sharper form of the displacement concern: for the hazard most like ours, their design does detect the relocation that our rings do not. We do not resolve that disagreement by appeal to precision. The difference we would point to is the conflict technology rather than the hazard. Their combatants are mobile and can reallocate across districts when a shock lowers the local prize; communal conflict is between residents of the same village and does not travel with them.

J.6 Same-Earthquake Local Design: Matched Detail

This appendix expands the same-earthquake local design of Section 5.6.1. It holds the *earthquake* and the *survey wave* fixed, but not region, soil, topography, or market access, since a ShakeMap footprint can span very different terrain; it is a same-shock comparison, not a matched-neighbour comparison. Second, the control villages also felt the earthquake, so the design estimates the incremental effect of crossing the damage threshold relative to felt shaking, not the effect of an earthquake relative to none. The composite estimate is -0.0153 , significant at 1 percent under kabupaten clustering, with a within-pre-period placebo of $p = 0.06$ that does not clear conventional levels but is not a clean pass either. To probe mean reversion, exact-matching each treated village, within its earthquake, to felt-controls that share its last-pre-wave conflict value, its full pre-period conflict path, and coarsened geography (population tercile, paved road, police post) retains 99

percent of treated cells, so the restriction has little bite, and the matched composite is similar, if slightly larger, than the unmatched one (-0.016 , earthquake-clustered $p < 0.05$); a narrow intensity band, treated at MMI [6, 7) against controls at [5, 6), gives -0.009 ($p = 0.02$). Because *warga* is rarer still, the local design does not resolve the definition-consistent outcome any more cleanly than the pooled estimator does.

Table 30: Local Comparison: Villages Shaken Above vs Below the Damage Threshold by the Same Earthquake

Takeaway: a same-earthquake felt-intensity comparison corroborates the sign of the composite effect; we do not treat it as independent identification.

Outcome	Village cluster	Earthquake-event cluster (24)	Kabupaten cluster
Composite (any communal)	-0.0153^{***} (0.0039)	-0.0153 (0.0093)	-0.0153^{***} (0.0058)
<i>Warga</i> (definition-consistent)	-0.0050^* (0.0030)	-0.0050 (0.0065)	-0.0050 (0.0035)
<i>Pre-trend diagnostics (village-clustered)</i>			
Pre-lead (t_{-2} , base t_{-1})	composite $+0.015$ ($p = 0.03$); <i>warga</i> $+0.007$ ($p = 0.20$)		
Last-pre-wave baseline only	composite -0.0100^{**} ($p = 0.03$); <i>warga</i> -0.0032 ($p = 0.38$)		
Pre-window placebo (t_{-1} vs t_{-2})	composite $p = 0.06$; <i>warga</i> $p = 0.20$		

Notes: Each of 24 earthquakes forms a stack in which treated villages (mean MMI $\geq$ VI) are compared to villages that felt the *same* earthquake at sub-damaging intensity (mean MMI in [4, 6)) and are not themselves MMI $\geq$ VI-treated as of that event (a small share cross the threshold in a later wave of the window; excluding them moves the estimate to -0.0151 , immaterially), in event-relative time (a $[-2, +2]$ -wave window), with event $\times$ village and event $\times$ wave fixed effects. This holds the earthquake and the survey wave fixed, but not region, soil, or topography, since treated and felt-control villages differ; it identifies the incremental effect of shaking crossing the damage threshold within a common earthquake. $N = 149,130$ stacked village-waves (5,275 treated and 52,564 felt-control pre-treatment cells). The same coefficient is shown under three clustering choices. The composite effect is significant at the 1 percent level under village clustering and at the 1 percent level under kabupaten clustering, but is not significant under the most demanding 24-cluster earthquake-level inference, and it survives the pre-trend diagnostics only in part: the composite pre-lead is $+0.015$ ($p = 0.03$) and the pre-window placebo $p = 0.06$, neither of which is a clean pass, alongside a last-pre-wave-only baseline that leaves it at -0.0100 ($p = 0.03$). The definition-consistent *warga* estimate is negative and significant under village clustering, but we do *not* read it as overturning its pooled CS(2021) null: it is imprecise under event-level inference and falls to -0.0032 ($p = 0.38$) once only the last pre-wave anchors the baseline, so its village-clustered significance is sensitive to the pre-period. Across the 24 event-specific local estimates, 14 are negative (mean -0.014), though the equal-weight average remains imprecise. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

J.7 Region-Type Check: Descriptive Detail

This appendix expands the region-type check of Section 5.6. Of the 43,351 in-sample villages, 6,123 (14 percent) ever report communal conflict, with the highest rates in the known communal-conflict provinces (Maluku, 55 percent of villages;

North and Central Sulawesi, 20 to 32 percent; East Nusa Tenggara, 30 percent) but substantial numbers also in populous Java (roughly 10 to 15 percent); Aceh, whose conflict was separatist rather than communal, is low (5 percent), a sign the outcome captures communal rather than armed-separatist violence. Adding province $\times$ wave fixed effects attenuates the main absorbing effect from -0.0186 to -0.0120 (Table 31); the within-province effect remains significant at 1 percent under village clustering ($p = 0.002$) but only marginal under kabupaten clustering ($p = 0.06$), and the *warga* margin does not survive province trends.

Table 31: Region-Type Check: Main Effect under Province-Specific Trends
Adding province $\times$ wave fixed effects attenuates the effect by about 40 percent; a within-province component persists but is smaller and less robust.

Specification	$\hat{D}_{\text{treat}}$	SE	p
Village + wave FE (baseline)	-0.0186^{***}	0.0032	< 0.001
+ Province $\times$ wave FE	-0.0120^{***}	0.0039	0.002
same, kabupaten-clustered	-0.0120	0.0063	0.06
+ Province $\times$ wave FE, <i>warga</i> outcome	-0.0015	0.0031	0.64

Notes: Absorbing first-exposure treatment on the in-sample native-2005 panel ($N = 303,457$), no controls, no conditioning on any realized outcome. The baseline row carries village and wave fixed effects; the remaining rows add province $\times$ wave fixed effects, allowing each of Indonesia's provinces its own conflict trend, so the effect is identified from within-province variation across villages. Village fixed effects already absorb region *levels*; this tests region-specific *trends*. Of the 43,351 in-sample villages, 6,123 (14 percent) ever report communal conflict; these are patterned by region, with the highest rates in the known communal-conflict provinces (Maluku, North and Central Sulawesi, and East Nusa Tenggara) but substantial numbers also in populous Java; Aceh, whose conflict was separatist rather than communal, is comparatively low, a reassuring sign that the outcome captures communal rather than armed-separatist violence. Under province $\times$ wave trends the effect changes to -0.0120 , staying significant at the 5 percent level under village clustering; it is only marginal under the more conservative kabupaten clustering, and the definition-stable *warga* margin does not survive. The decline is therefore partly regional, with a smaller within-province component surviving. Because these fixed effects absorb most of the regional variation from which communal conflict is identified, this specification is a more demanding within-province estimate. ** $p < 0.05$, *** $p < 0.01$.

J.8 Treatment-Outcome Timing: Full Audit

This appendix expands the treatment-outcome timing check of Section 5.6. Because PODES conflict at wave t covers the twelve months before May enumeration, an earthquake that falls *inside* that window gives a partial-exposure $t = 0$ outcome, mixing pre- and post-shock months, while a shock that predates the window gives a clean, fully-post outcome. We audit every earthquake by exact date (Table 32, Panel A): among the 19 stacked earthquakes, 73.7 percent are clean-post, and the partial-exposure share is dominated by the 2005-wave mega-events (the December 2004 Aceh tsunami and March 2005 Nias earthquake), the early cohorts we already treat separately, while every major earthquake that carries the effect (Padang 2009, Pidie Jaya 2016, Lombok and Palu 2018) is clean-post. Restricting the estimate to clean-post cohorts, which drops 54,628 partial-exposure village-waves, does not weaken the result: if anything it is slightly larger (-0.0206 versus -0.0186 , Panel B, a within-noise move consistent with attenuation of the partial-exposure $t = 0$ outcome), and the definition-stable *warga* outcome is significant at 1 percent (-0.0072 , standard error 0.0027). This restriction excludes the partial cohorts, chiefly the 2004–05 Aceh and Nias mega-events, rather than re-timing them to a later clean wave, so it changes cohort composition; re-timing is the stricter alternative we do not implement.

Table 32: Treatment-Outcome Timing: Clean-Post versus Partial Exposure
Ninety percent of stacked-design villages are clean-post, and restricting to clean-post cohorts strengthens the effect.

<i>Panel A. Onset-wave exposure classification (share of earthquakes)</i>		
	Clean-post	Partial-exposure
All events (≥ 20 first-treated, 44 total)	75.0% (33 events)	25.0% (11 events)
The 19 stacked earthquakes	73.7% (14 events)	26.3% (5 events)
The <i>all events</i> universe is the 44 earthquakes that first expose at least 20 sample villages, a broader set than the stacked-design universe used elsewhere because it also counts the always-treated 2005-wave events. Partial exposure is dominated by the 2005-wave mega-events (the December 2004 Aceh tsunami and March 2005 Nias earthquake, together 8,663 villages), whose outcome window straddles the shock; these are the early cohorts the paper already treats separately.		
<i>Panel B. Absorbing effect under clean-post timing</i>		
	Composite	Warga
Full sample (benchmark)	−0.0186*** (0.0032)	
Drop 2005-wave Aceh/Nias mega-events	−0.0185*** (0.0032)	
Clean-post cohorts only	−0.0206*** (0.0035)	−0.0072*** (0.0027)

Notes: Absorbing first-exposure treatment, village and wave fixed effects, village-clustered standard errors. A village's onset wave is *partial-exposure* if its earthquake falls inside the wave's 12-month conflict-reporting window ($\text{May}[t-1], \text{May}[t]$), so the $t = 0$ outcome mixes pre- and post-shock months; *clean-post* if the shock predates the window. Restricting to clean-post cohorts drops 54,628 partial-exposure village-waves and, if anything, leaves the composite effect slightly larger (−0.0206 vs −0.0186), a within-noise move consistent with attenuation of the partial-exposure $t = 0$; the definition-stable *warga* outcome becomes marginally significant ($p = 0.01$) once timing is clean. This restriction excludes the partial cohorts rather than re-timing them to a later clean wave, so it changes cohort composition. The full per-earthquake timing audit (date, pre-wave, first fully-post wave, months to the outcome window) is in the replication archive. * $p < 0.1$, *** $p < 0.01$.

Table 33: Benchmark TWFE: Survey-Window-Aligned MMI Exposure and Communal Conflict
2005-boundary panel, medium and high hazard zones

	Outcome: any communal conflict (binary)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ever exposed $\text{MMI} \geq \text{VI}$ (absorbing)	-0.0186*** (0.0032)	-0.0202*** (0.0033)						
MMI (max, window $t-1$)			0.0001 (0.0002)					
$\text{MMI} \geq \text{VI}$ (contemporaneous)								-0.0012 (0.0017)
$\text{MMI} \geq \text{VI}$ (window $t-1$)				-0.0055** (0.0026)	-0.0056** (0.0026)			-0.0056** (0.0026)
$\text{MMI} \geq \text{VI}$ (window $t-2$)					-0.0229*** (0.0035)			
$\text{MMI} \geq \text{VI}$ (window $t-3$)					-0.0010 (0.0018)			
$\text{MMI} \geq \text{V}$ (window $t-1$)						0.0026 (0.0018)		
$\text{MMI} \geq \text{VII}$ (window $t-1$)							-0.0057 (0.0035)	
Observations	303,457	292,712	292,712	292,712	292,712	292,712	292,712	292,712
Adj. R-squared	0.073	0.072	0.071	0.071	0.071	0.071	0.071	0.071

Standard errors clustered at village level in parentheses. Columns (1) and (2) report the primary absorbing specification: a village is treated from the first survey wave following its first $\text{MMI} \geq \text{VI}$ exposure onward. Columns (3) to (8) report episodic exposure in a given window. Earthquake exposure is spatially clustered, so village-level clustering may understate uncertainty. Re-clustering at the kabupaten level (350 clusters) leaves every point estimate unchanged and gives, for column (2), $\text{SE} = 0.0064$, $t = -3.18$, $p = 0.002$; for column (4), $\text{SE} = 0.0034$, $t = -1.64$, $p = 0.102$. The absorbing estimate is the one that survives this and province-level clustering; see the clustering robustness table. All specifications include village and survey-wave fixed effects, plus seven baseline village characteristics interacted with wave dummies: household electricity, paved road, internet access, police post, health centre, primary schools, and distance to the district capital. Baseline values are taken from PODES 2003 (enumerated August 2002), before the sample period, so they cannot themselves respond to exposure. Lagged rather than baseline covariates would be bad controls: exposure raises primary schools and paved roads and lowers police presence (see the mechanism table on village characteristics). Exposure windows are May-to-May 12-month blocks aligned to PODES enumeration; window $t-1$ lies fully before the conflict reference window. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

J.9 Benchmark TWFE: Distributed-Lag Estimates

J.10 Goodman–Bacon Decomposition

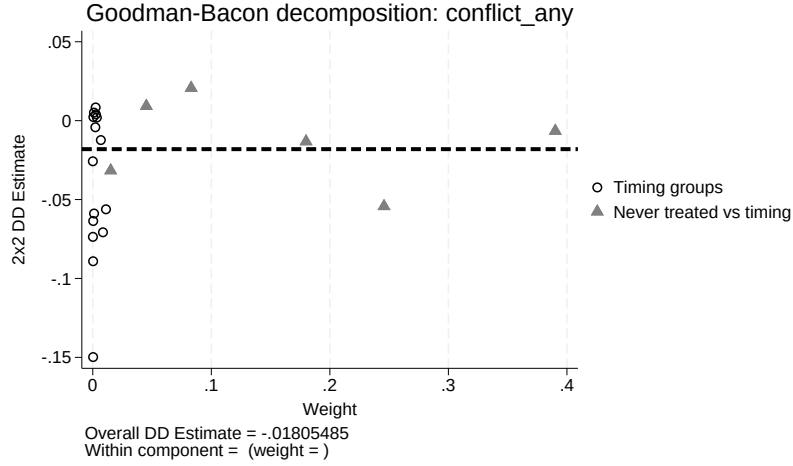


Figure 9: Goodman-Bacon Decomposition of TWFE Absorbing Estimate

Note: x -axis = weight on each 2×2 DiD; y -axis = DiD coefficient. Dot size proportional to weight. Re-measured 2026-08-17 on the current panel: 73.1% of weight from never-treated vs. timing comparisons, 3.1% from timing vs. timing, and 23.8% from the forbidden already-treated comparisons, which return -0.0205 against -0.0173 for the clean comparisons. Sample: the 2005-boundary panel. D_{treat} is absorbing, window-aligned first exposure, balanced panel $N = 303,457$. Source: [Goodman-Bacon \(2021\)](#).

J.11 Episodic Estimand: dCdH Non-Absorbing Treatment

Table 34 reports the de Chaisemartin and D’Haultfoeuille (2026) non-absorbing interval-treatment estimate ($D_{interval}$) summarized in Section 5. Panel B reports the placebo (pre-trend) coefficients; the first is individually positive and significant, so we report it rather than rely on the joint test alone.

Table 34: dCdH Non-Absorbing Treatment Estimate
2005-boundary panel, medium and high hazard zones

	TWFE	dCdH
	(1)	(2)
<i>Panel A: Treatment effects</i>		
$D_{\text{interval},it}$	-0.014*** (0.003)	
Effect ₁		-0.015*** (0.004)
Effect ₂		-0.008* (0.004)
Effect ₃		-0.005 (0.004)
<i>Panel B: Pre-trend (placebo) tests</i>		
Placebo ₁		0.010** (0.005)
Placebo ₂		-0.001 (0.006)
Placebo joint p -value		0.056
Effects joint p -value		0.003
N switchers (Effect ₁)		3,868
Observations	292,712	249,237
Village FE		Yes
Year FE		Yes
Controls	Yes	No [†]

Notes: Outcome: binary indicator for any communal conflict (`conflict_any`). Treatment: $D_{\text{interval},it} = 1$ if any earthquake with $\text{MMI} \geq \text{VI}$ occurred in $[\text{enum}(t-1) + 1\text{mo}, \text{enum}(t)-1\text{yr}-1\text{mo}]$; non-absorbing (can switch $0 \rightarrow 1 \rightarrow 0$). Source: USGS ShakeMap event-level rasters; PODES 2005–2024 (7 waves). Sample: KRB Medium/High seismic hazard, native 2005 village boundaries (nb2005). Column (1) TWFE: `reghdfe` with baseline covariates; SE clustered at village level. Column (2) dCdH: `did_multiplegt_dyn` (de Chaisemartin & D’Haultfoeuille, 2026), $\text{Effect}_\ell = \text{ATT}$ for switchers ℓ waves after first switch; Placebo_ℓ tests parallel trends ℓ waves before first switch. [†] Controls omitted in dCdH: the estimator’s switching design compares consecutive waves, and all Effect₁ comparisons require wave 2005 as baseline. TWFE with controls (−0.013***) is the controls-adjusted benchmark. * $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$

J.12 Dropping the G6 Cohort and Aceh Province

The G6 cohort is the only positive-ATT cohort (Section 5), an artifact of a floor effect in post-conflict Aceh, where the last pre-treatment conflict rate was near zero. Table 35 confirms that this cohort neither drives the headline estimate nor holds it back appreciably: dropping G6 leaves the composite ATT at -0.0172 against -0.0150 on the full sample, dropping Aceh province entirely gives -0.0157 , and dropping both gives -0.0170 , each significant at the 1 percent level. The average pre-trend lead remains insignificant in every restriction, and the definition-consistent *warga* outcome moves in the same direction, reaching conventional significance once G6 is dropped (-0.0074).

Table 35: Robustness: Dropping the G6 Cohort and Aceh Province
Baseline: nb2005, window-aligned first exposure, CS(2021) with baseline controls. Take-away: the headline estimate is stable when the positive-outlier G6/Aceh cohort is removed, so the outlier neither drives nor materially attenuates the result.

Specification	Composite (<i>conflict_any</i>)		<i>Warga</i>	
	CS(2021) ATT	Pre_avg (<i>p</i>)	CS(2021) ATT	<i>N</i>
Main (full sample)	-0.0150^{***}	-0.0077 (0.33)	-0.0060^*	227,045
Drop G6 cohort	-0.0172^{***}	-0.0088 (0.30)	-0.0074^{**}	225,757
Drop Aceh province	-0.0157^{***}	-0.0083 (0.32)	-0.0059^*	217,119
Drop Aceh + G6	-0.0170^{***}	-0.0088 (0.30)	-0.0070^{**}	216,650

Each row re-estimates the primary CS(2021) specification (doubly-robust weighting, not-yet-treated controls, *long2*, baseline covariates) on a restricted sample. G6 is the Aceh Pidie Jaya cohort, the only cohort with a positive ATT; Aceh province is identified by the first two digits of the village identifier. Pre_avg is the aggregate pre-treatment event-study coefficient (a conditional parallel-trends test). $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

J.13 Balancing on Pre-Treatment Growth Slopes

Section 5.6.2 shows that villages later struck by earthquakes trend differentially before exposure on several margins, so the primary specification relies on conditional rather than unconditional parallel trends. Because conditioning on covariate *levels* cannot by itself remove a differential *slope*, we add each village's pre-treatment growth in log population and log households (measured over 2005–2008 and held

time-invariant) directly to the control set. Population growth itself is not a source of concern: staggered-treated villages grew by 5.95 log points against 6.27 for the never-treated, a difference that is small, of the opposite sign to the one the concern anticipates, and statistically indistinguishable from zero ($p = 0.49$; households 10.30 against 10.24, $p = 0.91$). Conditioning on these slopes leaves the estimate essentially unchanged (Table 36): the composite ATT is -0.0148 versus the headline -0.0150 , with an insignificant average pre-trend lead (-0.0076 , $p = 0.33$), and the definition-consistent *warga* outcome is likewise stable.

J.14 Cohort and Event Dependence: Leave-One-Out

Table 37 documents the earthquakes behind the staggered variation, the input to the stacked event design of Section 5.6.1: 19 distinct events, each first exposing at least 20 in-sample villages, spread across 2006–2023 and from Sumatra to Maluku.

Table 36: Robustness: Conditioning on Pre-Treatment Growth Slopes
2005-boundary panel, window-aligned first exposure, Callaway-Sant’Anna

Control set	Composite (<i>conflict_any</i>)		<i>Warga</i>		<i>N</i>
	CS(2021) ATT	Pre_avg (<i>p</i>)	CS(2021) ATT		
Baseline controls (PODES 2003 levels)	-0.0150***	-0.0077 (0.33)	-0.0060*		227,045
+ pre-treatment growth slopes	-0.0148***	-0.0076 (0.33)	-0.0058*		227,038

The second row adds each village’s pre-treatment growth in $\ln(\text{population})$ and $\ln(\text{households})$, measured 2005–2008 and held time-invariant, to the baseline control set (available for all 43,351 villages). Conditioning on these slopes directly is a regression adjustment for differential pre-treatment trajectories; it addresses the same concern as reweighting or matching on trends, though it is not equivalent to entropy balancing or coarsened exact matching. The estimate is Estimates are Callaway-Sant’Anna (2021) group-time average treatment effects, doubly-robust with not-yet-treated controls, with analytical influence-function standard errors clustered at the village level. essentially unchanged. Pre_avg is the aggregate pre-treatment event-study coefficient. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 37: Earthquakes Behind the Staggered Variation (Event Audit)*Takeaway: the identifying variation comes from 19 distinct earthquakes, not a handful.*

Earthquake (region)	Date	Max MMI (village)	First-treated villages	Onset PODES wave
Pundong	May 2006	7.5	314	2008
Bukittinggi	Mar 2007	6.8	157	2008
Sungai Penuh	Sep 2007	6.9	306	2008
Gorontalo	Nov 2008	7.0	113	2011
Sarangani	Feb 2009	7.6	63	2011
Banjar	Sep 2009	6.5	85	2011
Pariaman	Sep 2009	8.1	824	2011
Reuleuet	Dec 2016	7.5	126	2018
Labuan Lombok	Aug 2018	8.1	353	2021
Palu	Sep 2018	8.2	252	2021
Sukabumi	Nov 2022	8.3	149	2024
Tanimbar	Jan 2023	7.4	36	2024
7 further events (≥ 20 villages each)		—	231	2008–2024
44 smaller events (< 20 villages each)		—	198	2008–2024
Total (19 events with ≥ 20 villages, 2008–2024 onset)			3,207	

Notes: Each village is assigned to the earliest earthquake exposing it to mean $\text{MMI} \geq \text{VI}$ over 1990–2024. The table lists the 12 largest events (selected by first-treated-village count, shown here in chronological order), 7 further events with at least 20 in-sample first-treated villages, and 44 smaller events; the 19 events with at least 20 in-sample villages account for 3,009 of the 3,207 staggered-treated villages (2008–2024 onset). Region labels are auto-derived from the nearest named place in the USGS ShakeMap catalog (see `gis/13b_event_region_labels.py` for the same transformation applied to Figure 6) and are approximate; all counts, dates, MMI values, and onset waves are exact. Source: USGS ShakeMap event rasters, native 2005 boundaries.

Is the result driven by one cohort or one earthquake? Group-level ATTs are heterogeneous (Section 5): the largest cohort effects are the more recent G7 and the Padang 2009 cohort G4, and G6 is positive in the no-controls specification. Table 38 answers the dependence question directly, re-estimating the no-controls CS(2021) ATT while dropping each treatment cohort one at a time and re-weighting the six cohort ATTs equally instead of by cohort share. This is a leave-one-cohort-out check, not a clean leave-one-event-out check: because a single first-exposure cohort can bundle more than one earthquake (the 2018 Lombok and Palu events both fall in G7), dropping a cohort can remove more than one event, and isolating individual events would require the event-level design we discuss as an extension. What the exercise does establish is that no single cohort drives the sign.

Table 38: Leave-One-Cohort-Out

2005-boundary panel, no controls so that all six cohorts are estimable. Takeaway: every re-estimate stays negative, so no single cohort drives the result.

Sample	ATT	SE	p	N
Full sample (no-controls benchmark)	−0.0150***	(0.0041)	0.000	234,465
<i>Leave-one-cohort-out</i>				
Drop G3 (2008)	−0.0156***	(0.0053)	0.003	228,305
Drop G4 (2011; Padang cohort)	−0.0194***	(0.0052)	0.000	226,450
Drop G5 (2014)	−0.0136***	(0.0041)	0.001	234,206
Drop G6 (2018; positive cohort)	−0.0171***	(0.0043)	0.000	233,044
Drop G7 (2021)	−0.0103**	(0.0044)	0.018	229,418
Drop G8 (2024)	−0.0152***	(0.0042)	0.000	232,918
Equal-weight cohort average	−0.0182***	(0.0043)	0.000	—

Notes: CS(2021) simple ATT (Callaway–Sant’Anna, doubly-robust, not-yet-treated controls, wave-ordinal time), outcome: any communal conflict, no controls so that all six cohorts are estimable; the with-controls headline ATT is reported in Table 3 and the no-controls full-sample benchmark here is −0.0150***. Each row drops one treatment cohort at a time (the treated villages of that cohort are removed). This is a leave-one-cohort check, not a clean leave-one-event check: a single first-exposure cohort can bundle more than one earthquake (the 2018 Lombok and Palu events both fall in G7), so dropping a cohort can remove more than one event. The equal-weight cohort average is the unweighted mean of the six cohort ATT(g)s (csdid_estat group’s GAverage row on the same full-sample no-controls model), removing any influence of cohort-share weighting. SE clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Every estimate is negative, ranging from −0.0103 to −0.0194 around the no-controls full-sample benchmark of −0.0150. Dropping the large Padang cohort (G4) *strengthens* the estimate to −0.0194 ($p < 0.001$), and dropping the positive G6 cohort strengthens it to −0.0171; the weakest drop is G7, at −0.0103, still significant at 5 percent. The equal-weight cohort average (−0.0182) shows the result is not an artifact of cohort-share weighting. No single cohort overturns the finding, and none carries it.

J.15 Placebo Tests and Randomization Inference

Four further exercises probe whether the estimate could be an artifact of chance assignment or of shaking too weak to cause damage.

First, spatial randomization inference. Because earthquakes strike spatially clustered sets of villages rather than villages drawn independently, a design-based placebo must relocate the shocks while preserving their geography. For each of 500 placebo draws we move every earthquake footprint to a random location among the never-treated villages, assigning placebo treatment to the nearest villages so as to approximately preserve each footprint’s size and compactness (relocated footprints occasionally overlap, truncating a draw by about 16 percent on average), with the earthquake’s true onset timing, and re-estimate the absorbing coefficient on the never-treated sample so that the true suppression cannot leak into the placebo control pool. The null’s standard deviation (0.0085) is 2.7 times the village-clustered standard error (0.0032): the spatial null concedes that the uncertainty from a small number of large footprints is far greater than village clustering implies, which is the few-shocks concern made quantitative, and the actual estimate clears it anyway. And the distribution is mildly right-shifted (mean +0.0029), which we attribute to the placebo pool rather than to any mechanical effect: uniform-in-area placement oversamples sparser never-treated regions whose conflict levels and trends differ from the denser struck areas. The actual estimate (−0.0186) nonetheless lies below *every* placebo draw; with $k = 0$ of $B = 500$ draws at least as negative, the randomization-inference convention $p = (k + 1)/(B + 1)$ gives a one-sided $p = 0.002$, which is the grid floor at this B . The result does not rest on the shift: recentering the null on its own mean, the actual deviation is exceeded by only 8 of the 500 draws (recentred two-sided $p = 0.018$). No random spatial placement of the shocks reproduces the observed suppression.

Second, for completeness we also permute the treatment cohort assignment (the first-exposure wave) across villages 2,000 times, ignoring geography. The placebo distribution is centred at zero (mean −0.00004, SD 0.0025), and the actual estimate is more extreme than all 2,000 draws ($p < 0.0005$). We report this cohort permutation only as a complement: because it scatters treatment across space and destroys the spatial structure of real earthquakes, it is not design-based evidence

on its own, and we lean on the spatial version above.

Third, we run a wild cluster bootstrap for the deliberately conservative province-level clustering (28 clusters), using Webb weights and 9,999 replications (Roodman, Nielsen, MacKinnon, & Webb, 2019): the bootstrap p -value is 0.040 for the two-way fixed-effects composite estimate, 0.16 for the pooled CS(2021) ATT, and 0.255 for the definition-consistent *warga* outcome, so under this most demanding few-cluster inference the composite clears 5 percent while the *warga* arm does not, as reported in Table 17. Province-level inference with only 28 clusters is the single most demanding test the estimate faces; the composite still clears 5 percent under it while *warga* does not, and every less extreme clustering level (village and kabupaten) keeps the composite effect significant.

Fourth, a low-intensity exposure check, which we are no longer able to run. The design constructs pseudo first-exposure cohorts from events with mean MMI in the felt-but-non-damaging $[3, 5)$ band, restricted to villages that never experience any event of MMI V or above. That restriction was workable when the event catalogue began in 2000. With the catalogue extended back to 1990, as Section 3 explains it must be for the absorbing cohort assignment to be correct, almost every hazard-zone village has at some point reached MMI V, and the felt-but-undamaged comparison group collapses to 791 village-waves. Estimates on that sample are uninformative (-0.0018 , $p = 0.91$ under two-way fixed effects). We therefore withdraw the low-intensity check rather than report it on a catalogue vintage inconsistent with the rest of the paper, and we make no claim about whether felt-but-non-damaging shaking moves conflict. The question of whether the effect is specific to the MMI VI damage threshold is instead addressed by the within-earthquake band comparison of Section 5.6, which contrasts villages at MMI $[6, 7)$ with villages at $[5, 6)$ struck by the same event and does not depend on the catalogue start year.

J.16 Alternative MMI Thresholds

Table 39: Robustness: Alternative Shaking Thresholds

Takeaway: the effect is negative and significant under the absorbing design at all three thresholds, but the ladder is not monotone. The TWFE estimate is largest at $MMI \geq VI$, and the CS estimate at $MMI \geq VII$ is imprecise because that cohort is small. We draw no dose-response conclusion from this table.

	(1) MMI $\geq V$ (weak)	(2) MMI $\geq VI$ (strong, main)	(3) MMI $\geq VII$ (very strong)
TWFE D_{treat} (absorbing)	−0.0063*** (0.0021)	−0.0198*** (0.0033)	−0.0174*** (0.0035)
CS(2021) overall ATT	−0.0076** (0.0037)	−0.0150*** (0.0043)	−0.0054 (0.0047)
Ever-treated villages	8,085	3,207	1,436
Village + wave FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Notes: Outcome: any communal conflict. Window-aligned first exposure at each threshold (cohort = first PODES wave enumerated after the first event above threshold); always-treated (first observed 2005) excluded. Controls are baseline (PODES 2003, enumerated August 2002) village characteristics interacted with wave dummies for the TWFE rows, and the corresponding level baseline for CS(2021)’s doubly-robust nuisance model. SE clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 39 shows results for $MMI \geq V$ (broader, more frequent treatment) and $MMI \geq VII$ (narrower, more destructive). Under the absorbing first-exposure design, $MMI \geq V$ yields −0.0063*** and $MMI \geq VII$ yields −0.0174***, both consistent in sign with the main finding but neither larger than the −0.0198*** at $MMI \geq VI$. Signs are negative and significant at all three thresholds, so the result is not an artifact of the MMI VI cutoff.

J.17 Exposure Construction: Raster-to-Village Aggregation

The treatment assigns each village the *mean* MMI across raster pixels intersecting its polygon, and a referee may reasonably ask whether the result depends on that aggregation rule: a mean can dilute localized strong shaking in large polygons, and a max can overstate it. We re-ran the zonal statistics for all 1,423 event ShakeMaps under four aggregation rules (pixel mean, pixel max, pixel median, and the raster value at the village’s representative point), rebuilt the window-aligned first-exposure cohorts under each, and re-estimated the design (Table 40). As a further check, we drop the largest 5 percent of villages by polygon area (above 56 km²), where within-polygon aggregation matters most. The estimates are stable across every construction: TWFE ranges from -0.0073 to -0.0209 , all significant at the 1 percent level, and the CS(2021) ATT from -0.0059 to -0.0169 , negative throughout and significant in four of the five constructions, the pixel-maximum rule being the exception at 10 percent. This panel assigns cohorts from its own windowed construction rather than from the production first-exposure file, so its levels sit below the headline and the comparison to read is the spread across rules, not any single row. Measurement error in the raster-to-village mapping is not driving the result.

Table 40: Robustness: Alternative Raster-to-Village Exposure Constructions
Outcome: any communal conflict; 2005-boundary panel, window-aligned absorbing first exposure, no covariates

Exposure construction ($\geq VI$)	TWFE	(SE)	CS(2021) ATT	(SE)
Pixel mean within boundary	−0.0110***	(0.0025)	−0.0100***	(0.0035)
Pixel max within boundary	−0.0073***	(0.0023)	−0.0059*	(0.0032)
Pixel median within boundary	−0.0107***	(0.0025)	−0.0096***	(0.0035)
Value at representative point	−0.0101***	(0.0024)	−0.0086**	(0.0035)
Drop largest 5% of villages by area	−0.0209***	(0.0034)	−0.0169***	(0.0043)

Zonal statistics re-run over all 1,423 event ShakeMaps at native 2005 boundaries (1.78 million village-event cells, zero events missing). Each row rebuilds first-exposure cohorts from the stated aggregation rule and re-estimates the absorbing TWFE and CS(2021) specifications without covariates. TWFE: village and wave fixed effects, village-clustered SEs, $N = 221,900$ – $277,200$. CS(2021): doubly-robust weighting, not-yet-treated controls, `long2`, analytical SEs. This panel assigns cohorts from its own windowed construction rather than from the production first-exposure file, so its levels are not directly comparable to the headline estimates; the informative content is the *spread* across aggregation rules. Across those rules the TWFE estimate lies between -0.0073 and -0.0209 and the CS estimate between -0.0059 and -0.0169 , so the sign and broad magnitude are not an artifact of the raster-to-village aggregation rule.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

J.18 SUTVA: Spatial Spillovers

Earthquakes do not stop at village boundaries, so never-treated control villages located near treated villages could be contaminated by spillovers, biasing the never-treated counterfactual that carries most of the identification weight (Section 5). We test this two ways using village-centroid distances and a KD-tree neighbour search. First, we add each village’s neighbours’ treated share within 10, 25, and 50 km as a spatial lag alongside the direct treatment indicator. The direct effect is stable, and if anything larger, after controlling for the lag (`conflict_any`: the absorbing -0.0203 moves to between -0.0237 and -0.0243 across radii; `conflict_warga` is essentially unchanged, near -0.009). The spillover coefficient itself is small and positive but imprecisely estimated at every radius ($+0.006$ to $+0.014$, all $p > 0.10$; the 95% confidence intervals reach up to about $+0.03$), so we do not read it as a precise zero. The identification-relevant fact is instead that the direct effect is stable, and if

anything larger, once the spatial lag is included, the pattern we would not observe if positive spillover into the controls were biasing the direct estimate toward zero. Second, we drop never-treated villages that have any treated neighbour within 25 or 50 km (a “donut” around treated villages) and re-estimate: the direct TWFE estimate is essentially unchanged, and CS(2021) on the 25 km donut sample yields ATT -0.0057 (negative but imprecise, $p = 0.236$) with an average-lead pre-trend that does not reject (Pre_avg $+0.0034$, $p = 0.559$). The composite joint Wald test and the covariate-free anchors remain the specifications in which a pre-trend signal appears. Excluding potentially contaminated controls does not overturn the estimate, the pattern we would expect if spillover contamination were not a material threat to identification here.

Table 41 summarizes three further checks on the absorbing specification under wave fixed effects. Province $\times$ Wave fixed effects, which the rule of Section 4.1 selects as the preferred specification, move the coefficient from -0.0198^{***} to -0.0148^{***} ; it remains significant at 1 percent, so the result is not an artifact of province-level shocks coinciding with earthquake timing. The balanced-panel restriction (villages present in all seven waves) leaves the estimate unchanged, because the absorbing design’s sample is already balanced; no observations are dropped by the restriction. Always-treated villages are excluded from the estimation sample by construction under the windowed first-exposure design (any village already exposed at the 2005 survey baseline is dropped from g), so there is no separate ablation to report for that concern. Finally, a forward-lag placebo that leads D_{treat} by one wave finds a lead coefficient of -0.0045 ($t = -0.84$, $p = 0.40$), against a contemporaneous effect of -0.0220^{***} estimated jointly with it. The lead is a clean null, so the design shows no sign of anticipatory adjustment on this margin. It is, however, specific to the composite outcome and to particular cohorts rather than a systematic anticipatory pattern: it is insignificant on the definition-consistent *warga* outcome ($t = -1.6$), reverses sign and turns positive once the 2021 cohort is excluded, and does not appear on assault ($t = 0.8$), the individual-crime placebo. This mirrors the split in the pre-trend tests, where the composite pre-period is not clean while *warga* passes; we retain it as a disclosed caveat and lean on the definition-stable outcome and the composition contrast accordingly. The HonestDiD bounds in Section 5.6.2

quantify exactly how much residual violation of this kind the estimate can absorb before the sign is no longer identified.

J.19 Additional Checks: Coefficient Stability and Spatial Inference

Coefficient stability under [Oster \(2019\)](#), computed for the absorbing window-aligned treatment ($\hat{\beta} = -0.0203$, $R^2 = 0.243$ with full controls): adding the full control set moves the coefficient *away* from zero, from -0.001 to -0.020 . At Oster’s recommended bound $R_{\max}^2 = 1.3 \times R^2 = 0.316$ the implied $\delta = -3.9$ for the composite outcome (and $\delta = -0.37$ at the extreme bound $R_{\max}^2 = 1$). A negative δ means that proportional selection on unobservables would have to run opposite in sign to the observed selection on covariates in order to explain the effect away: the observables, if anything, pull the coefficient away from zero rather than toward it. Like a large positive δ , this sign pattern indicates that the estimate is not an artifact of the kind of selection the observables reveal.

Conley ([Conley, 1999](#)) spatial-HAC standard errors, computed with a Bartlett kernel via a scalable KD-tree implementation on the subset of villages with matched centroids ($N = 190,279$, about 84 percent of the panel; underlying coefficient -0.0269), give $SE = 0.0080\text{--}0.0095$ across distance cutoffs of 50–150 km ($t \approx -2.8$ to -3.3 ; $p < 0.01$ at every cutoff, both outcomes). Alternative clustering brackets these: kabupaten-level clustering keeps the estimate significant, while province-level clustering (few clusters) is the conservative bound, at which a wild-cluster bootstrap still rejects at 5 percent for the two-way fixed-effects estimate ($p = 0.040$) but not for the pooled CS(2021) ATT ($p = 0.16$). The ordering (village-clustered $<$ Conley $<$ kabupaten $<$ province) is exactly as spatial correlation predicts, and the effect remains significant under the distance-based correction appropriate for a localized physical shock, though it is sensitive to the most conservative few-cluster inference.

Table 41: Additional Robustness Checks: nb2005 Baseline (Primary)

Specification	$\hat{\beta}$	SE	N	Note
<i>Panel A: Province $\times$ Wave Fixed Effects</i>				
Village + wave FE (baseline)	-0.0198***	(0.0033)	227,045	Headline specification Absorbs province-level shocks
Village + province $\times$ wave FE	-0.0148***	(0.0040)	227,045	
<i>Panel B: Balanced Panel</i>				
Unbalanced (baseline)	-0.0198***	(0.0033)	227,045	All villages 0 obs dropped for balance
Balanced (7 waves)	-0.0198***	(0.0033)	227,045	
<i>Panel C: Always-Treated Villages</i>				
Under the absorbing windowed-first-exposure design, villages already exposed at the survey baseline ($g_wave = 2$, i.e., first exposure at or before the 2005 wave) are excluded from g by construction, so there is no separate always-treated group left to drop as an ablation; this concern is structural to the design rather than a further robustness check.				
<i>Panel D: Forward-Lag Placebo (No Anticipation)</i>				
D_{treat} (current)	-0.0220***	(0.0042)	227,045	Headline treatment, jointly estimated with lead $t = -0.84, p = 0.401$ (see notes)
$F1_D_{\text{treat}}$ (future, placebo)	-0.0045	(0.0053)	227,045	
<i>Notes:</i> Baseline: absorbing two-way fixed-effects LPM with village and wave fixed effects, matching Table 4. Outcome: any communal conflict. Sample: KRB Medium/High seismic hazard zones, nb2005 windowed first-exposure design. Panel A replaces wave FE with province $\times$ wave FE to absorb regional time trends. Panel D leads D_{treat} by one wave to test for anticipation. Standard errors clustered at village level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.				

J.20 Rare-Outcome Functional Form: Conditional Logit

Communal conflict is a rare binary outcome (2.6 percent pooled), so a linear probability model could in principle misbehave near the boundary. We re-estimate the absorbing design with a conditional (fixed-effects) logit, which conditions out the village effects without the incidental parameters problem and uses only villages whose outcome varies over time. The effect holds in the odds metric: first exposure reduces the odds of reported communal conflict by about a quarter without controls (odds ratio 0.75, $\hat{\beta} = -0.29$, cluster-robust $z = -3.14$, $N = 38,871$ observations in 5,553 switching villages) and by about a third with the standard baseline covariates (odds ratio 0.62, $z = -4.5$). The definition-consistent *warga* outcome is a null without controls (odds ratio 0.93, $z = -0.6$) and negative with controls (0.74, $z = -2.2$), so the *warga* margin again carries its signal only conditionally, which we disclose. A fixed-effects Poisson model, an alternative rare-outcome specification, yields an incidence-rate ratio of 0.79 ($z = -2.9$). The LPM point estimates are therefore not an artifact of the linear functional form; the nonlinear models agree on sign.

J.21 Robustness to the Control Set: Bad-Control and Baseline-Covariate Checks

Table 42 accompanies Section 5.6.4 and re-estimates the headline ATT under alternative covariate sets, confirming that the result does not depend on the potentially endogenous infrastructure controls.

K Communication Infrastructure: Pre-Trend-Failed Outcomes

Two outcomes from the communication-infrastructure battery in Section 5.7.3 are reported here. The absence of any mobile signal is essentially unchanged after exposure, while any street lighting rises by 6.1 percentage points, but in both cases the aggregate pre-treatment event-study mean is itself large and significant, so treated and comparison villages were already diverging before first exposure. These

Table 42: Robustness to the Control Set: Bad-Control and Baseline-Covariate Checks
Outcome: any communal conflict (and warga); 2005-boundary panel, window-aligned absorbing first exposure

Control set	Composite (<i>conflict_any</i>)			Warga		Observations
	CS(2021)	ATT	(SE)	Pre_avg (<i>p</i>)	CS(2021) ATT	
No covariates (anchor)	-0.0150***		(0.0041)	+0.0107 (0.01)	-0.0071**	234,465
Baseline PODES 2003 levels (headline)	-0.0150***		(0.0043)	+0.0159 (0.00)	-0.0060*	227,045
Baseline 2005 levels	-0.0148***		(0.0043)	+0.0151 (0.00)	-0.0070**	211,981
Lagged set, paved road dropped	-0.0155**		(0.0061)	-0.0121 (0.07)	-0.0042	189,877
Geographic only (distance)	-0.0161***		(0.0056)	-0.0082 (0.17)	-0.0051	191,923

Each row re-estimates the primary Callaway-Sant'Anna specification with a different covariate set on the same panel. The headline row uses PODES 2003 baseline levels, enumerated August 2002 and therefore predetermined with respect to every exposure in the sample. The composite estimate is stable across all five sets, from -0.0148 to -0.0161; what varies is the pre-treatment average, which is significantly positive under the level sets and insignificant under the lagged ones. Estimates are Callaway-Sant'Anna (2021) group-time average treatment effects, doubly-robust with not-yet-treated controls, with analytical influence-function standard errors clustered at the village level. Sample sizes differ because lagged covariates are unavailable in the first wave. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

outcomes move in directions consistent with greater remoteness or reconstruction, but their pre-trends caution against causal interpretation. The positive street-lighting coefficient is most plausibly attributed to post-disaster reconstruction and expanded state presence in exposed villages rather than to any earthquake effect on communication capacity. Its pre-trend does pass, unlike the mobile-signal series, but a reconstruction response is not a capacity mechanism, so we do not read it as one.

Table 43: Communication-Infrastructure Outcomes

2005-boundary panel, window-aligned absorbing first exposure, no controls

Outcome	Treated pre-mean	CS(2021) ATT	(SE)	p	Pre_avg (p)
No mobile signal	0.114	−0.0003	(0.0064)	0.964	+0.0268 (0.000)
Any street lighting	0.714	+0.0458***	(0.0086)	0.000	+0.0057 (0.487)

No mobile signal is the indicator that a village has no mobile-phone reception. Street lighting is built from PODES items r502a and r502b, whose 2018 redesign produces a level break; its own pre-treatment aggregate is insignificant. Callaway–Sant’Anna, doubly-robust, not-yet-treated controls, no covariates. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

L Social Capital: Pre-Trend-Failed Outcomes

Three further outcomes from the social-capital battery in Section 5.7.3 are reported here. The presence of any active sports-activity group falls after exposure, its count moves in the opposite direction, and the inverse-hyperbolic-sine of total village community institutions declines in the 2018–2024 panel, but in each case the aggregate pre-treatment event-study mean is itself significant, so treated and comparison villages were already diverging before first exposure. These outcomes move in directions consistent with either organisational disruption or measurement breaks across the 2018 questionnaire redesign, but their pre-trends caution against causal interpretation.

Table 44: Social-Capital Outcomes with Failed Pre-Trends*Baseline: 2005-boundary panel, window-aligned absorbing first exposure, no controls*

Outcome	Treated pre-mean	CS(2021) ATT	(SE)	p	Pre_avg (p)
Sports-activity group (any)	0.943	+0.0003	(0.0049)	0.949	+0.0002 (0.950)
Sports groups (count)	3.717	+0.1044**	(0.0455)	0.022	+0.1133 (0.000)
Karang taruna (any)	0.857	+0.0006	(0.0114)	0.956	−0.0100 (0.115)
Village institutions (ihs)	—	−0.0077	(0.0244)	0.751	+0.2106 (0.001)

Social-capital outcomes from PODES. The sports-group count and the village-institution index both fail their own pre-treatment test, so neither is read as evidence either way; they are reported because the mechanism discussion refers to them. Callaway–Sant’Anna, doubly-robust, not-yet-treated controls, no covariates. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

M Heterogeneity on the Definition-Stable Outcome and Group Structure

This appendix reports two heterogeneity exercises discussed in Section 5.7.5. Table 45 re-estimates the two surviving composite moderators, police monitoring and religious polarization, on the definition-stable *warga* outcome alongside the composite outcome: the *warga* response concentrates where religious polarization is high but not where monitoring is high, so the imprecise pooled *warga* effect reflects heterogeneity rather than absence of an effect. Table 46 interacts first exposure with ethnic and religious polarization *and* fractionalization, separately and in a religious horse-race. Religious group structure moderates the effect while ethnic group structure does not, but for religion the polarization and fractionalization indices are nearly collinear and the horse-race cannot separate them, so the moderation is read as a group-structure channel rather than as polarization specifically.

Table 45: Composite versus Definition-Stable Outcome by Moderator (2005-Boundary Panel)

	Police (village cluster)		Relig. polar. (kab. cluster)	
	(1)	(2)	(3)	(4)
	Composite	Warga	Composite	Warga
D_{treat}	-0.0305*** (0.0045)	-0.0133*** (0.0035)	-0.0248*** (0.0065)	-0.0107*** (0.0038)
$D_{\text{treat}} \times \text{Police (std.)}$	-0.0075*** (0.0025)	-0.0034* (0.0020)		
$D_{\text{treat}} \times \text{Relig. polar. (std.)}$			-0.0175*** (0.0065)	-0.0076** (0.0037)
Observations	194,360	194,360	200,375	200,375
Adj. R^2	0.081	0.065	0.070	0.055

Standard errors in parentheses, clustered at village level (police) or kabupaten level (religious polarization). Absorbing windowed first-exposure treatment; village and wave fixed effects and the baseline covariate set throughout. Moderators standardized (mean zero, unit standard deviation).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 46: Polarization versus Fractionalization, Ethnic and Religious (2005-Boundary Panel)

	(1)	(2)	(3)	(4)	(5)
	Eth. P	Eth. F	Relig. P	Relig. F	Relig. race
D_{treat}	-0.0285*** (0.0085)	-0.0272*** (0.0076)	-0.0248*** (0.0065)	-0.0247*** (0.0065)	-0.0258*** (0.0067)
× Ethnic polarization (std.)	-0.0092 (0.0083)				
× Ethnic fractionalization (std.)		-0.0150** (0.0066)			
× Religious polarization (std.)			-0.0175*** (0.0065)		-0.1617* (0.0924)
× Religious fractionalization (std.)				-0.0165*** (0.0063)	0.1421 (0.0897)
Observations	198,779	198,779	200,375	200,375	200,375
Kabupaten clusters	235	235	253	253	253
Adj. R^2	0.067	0.067	0.070	0.070	0.070

Kabupaten-clustered standard errors in parentheses. Absorbing windowed first-exposure treatment; village and wave fixed effects and the baseline covariate set throughout. Moderators standardized (mean zero, unit standard deviation). Column 5 is a religious horse-race with polarization and fractionalization jointly. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

N Covariate Trajectories: CS(2021) on Observables

Table 47: Covariate Event-Study Aggregates: Pre- and Post-Treatment Means
Baseline: 2005-boundary panel, CS(2021), always-treated cohort dropped

Covariate	Pre-treatment aggregate	p	Post-treatment aggregate	p
Any household electricity	−0.0156***	0.001	0.0027	0.448
Paved road access	0.0283***	0.007	−0.0591***	0.000
Internet available	−0.0666***	0.000	0.0332***	0.000
Police post present	−0.0044	0.596	−0.0088	0.123
Number of health centres	−0.0057	0.697	−0.0338**	0.022
Health centre present	0.0070	0.528	−0.0055	0.382
Number of primary schools	−0.2242***	0.000	−0.0776***	0.005
Distance to district capital	0.3853	0.458	1.6306***	0.000

Each row runs the primary Callaway–Sant’Anna specification with the covariate as the outcome and no controls, on the analysis sample with the always-treated cohort dropped. A significant pre-treatment aggregate indicates that the covariate itself trends differentially before exposure, which is why the design conditions on covariates rather than assuming unconditional parallel trends. Population and household counts cannot be tested here: BPS stopped disseminating them after the 2011 wave, so they are missing for four of the seven waves and the event-study aggregate is not estimable. Their pre-treatment growth is examined separately over 2005–2008 in Section J.13. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

O PGA-Derived MMI: An Independent Exposure Measure

ShakeMap’s Modified Mercalli Intensity blends instrumental ground motion with macroseismic (felt-report) data, so a natural question is whether our result depends on that construction. As an independent check we rebuild exposure from ShakeMap Peak Ground Acceleration (PGA) alone, which is purely instrumental, and convert it to MMI through the Wald et al. (1999) relationship, $\text{MMI} = 3.66 \log_{10}(\text{PGA}) - 1.66$, cross-validated against Worden, Gerstenberger, Rhoades, and Wald (2012).

Building this measure required correcting a unit error in the PGA raster-to-village pipeline. ShakeMap distributes some PGA grids in $\ln(g)$ and others linearly in $\%g$; the earlier pipeline exponentiated the log grids correctly but then treated the resulting values (in g) as $\%g$, understating PGA on those grids by a factor of one hundred, a constant $3.66 \log_{10}(100) \approx 7.3$ MMI-point deficit that rendered the derived intensities uninformative. After the fix, the PGA-derived MMI agrees closely with ShakeMap MMI, with a median discrepancy below one-tenth of an intensity unit for shaken villages, and the two exposure indicators correlate 0.79 at the village-wave level.

Re-estimating the main design on this independent measure, holding the window-aligned first-exposure timing and all else fixed, reproduces the result across every specification (Table 48): the primary CS(2021) estimate is -0.0129 ($p = 0.001$) against the ShakeMap-MMI -0.0150 , and the definition-stable *warga* outcome is likewise negative and significant. Both columns of that table are computed in a single run on the identical panel, so the comparison is exact rather than assembled across specifications. The communal decline is therefore not an artifact of ShakeMap’s intensity construction.

Table 48: Measurement Robustness: PGA-Derived MMI

An intensity measure built from instrumental PGA alone reproduces the result across every specification.

	ShakeMap MMI (primary)	PGA-derived MMI (Wald 1999)
CS(2021) ATT, composite	−0.0150*** (0.0043)	−0.0129*** (0.0039)
Absorbing TWFE, no controls	−0.0186*** (0.0032)	−0.0128*** (0.0030)
Absorbing TWFE, with controls	−0.0202*** (0.0033)	−0.0142*** (0.0030)
Definition-stable <i>warga</i> , TWFE controls	−0.0071*** (0.0026)	−0.0044* (0.0025)
Ever-treated villages	3,207	4,047

Notes: The right column replaces ShakeMap Modified Mercalli Intensity with an MMI derived independently from ShakeMap Peak Ground Acceleration through the [Wald et al. \(1999\)](#) relationship, $MMI = 3.66 \log_{10}(PGA) - 1.66$, keeping everything else fixed: the same window-aligned first-exposure timing, the same village and wave fixed effects, and village-clustered standard errors. First exposure is the first survey wave enumerated after a village’s first PGA-derived $MMI \geq VI$ event, exactly as for the ShakeMap-MMI treatment. The PGA and ShakeMap-MMI treatments identify nearly the same villages (4,047 vs 3,207 ever-treated; the two exposure indicators correlate 0.79 at the village-wave level), and every estimate reproduces the main negative effect at comparable or slightly larger magnitude, including the definition-stable *warga* outcome. The PGA-derived CS(2021) estimate is significant ($p = 0.001$) against $p = 0.000$ for ShakeMap MMI, reflecting the wider standard error of the noisier instrumental measure rather than a smaller point estimate. This establishes that the result is not an artifact of ShakeMap’s MMI construction, which blends instrumental and macroseismic intensity, as opposed to the purely instrumental PGA. An earlier calendar-year-binned PGA measure produced a spurious positive coefficient, an instance of the same survey-window timing artifact documented in Section 5; it disappears once PGA exposure is aligned to the survey calendar.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

P SNPK/NVMS: Why It Cannot Corroborate the Village-Level Result

ACLED’s Indonesia coverage begins in 2015 and so cannot reach the earthquakes that identify most main-specification cohorts, including the 2009 Padang event behind the largest treated cohort. The SNPK/NVMS incident database ([Barron et al., 2016](#)), a newspaper-based violence series independent of PODES, covers 2005–2014 and is the one source whose window includes those cohorts. We attempted the check and report here why it does not deliver one.

Kabupaten-level analysis follows [Bazzi and Gudgeon \(2021\)](#), who use the same database at the same geographic level. Kabupaten codes are harmonized using a single fixed (2014-vintage) BPS crosswalk. SNPK’s own kabupaten codes are recorded in one fixed later vintage across all incident-years rather than the code administratively valid at the time of each incident: a post-split code already appears in the 2005 incident file, years before that split existed administratively. Matching against the 2014-vintage crosswalk achieves a 99.9 percent match rate versus 97.0 percent under a naive year-specific match. The sample runs 2005–2014 and is restricted to the 18 provinces with continuous SNPK coverage over the period, since SNPK expanded to national coverage in 2012. Treatment is kabupaten-level absorbing first exposure to $\text{MMI} \geq \text{VI}$ from event dates; 37 always-treated kabupaten (first exposure before 2005) are excluded, leaving 77 ever-treated kabupaten. Kecamatan-level harmonization, which would match the main design’s geography, was not feasible: the source MFD kecamatan crosswalk ships only as an archive format not extractable in this environment.

Restricting to the 18 coverage-constant provinces does not hold recorded incidence fixed. Between 2005–2009 and 2010–2014 the share of never-treated kabupaten recording at least one identity conflict rises from 0.269 to 0.440, a factor of 1.63; among ever-treated kabupaten it rises from 0.086 to 0.190, a factor of 2.21. Both groups rise, and they rise by similar proportions. What differs is the base: never-treated kabupaten enter the window with roughly three times the recorded incidence of ever-treated ones.

That difference alone is enough to generate a negative estimate in the absence

of any treatment effect. A common proportional increase in reporting raises a high-base group by more percentage points than a low-base group, and a difference-in-differences estimated in levels reads the resulting gap as a decline among the treated. The signature is visible across outcomes. The implied level change is -0.039 for any social conflict, whose control-to-treated base ratio is 1.60; -0.054 for vigilante violence, at a base ratio of 1.77; and -0.067 for identity conflict, at a base ratio of 3.14. The estimated effect is ordered by the base gap rather than by any feature of the outcome itself.

Estimating the same specification in proportional rather than level form settles the question. Replacing the linear probability model with Poisson on incident counts, holding the kabupaten and year fixed effects and the kabupaten-level clustering unchanged, reverses every sign. Identity conflict moves from -0.0818 (SE 0.0239) in levels to $+0.514$ (SE 0.452) in proportional form, any social conflict from -0.006 (SE 0.020) to $+0.293$ (SE 0.231), and vigilante violence from -0.005 (SE 0.034) to $+0.189$ (SE 0.383). No proportional estimate is distinguishable from zero, and none points in the direction the level estimates suggest. An estimate that changes sign with the scale on which it is measured is not evidence about behaviour.

The contrast with the main analysis is the point of reporting this at all. The village-level composition result is estimated on PODES, a census whose enumeration protocol does not expand mid-sample, and it survives the identical test: the outcome-interacted specification re-estimated as Poisson preserves both the sign of each response and the ranking between communal violence and property crime. SNPK does not survive it, because what moves over the SNPK window is how much violence gets recorded rather than how much occurs. That the reporting technology rather than the underlying behaviour can drive the result is not a concern peculiar to us. Studying Indonesian crime over the same period, [Hardiman \(2025\)](#) finds that newspaper-reported incidents show no relationship with income shocks while survey-based victimisation over the same shocks does, which is the same divergence between a media-driven series and an enumeration that we encounter here.

West Sumatra, home of the Padang cohort, lies outside the 18 coverage-constant

provinces in any case, and extending the panel to include it does not recover the check. The twelve Padang-cohort kabupaten record no identity conflict in any year of 2005–2014 and an unchanging value of the broader conflict measure throughout, leaving no variation from which to identify anything.

We therefore claim no SNPK corroboration of the village-level result. The gap this leaves is real and we state it plainly: no independent violence series covers the cohorts that identify the main estimates, and the village-level result rests on PODES together with the internal-validity evidence assembled above.

Q Event Study Figures

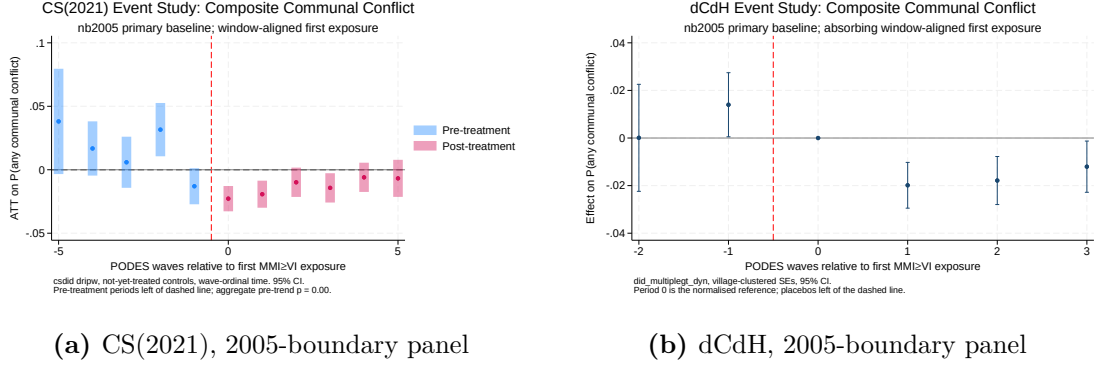


Figure 10: Event Study, 2005-boundary panel: Callaway–Sant’Anna and de Chaisemartin–D’Haultfoeuille

Note: Dynamic effects in event time, waves relative to first MMI $\geq$ VI exposure, with pointwise 95 percent confidence intervals. Outcome: any communal conflict. Panel (a) is Callaway–Sant’Anna (2021), doubly-robust, not-yet-treated controls, `long2` convention, $N = 227,045$. It is the conventional benchmark on the wave-fixed-effects panel, where its joint pre-trend test rejects, and it is shown here beside the episodic estimator; the main-text event study of Figure 1 instead uses Sun–Abraham under province-by-wave effects, for the reasons given in that caption. Panel (b) is the de Chaisemartin–D’Haultfoeuille non-absorbing interval estimator on the same panel. Under wave fixed effects the average pre-treatment lead is insignificant for both outcomes while the joint Wald test rejects; the preferred province-by-wave specification passes both (Section 5.6).

R HonestDiD Sensitivity Bounds

Table 49 reports the Rambachan and Roth (2023) HonestDiD sensitivity analysis applied to the Sun–Abraham interaction-weighted event study. We report it because its leads are tighter than the pooled estimators’, not because they are clean: the composite t_{-2} lead is +0.0138 and significant at 5 percent, which is

exactly the violation the sensitivity analysis is meant to price. The target is the immediate post-treatment effect t_0 (-0.0199^{***} for the composite outcome, -0.0075^* for *warga*). The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ allows post-treatment violations of parallel trends up to $\bar{M}$ times the largest pre-treatment violation; the breakdown value is the largest $\bar{M}$ at which the robust 95 percent identified set still excludes zero.

On a grid fine enough to locate it, the composite breakdown value lies between $\bar{M} = 0.40$, where the set is $[-0.040, -0.000]$, and $\bar{M} = 0.45$, where it is $[-0.042, +0.001]$: the sign survives post-treatment violations up to about two fifths of the largest observed pre-treatment deviation, not up to half. The definition-consistent *warga* target is far more fragile, breaking down between $\bar{M} = 0.05$ and $\bar{M} = 0.10$, consistent with its weaker precision; reporting only that it admits zero by $\bar{M} = 0.5$ would understate how early it fails. These are moderate bounds, not the near-unbounded robustness a perfectly clean pre-period would deliver. We report them as the quantitative statement of how much residual violation the result can absorb, alongside the not-yet-treated-only estimate. The doubly-robust property of the estimator is a separate safeguard, against misspecification of the nuisance models rather than against failure of conditional parallel trends itself.

The same analysis applied to the Callaway–Sant’Anna estimator on the wave-fixed-effects panel is less forgiving, though not as unforgiving as we previously reported. Under exact parallel trends the identified set for the aggregated composite ATT is $[-0.025, -0.008]$, simply the headline confidence interval, and it excludes zero; once a relative-magnitudes violation of $\bar{M} = 0.5$ is allowed it widens to $[-0.057, +0.024]$ and admits zero, so the breakdown value for that target lies between $\bar{M} = 0$ and $\bar{M} = 0.5$ rather than at zero. The target that isolates the immediate effect is more durable on the same panel, still excluding zero at $\bar{M} = 0.5$ ($[-0.042, -0.004]$). What binds on this panel is the smoothness restriction Δ^{SD} , which bounds how far the post-treatment trend may depart from a linear extrapolation of the pre-trend: it reaches zero at its mildest deviation ($[-0.041, 0.000]$). Reporting both restriction families follows the current difference-in-differences synthesis (Roth et al., 2023). Under the preferred specification the same target instead breaks down between $\bar{M} = 0.25$ and $\bar{M} = 0.50$, for the

anchoring reason set out in Section J.2. The gap from the Sun–Abraham bounds is mechanical, not evidential, and in two distinct senses. First, the two bound different target parameters on different comparison groups: the Sun–Abraham figure is an immediate effect estimated without covariates on never-treated controls, while the primary estimator conditions on covariates and uses not-yet-treated controls. Second, $\bar{M}$ is scaled by each estimator’s own largest pre-treatment deviation, so an estimator with smoother leads mechanically reports a larger $\bar{M}$ on identical data. Converting both to absolute permitted deviation removes the illusion: the primary estimator tolerates up to about 0.009 and the Sun–Abraham estimate about 0.009 to 0.017, so in outcome units the two are close to tied despite reported breakdown values that differ by a factor of two. $\bar{M}$ is thus not a common yardstick across estimators, and we do not treat the Sun–Abraham figure as evidence that its identifying assumption is stronger.

A further bound is worth reporting because it is the one place the sensitivity analysis does real work rather than restating an insignificant target. The effect one wave after exposure, -0.0173 ($p < 0.001$), is the event-study coefficient that survives a simultaneous band across the whole path. Applying the same analysis to it gives a breakdown between $\bar{M} = 0$ and $\bar{M} = 0.25$ under relative magnitudes, and no margin at all under smoothness. The reason is arithmetic rather than estimator choice: the largest deviation between consecutive pre-treatment leads is 0.037, about 2.2 times the effect being defended, so even a small multiple of that yardstick swamps the estimate.

Table 49: HonestDiD Relative-Magnitudes Identified Sets (Sun–Abraham t_0)

$\bar{M}$	Composite ($t_0 = -0.0199$)		Warga ($t_0 = -0.0075$)	
	lower	upper	lower	upper
0.0 (original)	−0.029	−0.010	−0.015	0.000
0.5	−0.044	0.003	−0.022	0.006
1.0	−0.061	0.021	−0.032	0.016
1.5	−0.081	0.040	−0.043	0.027
2.0	−0.100	0.059	−0.054	0.038

Notes: 95 percent robust identified sets under $\Delta^{RM}(\bar{M})$ (C-LF method, $\alpha = 0.05$), computed on the window-aligned Sun–Abraham event study (2 pre-periods, 4 post-periods) for the immediate effect t_0 , under wave fixed effects. Both outcomes admit zero at $\bar{M} = 0.5$. The corresponding analysis on the preferred province-by-wave specification, where the pre-treatment leads are smaller and the breakdown point correspondingly higher, is reported in Table 28.

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